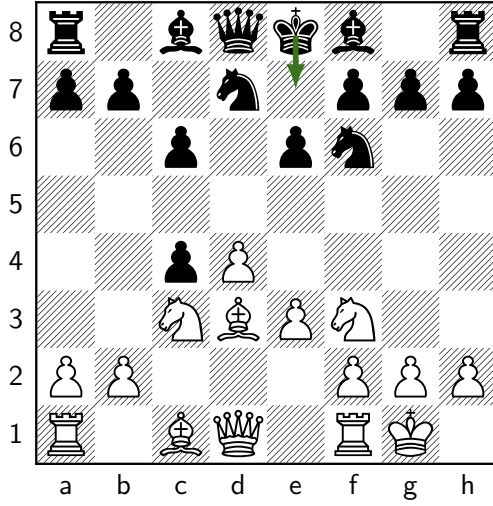


Melanie-BOTS (CCRL engines) — Semi-Slav

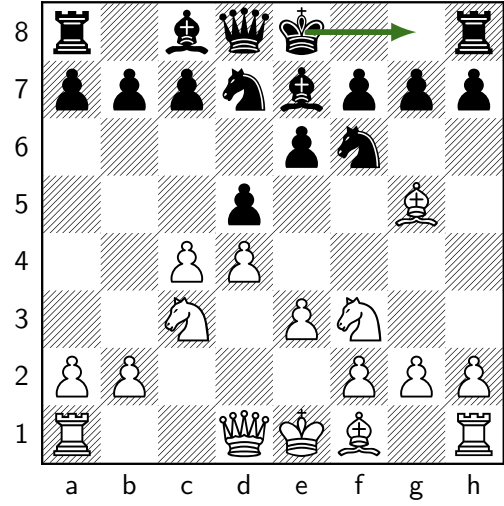
Melanie (Deep_Vision)

Abstract

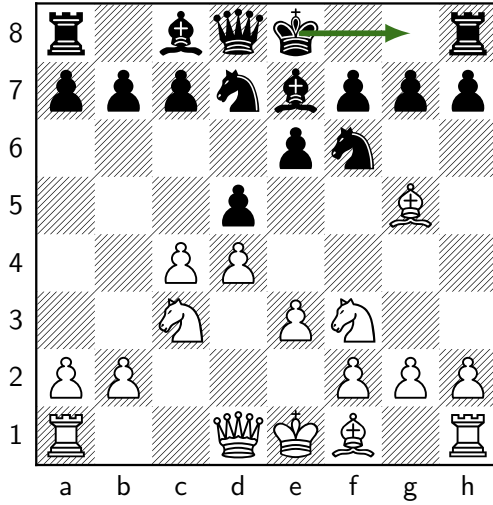
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



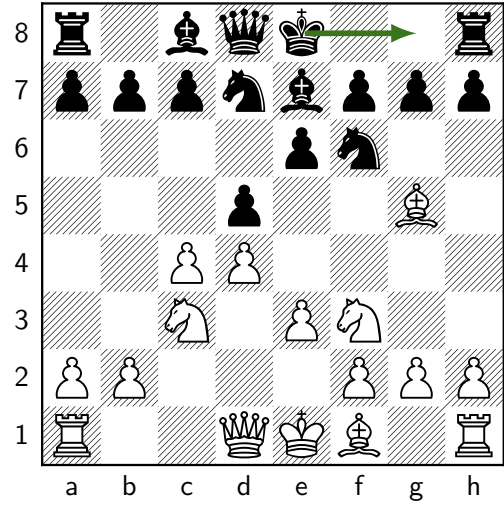
(a)



(b)

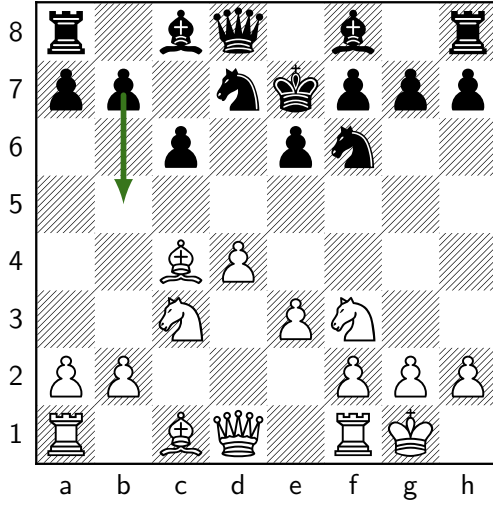


(c)

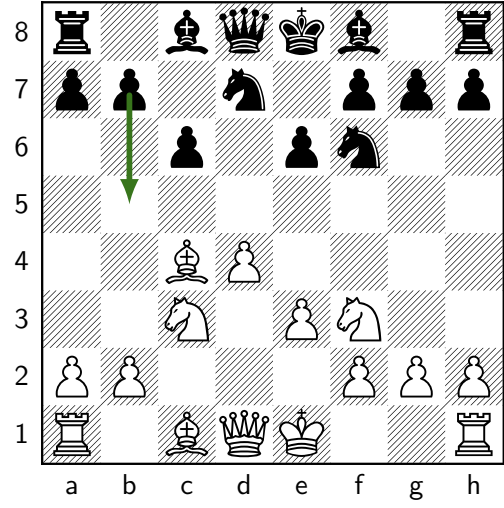


(d)

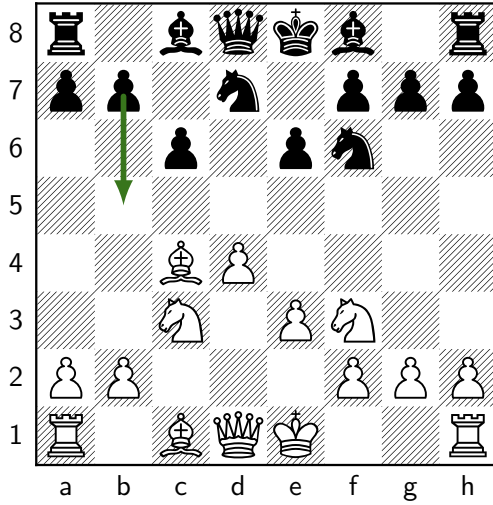
Figure 1: **Move 7, Black: Ke7**. (a) Our position; Melanie plays Ke7 (e8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#25` (ply 11) [structural distance $d=40$, QAP obj=25]: the master played 0-0; under μ our *Ke1* plays the role of that game's *Ke1*; cross-type roles: our *nc3* as their *Ra8*, our *Nd7* as their *Bc8*; transferred move e8e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2990` (ply 11) [structural distance $d=40$, QAP obj=25]: the master played 0-0; under μ our *Ke1* plays the role of that game's *Ke1*; cross-type roles: our *nc3* as their *Ra8*, our *Nd7* as their *Bc8*; transferred move e8e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2991` (ply 11) [structural distance $d=40$, QAP obj=25]: the master played 0-0; under μ our *Ke1* plays the role of that game's *Ke1*; cross-type roles: our *nc3* as their *Ra8*, our *Nd7* as their *Bc8*; transferred move e8e7 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Ke7 (summed QAP objective 243, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



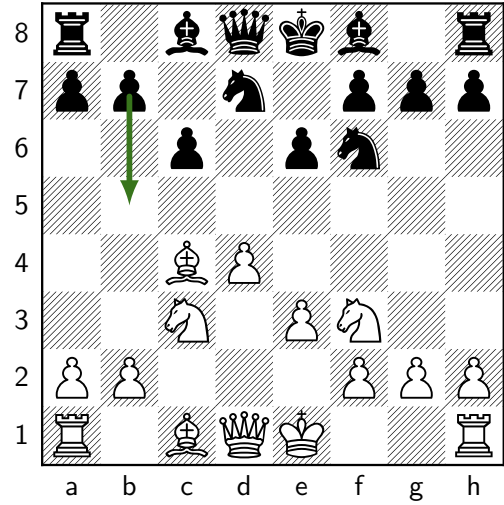
(a)



(b)

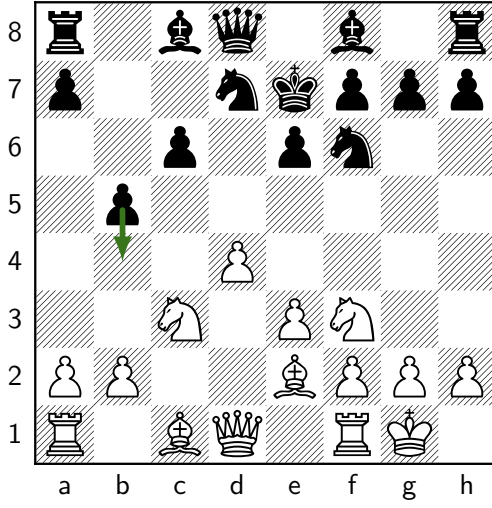


(c)

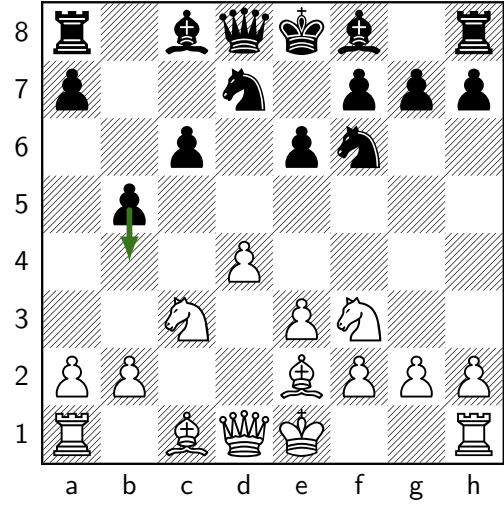


(d)

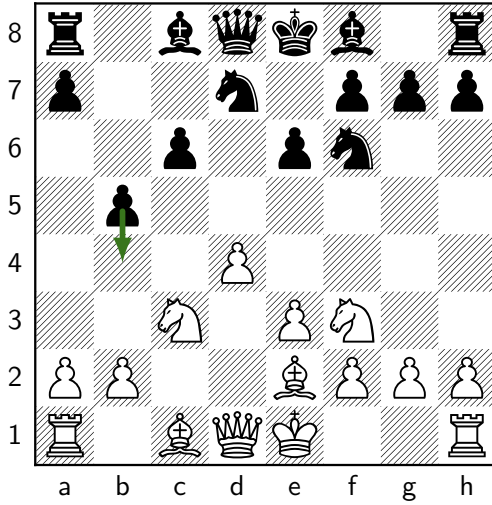
Figure 2: **Move 8, Black: b5.** (a) Our position; Melanie plays b5 (b7b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#119` (ply 13) [structural distance $d=56$, QAP obj=27]: the master played b5; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Bf8* as their *Qd8*, our *nf3* as their *Bf8*; transferred move b7b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#120` (ply 13) [structural distance $d=56$, QAP obj=27]: the master played b5; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Bf8* as their *Qd8*, our *nf3* as their *Bf8*; transferred move b7b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#280` (ply 13) [structural distance $d=56$, QAP obj=27]: the master played b5; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Bf8* as their *Qd8*, our *nf3* as their *Bf8*; transferred move b7b5 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to b5 (summed QAP objective 269, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



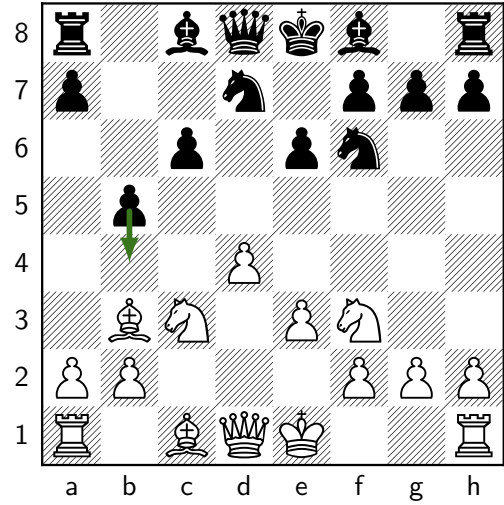
(a)



(b)

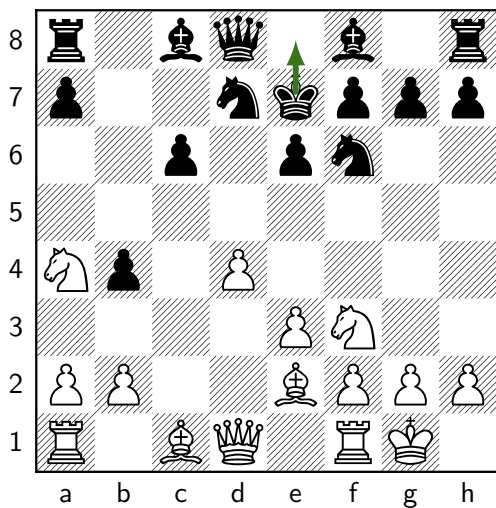


(c)

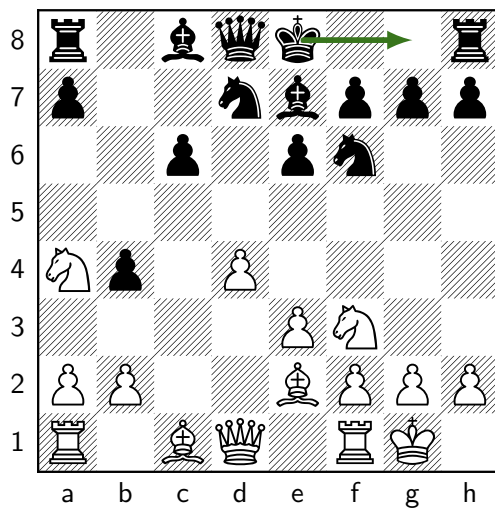


(d)

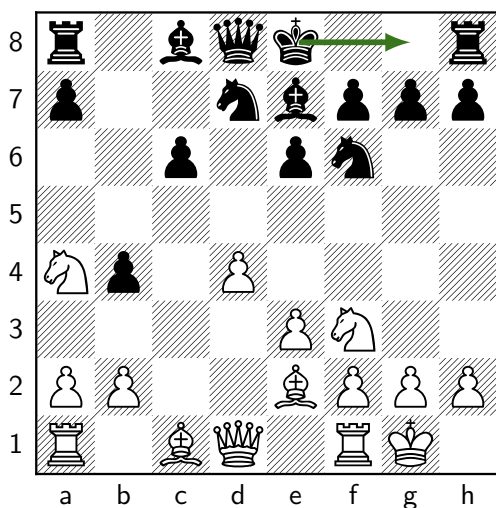
Figure 3: **Move 9, Black: b4.** (a) Our position; Melanie plays **b4** (**b5b4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2365` (ply 15) [structural distance $d=56$, QAP obj=53]: the master played **b4**; under μ our Pb_4 plays the role of that game’s Pb_4 ; cross-type roles: our pf_2 as their Bc_8 , our Bf_8 as their Qd_8 ; transferred move **b5b4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2366` (ply 15) [structural distance $d=56$, QAP obj=53]: the master played **b4**; under μ our Pb_4 plays the role of that game’s Pb_4 ; cross-type roles: our pf_2 as their Bc_8 , our Bf_8 as their Qd_8 ; transferred move **b5b4** (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3511` (ply 15) [structural distance $d=68$, QAP obj=34]: the master played **b4**; under μ our Pb_4 plays the role of that game’s Pb_4 ; cross-type roles: our Pc_6 as their Ra_8 , our Nd_7 as their Bc_8 ; transferred move **b5b4** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **b4** (summed QAP objective 175, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



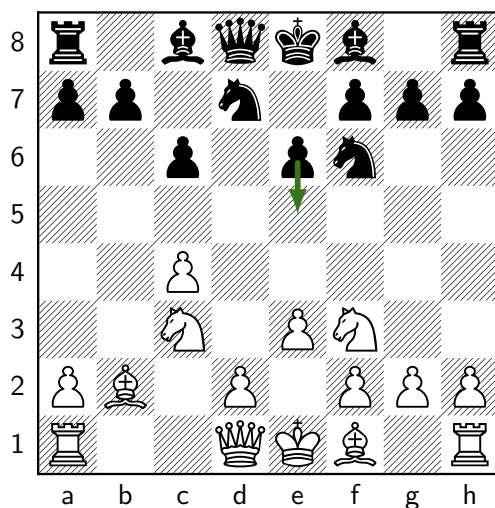
(a)



(b)

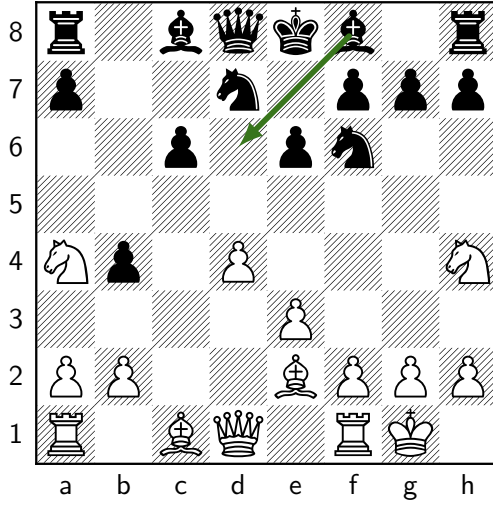


(c)

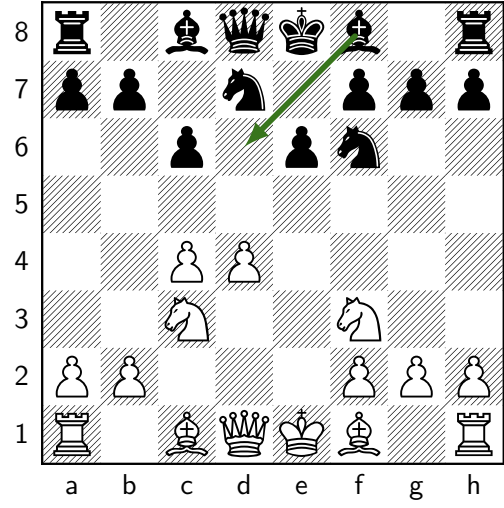


(d)

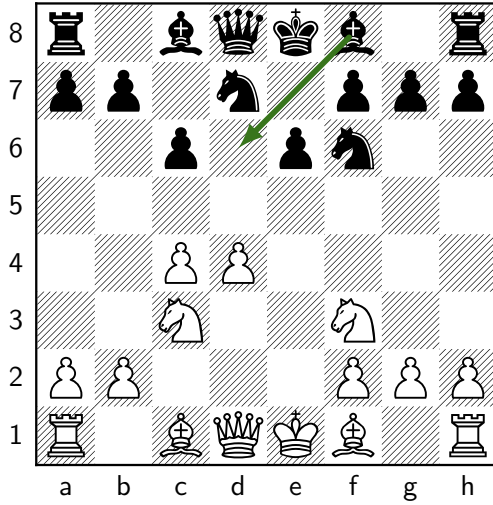
Figure 4: **Move 10, Black: Ke8.** (a) Our position; Melanie plays Ke8 (e7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2365` (ply 19) [structural distance $d=25$, QAP obj=8]: the master played 0-0; under μ our *Ke2* plays the role of that game’s *Ke1*; cross-type roles: our *Pb4* as their *Ra8*, our *Nd7* as their *Bc8*; transferred move e7e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2366` (ply 19) [structural distance $d=25$, QAP obj=8]: the master played 0-0; under μ our *Ke2* plays the role of that game’s *Ke1*; cross-type roles: our *Pb4* as their *Ra8*, our *Nd7* as their *Bc8*; transferred move e7e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#740` (ply 13) [structural distance $d=43$, QAP obj=8]: the master played e5; under μ our *Pe3* plays the role of that game’s *Pe3*; cross-type roles: our *na4* as their *Bc8*, our *Nf6* as their *Bf8*; transferred move e6e5. Synthesis: 2 retrieved analogs transferred to Ke8 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



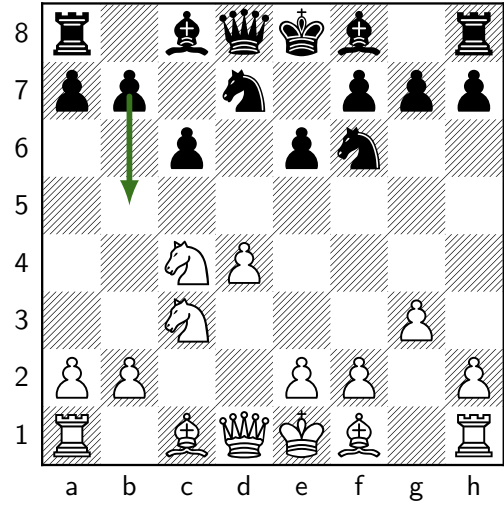
(a)



(b)

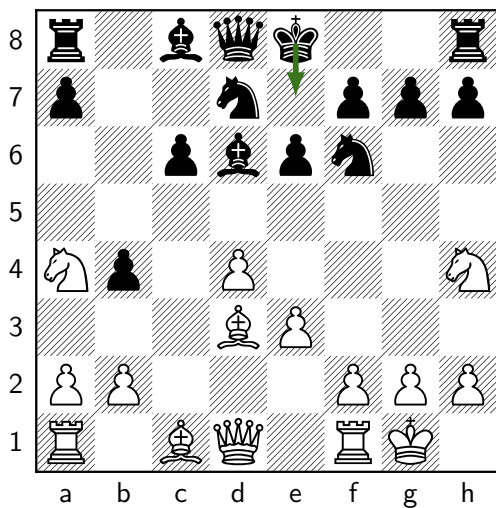


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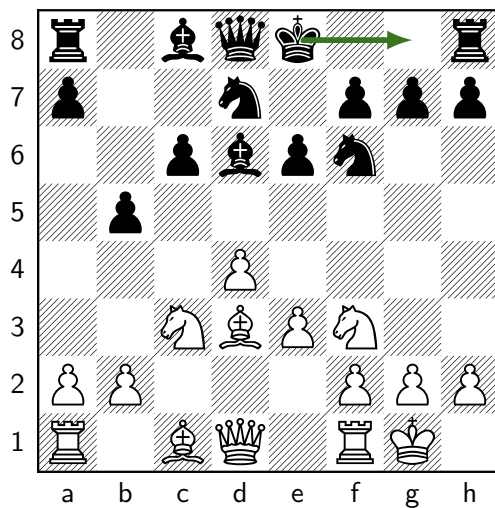


(d)

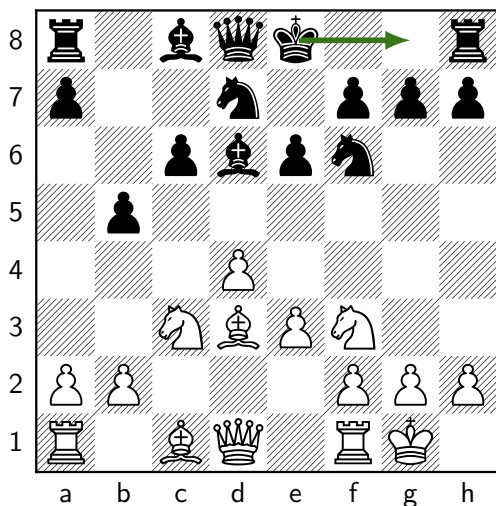
Figure 5: **Move 11, Black: Bd6.** (a) Our position; Melanie plays Bd6 (f8d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#884` (ply 13) [structural distance $d=69$, QAP obj=9]: the master played Bd6; under μ our *Bf1* plays the role of that game’s *Bf1*; cross-type roles: our *pf2* as their *Ra8*, our *na4* as their *Bc8*; transferred move f8d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#885` (ply 13) [structural distance $d=69$, QAP obj=9]: the master played Bd6; under μ our *Bf1* plays the role of that game’s *Bf1*; cross-type roles: our *pf2* as their *Ra8*, our *na4* as their *Bc8*; transferred move f8d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3238` (ply 13) [structural distance $d=66$, QAP obj=9]: the master played b5; under μ our *Pc3* plays the role of that game’s *Pb2*; cross-type roles: our *na4* as their *Ra8*, our *pd4* as their *Bf8*; transferred move c6c5. Synthesis: 2 retrieved analogs transferred to Bd6 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



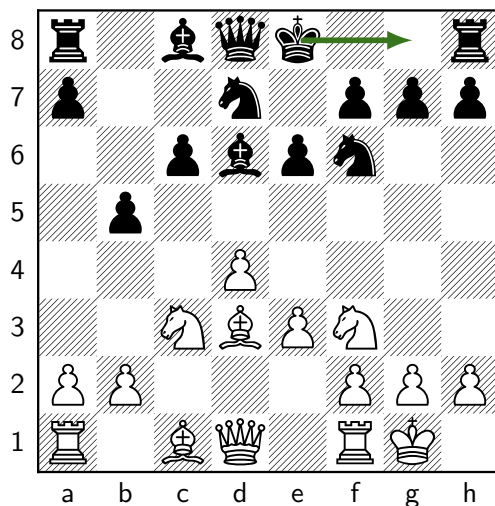
(a)



(b)

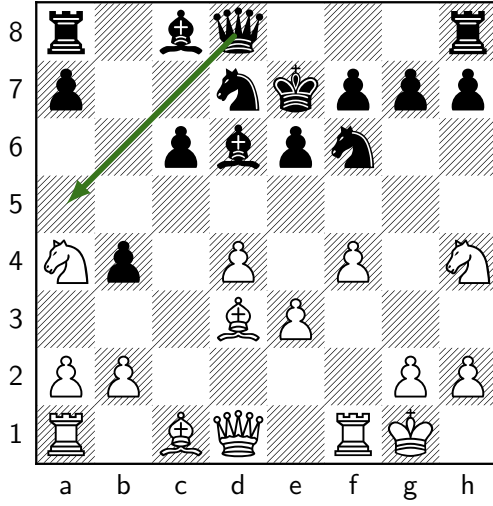


(c)

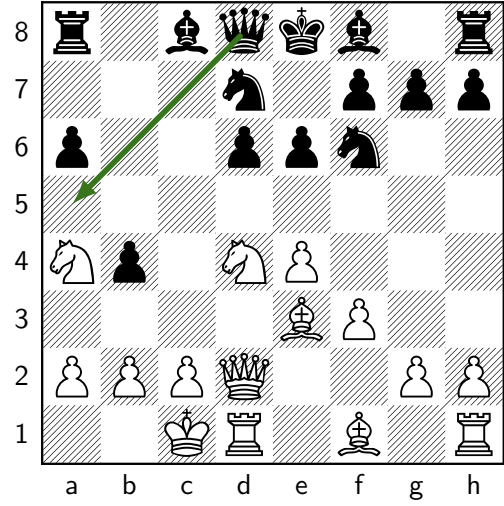


(d)

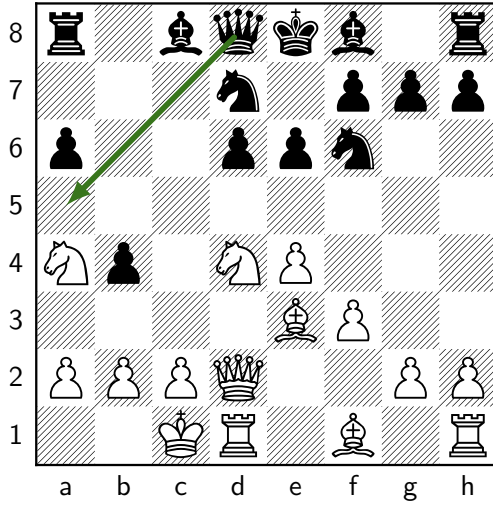
Figure 6: **Move 12, Black: Ke7.** (a) Our position; Melanie plays Ke7 (e8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3024` (ply 17) [structural distance $d=58$, QAP obj=49]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Nf6* as their *Pc6*, our *bd3* as their *nc3*; transferred move e8e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3025` (ply 17) [structural distance $d=58$, QAP obj=49]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Nf6* as their *Pc6*, our *bd3* as their *nc3*; transferred move e8e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2598` (ply 17) [structural distance $d=58$, QAP obj=49]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Nf6* as their *Pc6*, our *bd3* as their *nc3*; transferred move e8e7 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Ke7 (summed QAP objective 397, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



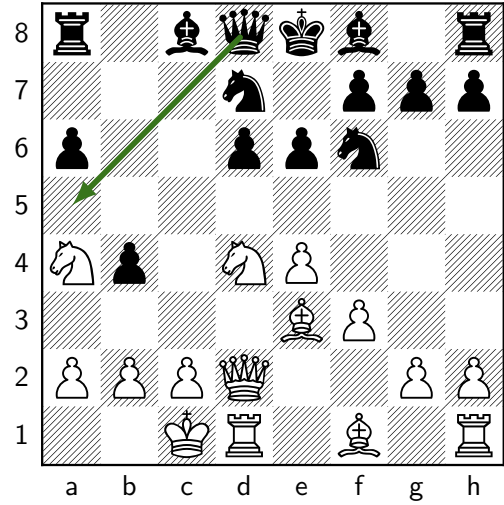
(a)



(b)

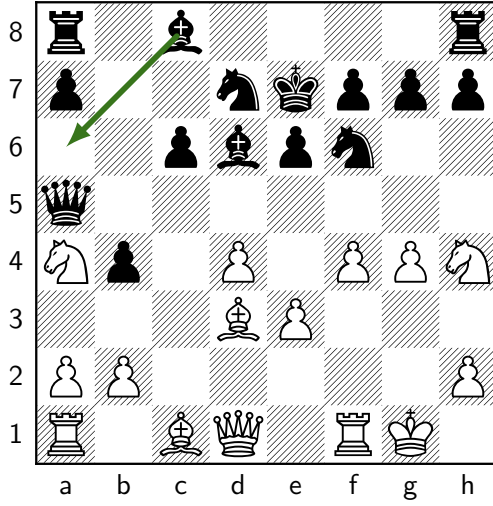


(c)

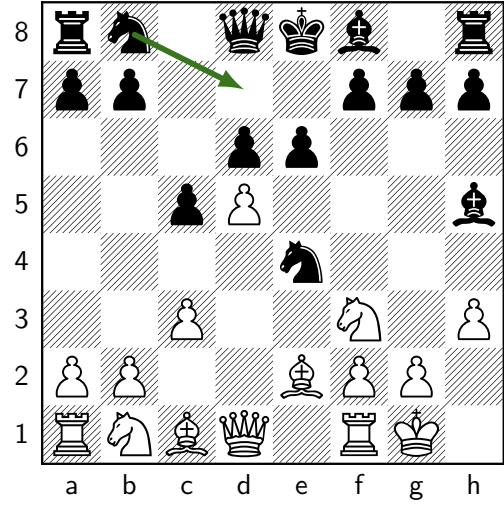


(d)

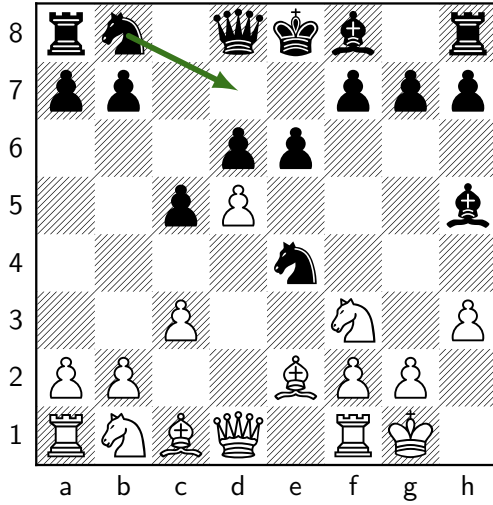
Figure 7: **Move 13, Black: Qa5.** (a) Our position; Melanie plays Qa5 (d8a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2844` (ply 19) [structural distance $d=79$, QAP obj=48]: the master played Qa5; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our pb2 as their Ra8, our Nd7 as their Rh8; transferred move d8a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2845` (ply 19) [structural distance $d=79$, QAP obj=48]: the master played Qa5; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our pb2 as their Ra8, our Nd7 as their Rh8; transferred move d8a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3776` (ply 19) [structural distance $d=79$, QAP obj=48]: the master played Qa5; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our pb2 as their Ra8, our Nd7 as their Rh8; transferred move d8a5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qa5 (summed QAP objective 286, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



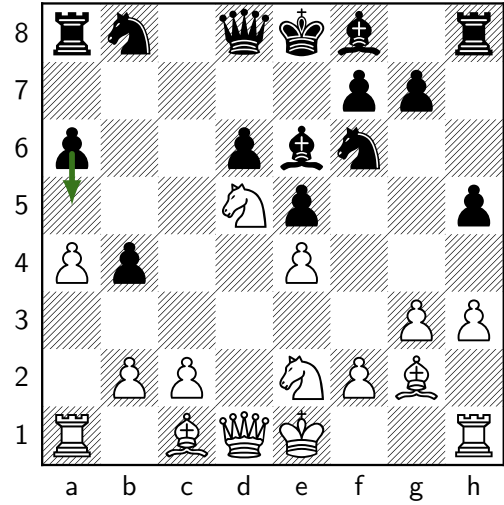
(a)



(b)

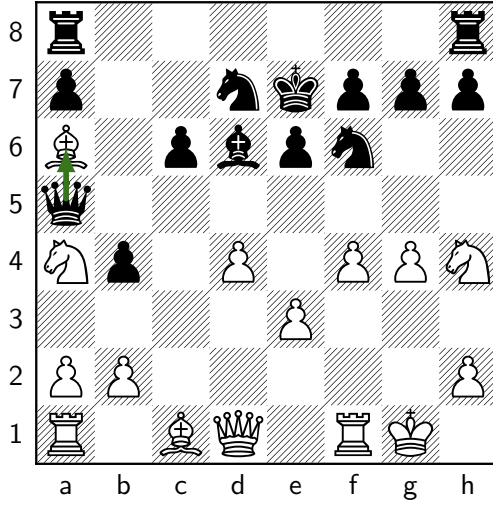


(c)

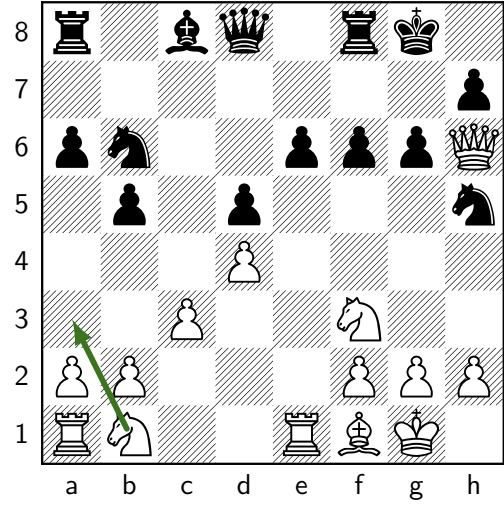


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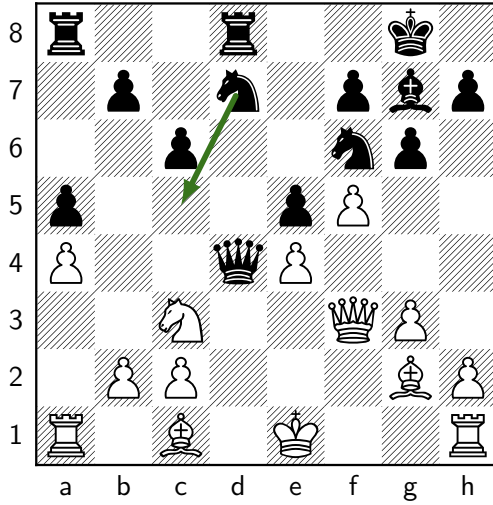
Figure 8: **Move 14, Black: Ba6.** (a) Our position; Melanie plays Ba6 (c8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#792` (ply 15) [structural distance $d=106$, QAP obj=13]: the master played Nd7; under μ our Bc1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Bd6 as their Qd8, our pf4 as their Bf8; transferred move c8a6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#793` (ply 15) [structural distance $d=106$, QAP obj=13]: the master played Nd7; under μ our Bc1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Bd6 as their Qd8, our pf4 as their Bf8; transferred move c8a6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1227` (ply 21) [structural distance $d=103$, QAP obj=31]: the master played a5; under μ our Pe3 plays the role of that game’s Pa3; cross-type roles: our Pg7 as their Ra8, our pd4 as their Nb8; transferred move e6e5. Synthesis: 2 retrieved analogs transferred to Ba6 (summed QAP objective 26, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



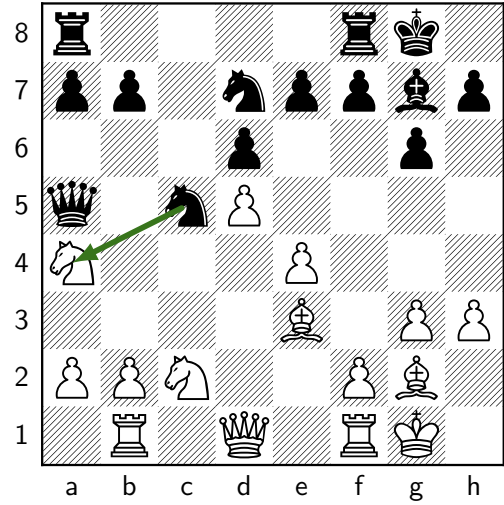
(a)



(b)

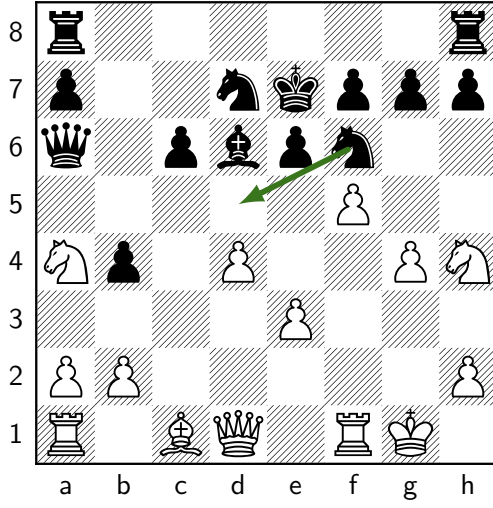


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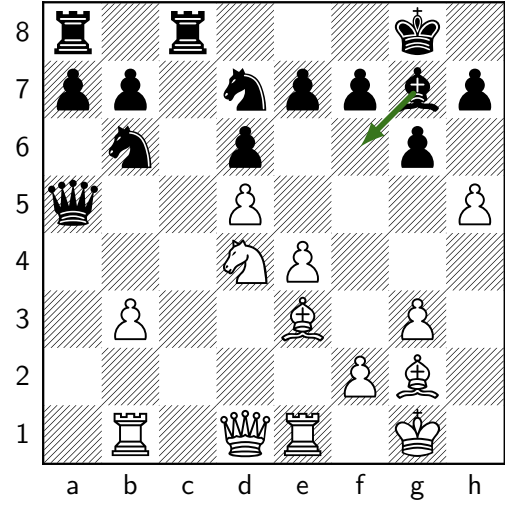


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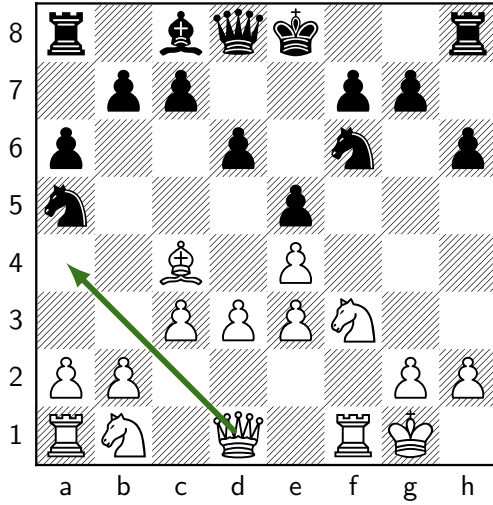
Figure 9: **Move 15, Black: Qxa6.** (a) Our position; Melanie plays Qxa6 (a5a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3525` (ply 30) [structural distance $d=108$, QAP obj=53]: the master played Nc3; under μ our Ph2 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Nd7 as their Re1, our Nf6 as their Bf1; transferred move h7h5. (c) Retrieved from `ccrl_top_p0003#410` (ply 41) [structural distance $d=91$, QAP obj=33]: the master played Nc5; under μ our Nf3 plays the role of that game’s Nd2; cross-type roles: our Bd6 as their Rd8, our nh4 as their Pf7; transferred move f6h5. (d) Retrieved from `ccrl_top_p0001#1038` (ply 31) [structural distance $d=97$, QAP obj=32]: the master played Nxa4; under μ our Qa4 plays the role of that game’s Nc4 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Ra8, our Bd6 as their Rf8; transferred move a5a4. Synthesis: no retrieved analog transferred to Qxa6 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



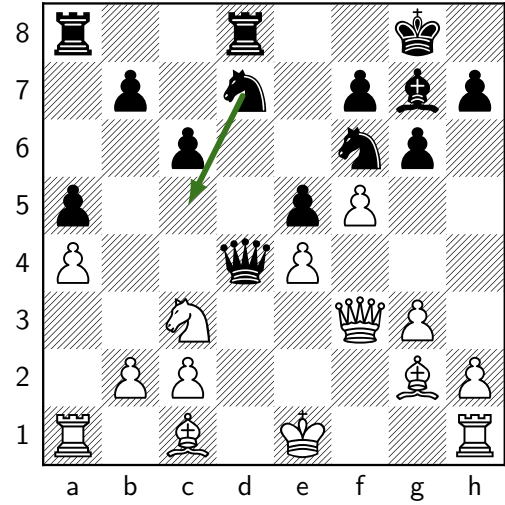
(a)



(b)

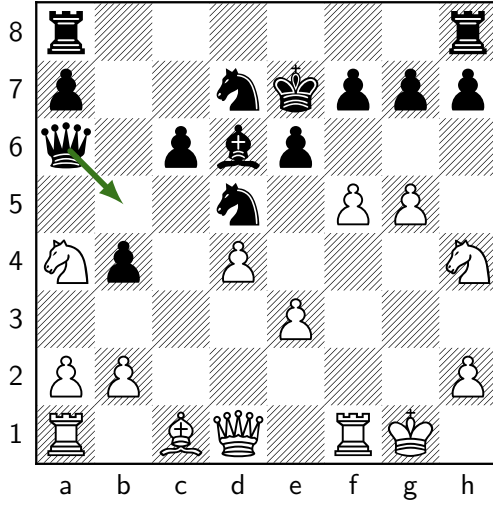


(c)

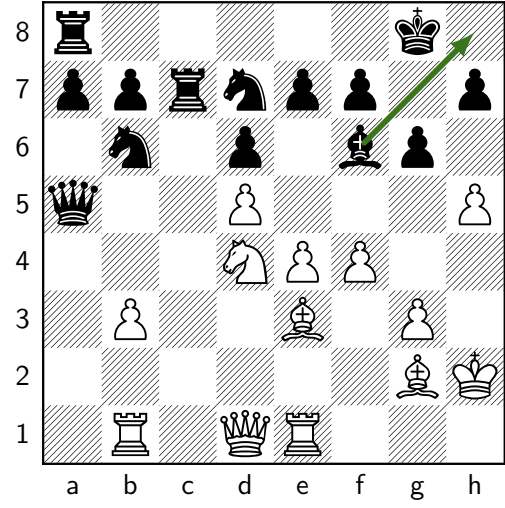


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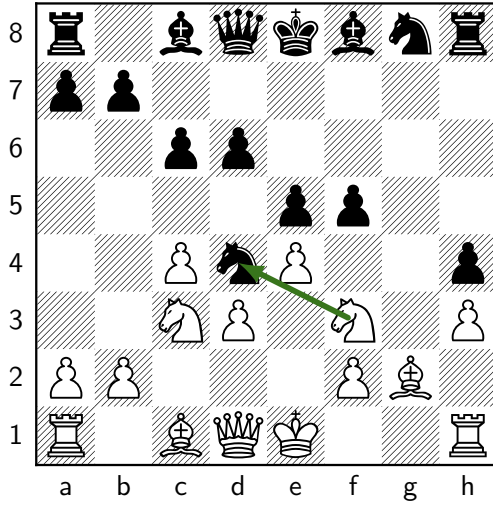
Figure 10: **Move 16, Black: Nd5.** (a) Our position; Melanie plays Nd5 (f6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1038` (ply 41) [structural distance $d=95$, QAP obj=61]: the master played Bf6; under μ our Nf3 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Pc6 as their Ra8, our Bd6 as their Rc8; transferred move f6d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2069` (ply 18) [structural distance $d=102$, QAP obj=35]: the master played Qa4+; under μ our Pb5 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our nh4 as their Ra1, our Rh8 as their Nb1; transferred move b4b3. (d) Retrieved from `ccr1_top_p0003#410` (ply 41) [structural distance $d=74$, QAP obj=34]: the master played Nc5; under μ our Nd2 plays the role of that game’s Nd2; cross-type roles: our Pb4 as their Ra8, our Bd6 as their Rd8; transferred move d7e5. Synthesis: 1 retrieved analog transferred to Nd5 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



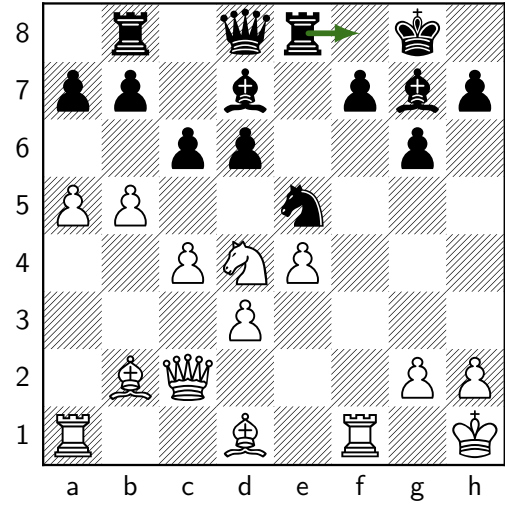
(a)



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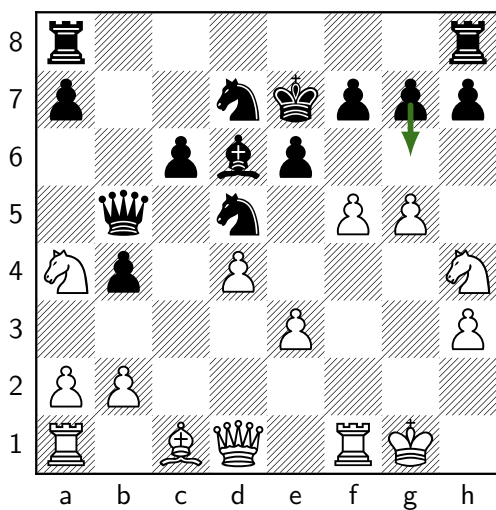


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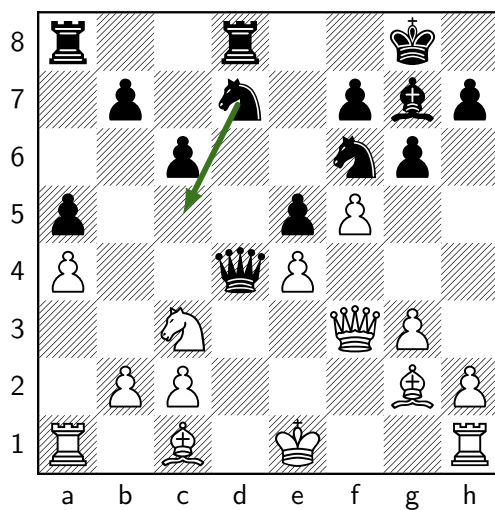


(d)

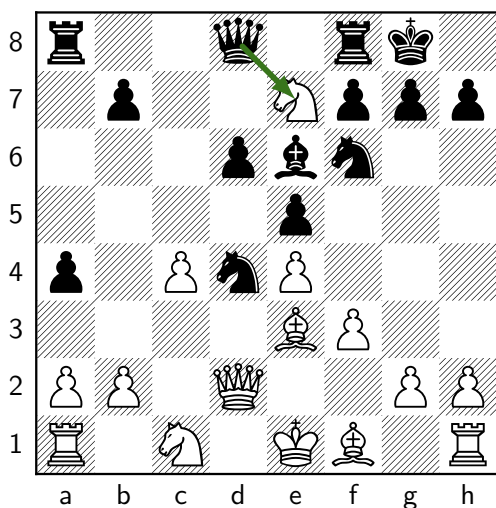
Figure 11: **Move 17, Black: Qb5.** (a) Our position; Melanie plays Qb5 (a6b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1038` (ply 45) [structural distance $d=107$, QAP obj=59]: the master played Bb8; under μ our $Qa3$ plays the role of that game’s $Bf3$ (a **cross-type** correspondence); cross-type roles: our $ph2$ as their $Ra8$, our $pb2$ as their $Rc7$; transferred move a6b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3365` (ply 20) [structural distance $d=113$, QAP obj=56]: the master played Nxd4; under μ our $Qa3$ plays the role of that game’s $Nf3$ (a **cross-type** correspondence); cross-type roles: our $Nd5$ as their $Bc1$, our $Pa7$ as their $Qd1$; transferred move a6a4. (d) Retrieved from `ccr1_top_p0001#1660` (ply 43) [structural distance $d=108$, QAP obj=49]: the master played Rf8; under μ our $Bd3$ plays the role of that game’s $Re1$ (a **cross-type** correspondence); cross-type roles: our $pb2$ as their $Qd8$, our $Ph7$ as their $Bd7$; transferred move d6b8. Synthesis: 1 retrieved analog transferred to Qb5 (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



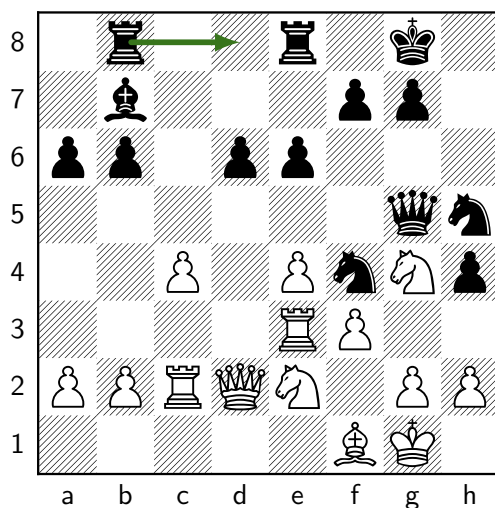
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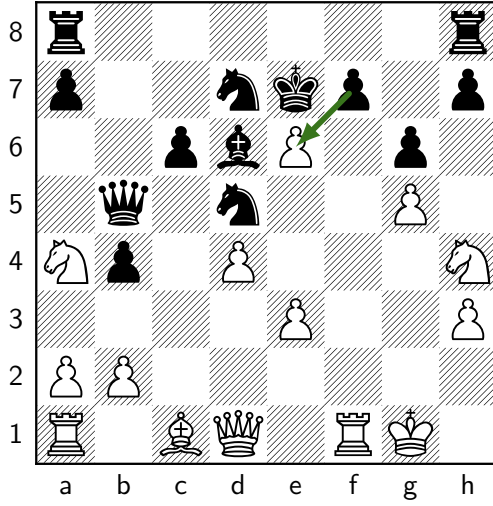


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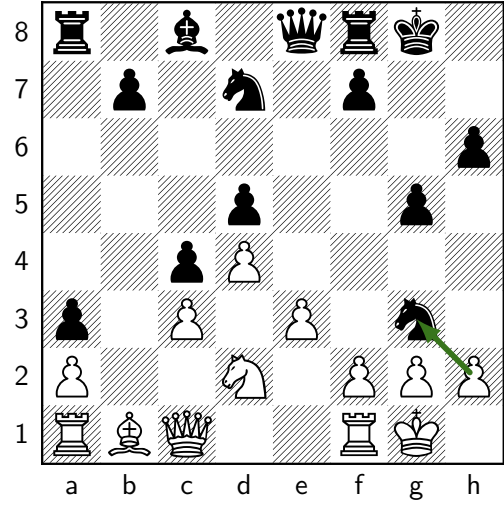


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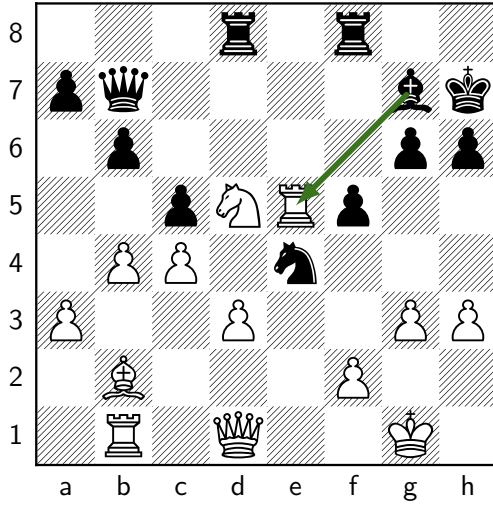
Figure 12: **Move 18, Black: g6**. (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#410` (ply 41) [structural distance $d=108$, QAP obj=43]: the master played Nc5; under μ our *Pg2* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *pb2* as their *Ra8*, our *Nd5* as their *Rd8*; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3005` (ply 25) [structural distance $d=115$, QAP obj=8]: the master played Qxe7; under μ our *Pg2* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Ra8*, our *Bd6* as their *Rf8*; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3008` (ply 49) [structural distance $d=109$, QAP obj=35]: the master played Rbd8; under μ our *Ph2* plays the role of that game’s *Rb1* (a **cross-type** correspondence); cross-type roles: our *Nd5* as their *Re8*, our *pb2* as their *Bb7*; transferred move h7h6. Synthesis: 2 retrieved analogs transferred to g6 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



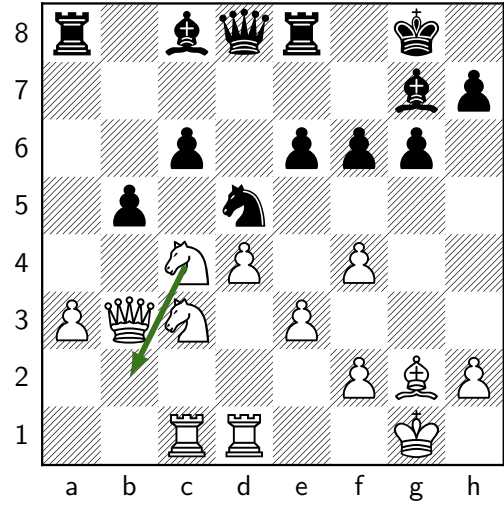
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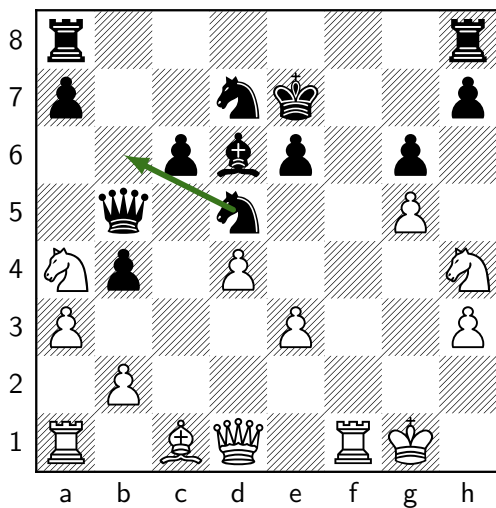


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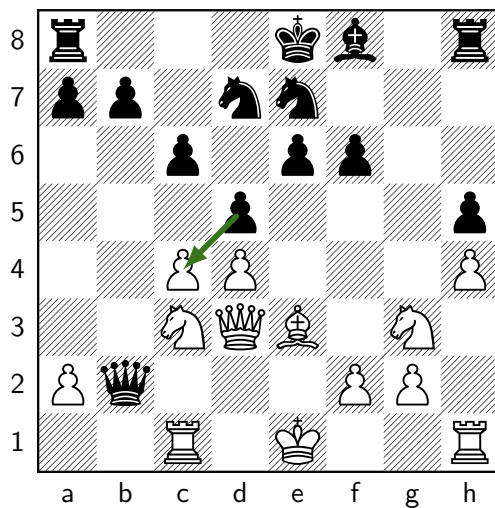


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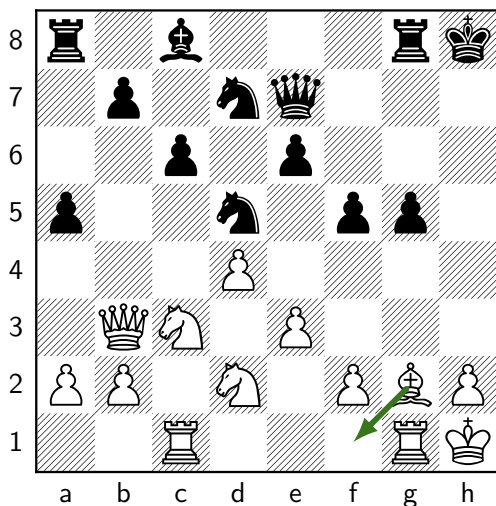
Figure 13: **Move 19, Black: fxe6.** (a) Our position; Melanie plays fxe6 (f7e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#764 (ply 36) [structural distance $d=111$, QAP obj=43]: the master played hgx3; under μ our Pf2 plays the role of that game’s Ph2; cross-type roles: our Pa7 as their Ra1, our nh4 as their Bb1; transferred move f7e6 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1225 (ply 47) [structural distance $d=108$, QAP obj=66]: the master played Bxe5; under μ our Pf2 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Qb5 as their Rd8, our Pc6 as their Rf8; transferred move f7f5. (d) Retrieved from ccr1_top_p0001#3659 (ply 38) [structural distance $d=106$, QAP obj=58]: the master played Nb2; under μ our Nd2 plays the role of that game’s Nc4; cross-type roles: our Nd5 as their Rc1, our Bd6 as their Rd1; transferred move d7b6. Synthesis: 1 retrieved analog transferred to fxe6 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



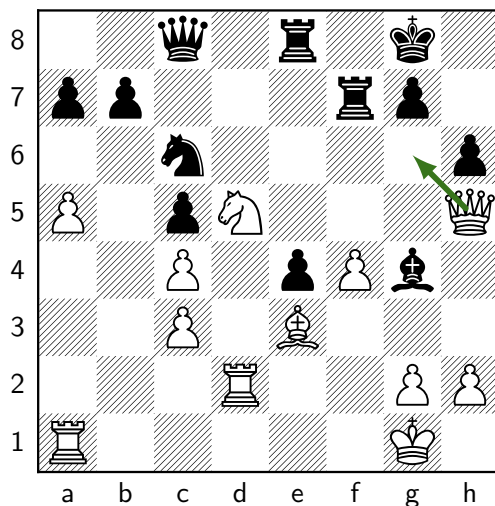
(a)



(b)



(c)



(d)

Figure 14: **Move 20, Black: N5b6.** (a) Our position; Melanie plays N5b6 (d5b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1166` (ply 27) [structural distance $d=102$, QAP obj=24]: the master played dxc4; under μ our Nd4 plays the role of that game’s Pd4 (a **cross-type** correspondence); cross-type roles: our pa3 as their Bf8, our bc1 as their Rh8; transferred move d5e3. (c) Retrieved from `ccr1_top_p0002#2892` (ply 36) [structural distance $d=98$, QAP obj=24]: the master played Bf1; under μ our Qb4 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Rc1, our Bd6 as their Rg1; transferred move b5b6. (d) Retrieved from `ccr1_top_p0002#2562` (ply 50) [structural distance $d=96$, QAP obj=16]: the master played Qg6; under μ our Qb4 plays the role of that game’s Qh5; cross-type roles: our pd4 as their Rd2, our Nd5 as their Be3; transferred move b5e2. Synthesis: no retrieved analog transferred to N5b6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

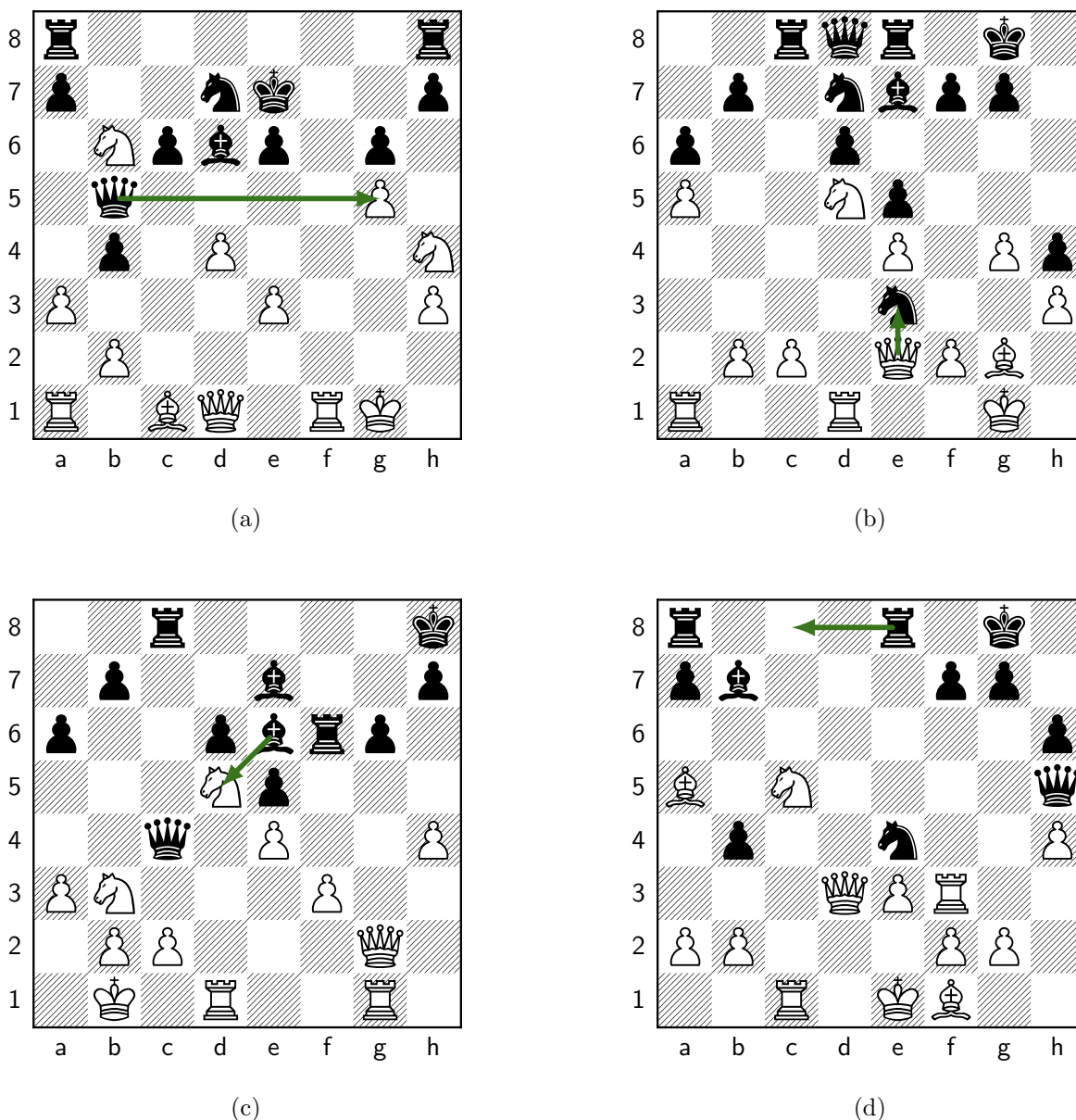
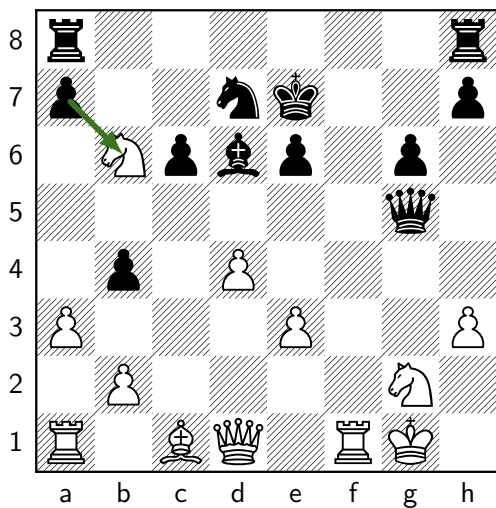
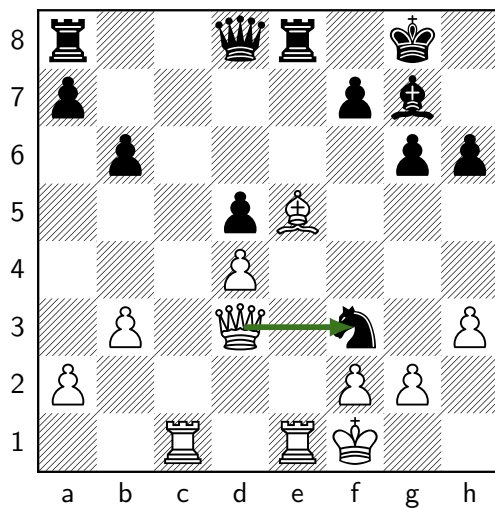


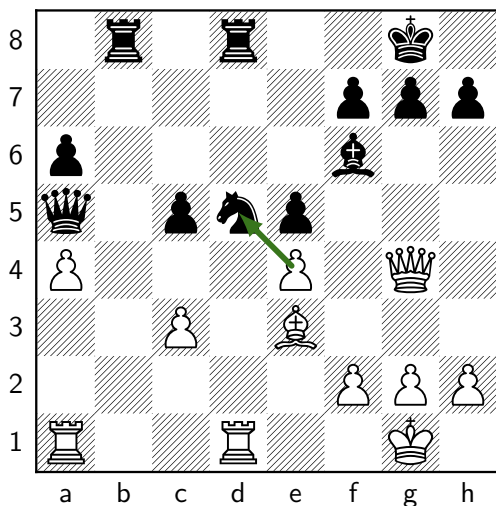
Figure 15: **Move 21, Black: Qxg5+.** (a) Our position; Melanie plays Qxg5+ (b5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1228` (ply 40) [structural distance $d=87$, QAP obj=127]: the master played Qxe3; under μ our $Qb4$ plays the role of that game’s $Qe2$; cross-type roles: our $ph3$ as their $Ra1$, our $Bd6$ as their $Pc2$; transferred move b5b6. (c) Retrieved from `ccr1_top_p0003#2539` (ply 45) [structural distance $d=92$, QAP obj=114]: the master played Bxd5; under μ our $Pa2$ plays the role of that game’s $Be3$ (a **cross-type** correspondence); cross-type roles: our $Pc6$ as their $Rc8$, our $Ra8$ as their $Be7$; transferred move a7b6. (d) Retrieved from `ccr1_top_p0001#2730` (ply 43) [structural distance $d=83$, QAP obj=78]: the master played Rec8; under μ our $Bd3$ plays the role of that game’s $Re1$ (a **cross-type** correspondence); cross-type roles: our $nh4$ as their $Ra8$, our $nb6$ as their $Bb7$; transferred move d6b8. Synthesis: no retrieved analog transferred to Qxg5+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



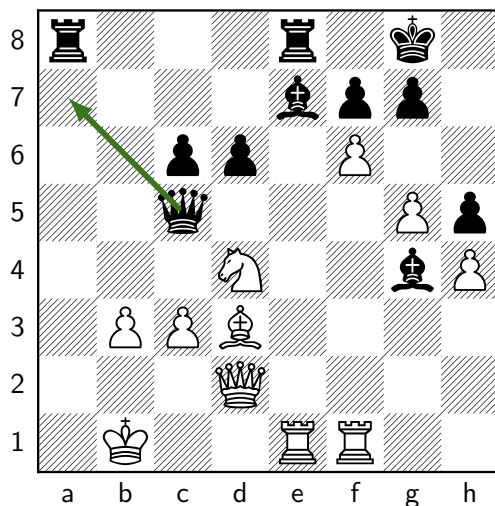
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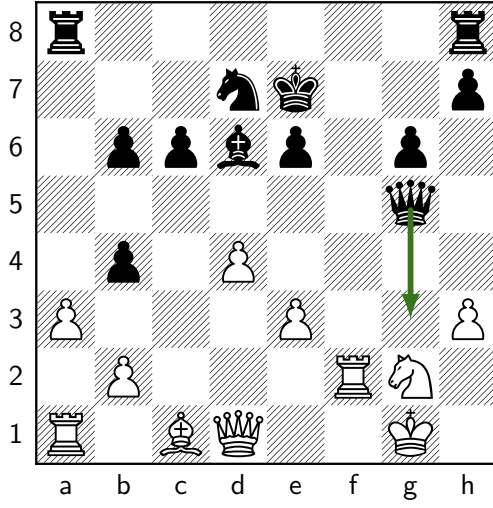


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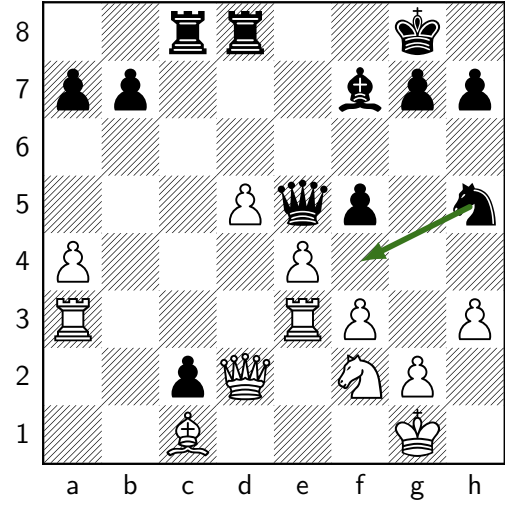


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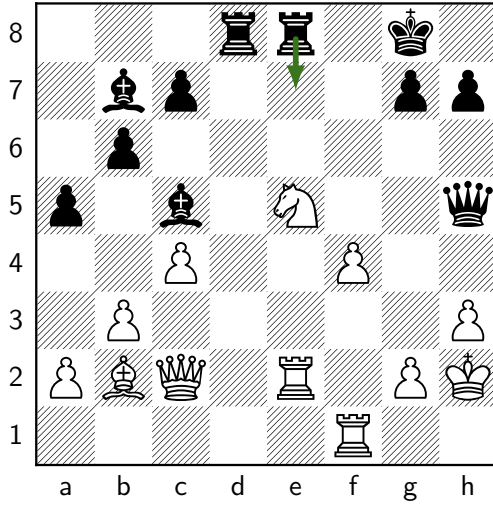
Figure 16: **Move 22, Black: axb6.** (a) Our position; Melanie plays **axb6** (**a7b6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2434` (ply 44) [structural distance $d=77$, QAP obj=87]: the master played **Qxf3**; under μ our *Pa2* plays the role of that game’s *Qd3* (a **cross-type** correspondence); cross-type roles: our *Nd7* as their *Pa2*, our *ng2* as their *Be5*; transferred move **a7b6** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#824` (ply 52) [structural distance $d=77$, QAP obj=81]: the master played **exd5**; under μ our *Pa2* plays the role of that game’s *Pe4*; cross-type roles: our *bc1* as their *Ra1*, our *Nd7* as their *Rd1*; transferred move **a7b6** (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1960` (ply 53) [structural distance $d=84$, QAP obj=53]: the master played **Qa7**; under μ our *Pa2* plays the role of that game’s *Qc4* (a **cross-type** correspondence); cross-type roles: our *Pb4* as their *Ra8*, our *Qg5* as their *Re8*; transferred move **a7b6** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **axb6** (summed QAP objective 221, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



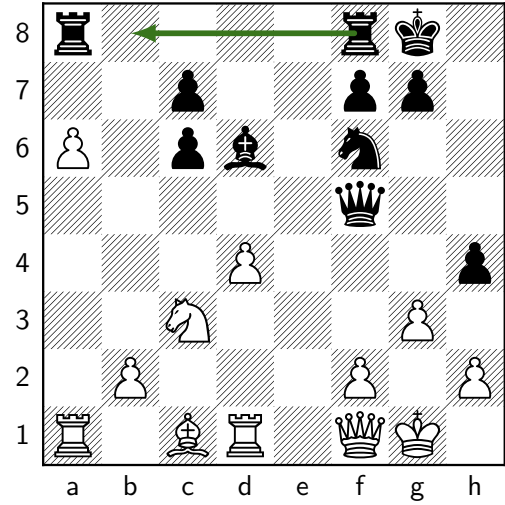
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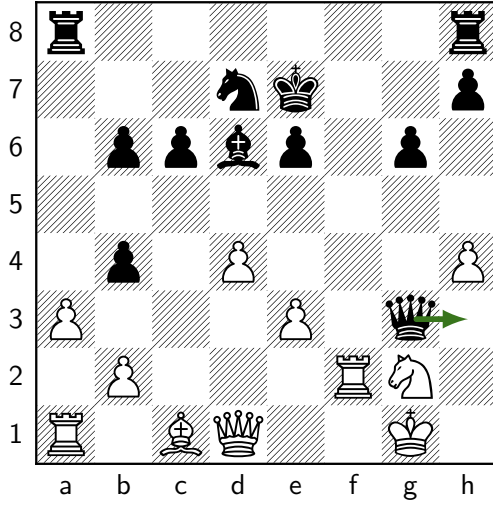


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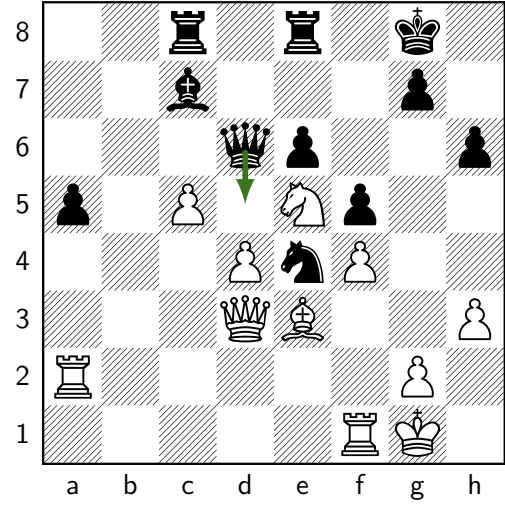


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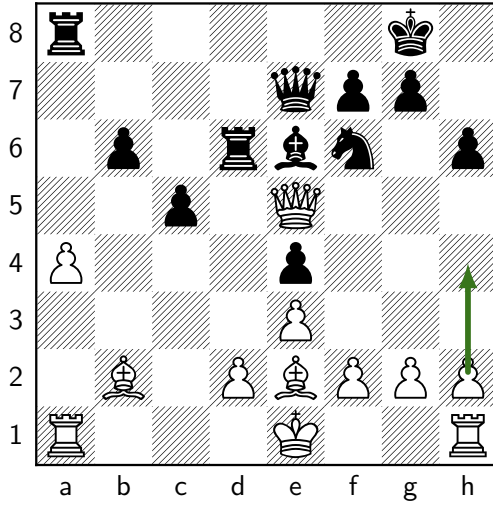
Figure 17: **Move 23, Black: Qg3**. (a) Our position; Melanie plays **Qg3** (g5g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2755` (ply 49) [structural distance $d=74$, QAP obj=108]: the master played **Nf4**; under μ our **Nd2** plays the role of that game’s **Nh4**; cross-type roles: our **Pg6** as their **Rc8**, our **Qg5** as their **Rd8**; transferred move **d7e5**. (c) Retrieved from `ccr1_top_p0003#2072` (ply 57) [structural distance $d=77$, QAP obj=83]: the master played **Re7**; under μ our **Qg4** plays the role of that game’s **Re1** (a **cross-type** correspondence); cross-type roles: our **Nd7** as their **Rd8**, our **Pb6** as their **Bb7**; transferred move **g5b5**. (d) Retrieved from `ccr1_top_p0003#1182` (ply 45) [structural distance $d=60$, QAP obj=73]: the master played **Rfb8**; under μ our **Bd3** plays the role of that game’s **Rf1** (a **cross-type** correspondence); cross-type roles: our **Qg5** as their **Ra8**, our **Rh8** as their **Pc7**; transferred move **d6c7**. Synthesis: no retrieved analog transferred to **Qg3** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



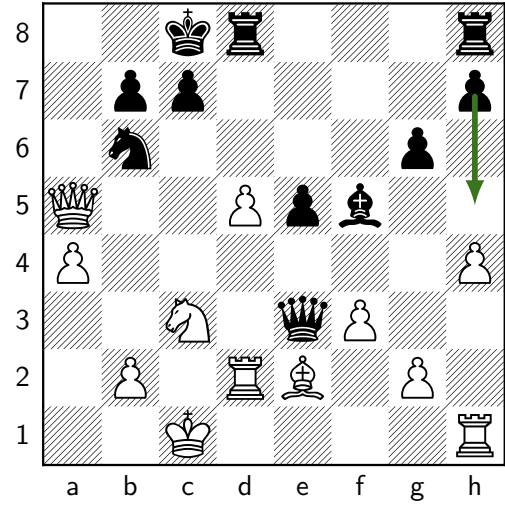
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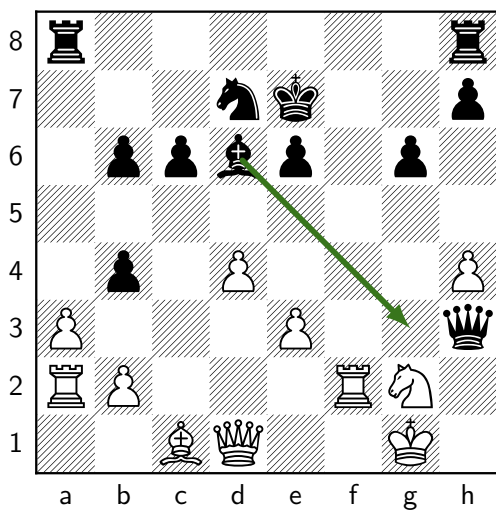


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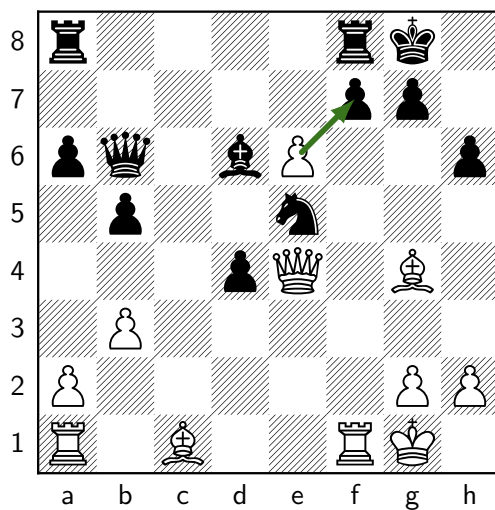


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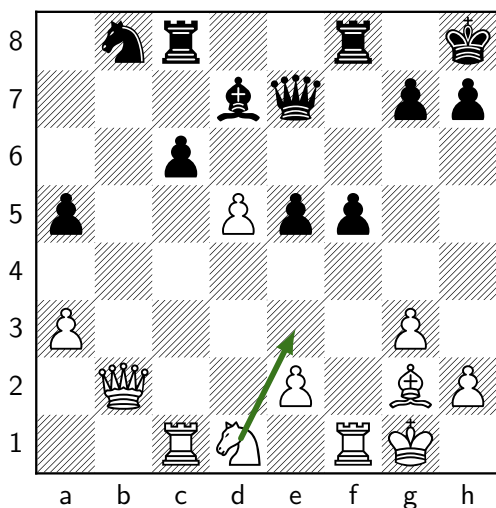
Figure 18: **Move 24, Black: Qh3**. (a) Our position; Melanie plays Qh3 (g3h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1280` (ply 57) [structural distance $d=84$, QAP obj=28]: the master played Qd5; under μ our Qg6 plays the role of that game’s Qd3; cross-type roles: our pd4 as their Re8, our Ra8 as their Bc7; transferred move g3h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#464` (ply 44) [structural distance $d=75$, QAP obj=157]: the master played h4; under μ our Pg3 plays the role of that game’s Ph2; cross-type roles: our Ph7 as their Rh1, our Pb4 as their Be2; transferred move g6g5. (d) Retrieved from `ccr1_top_p0002#909` (ply 43) [structural distance $d=69$, QAP obj=104]: the master played h5; under μ our Nd2 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our Bd6 as their Rd8, our Pg6 as their Nb6; transferred move d7e5. Synthesis: 1 retrieved analog transferred to Qh3 (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



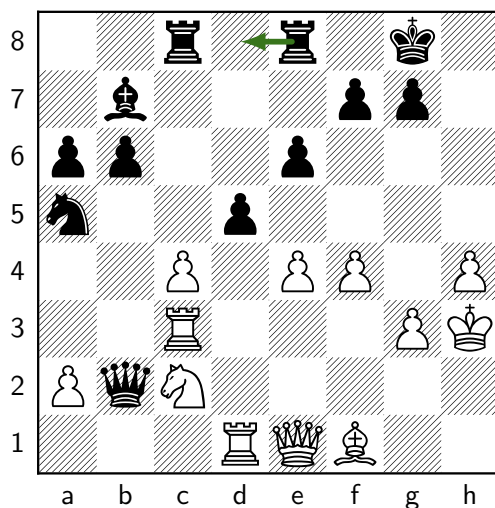
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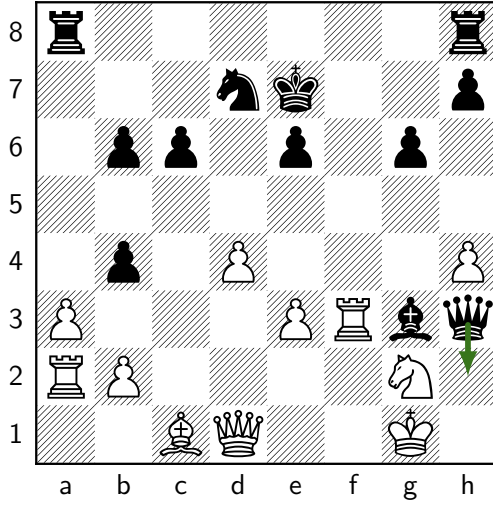


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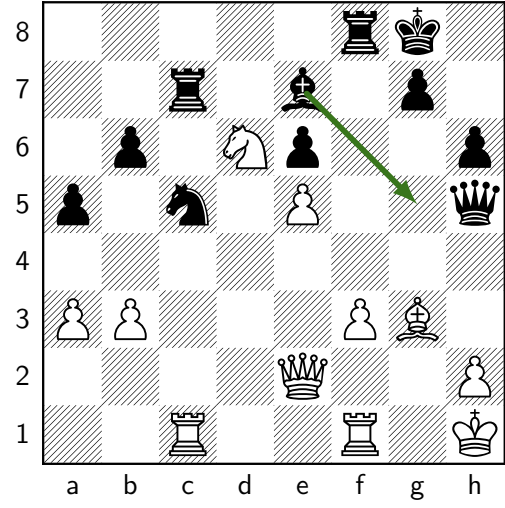


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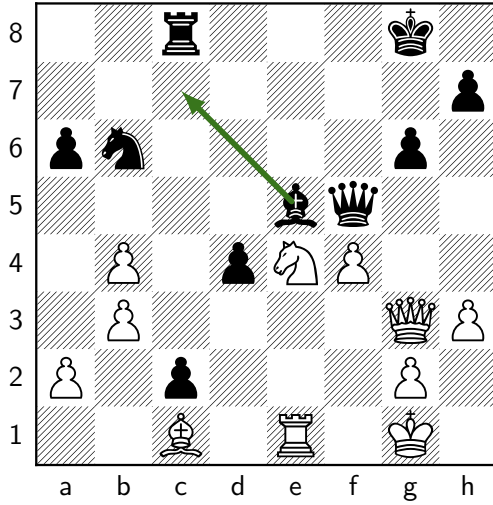
Figure 19: **Move 25, Black: Bg3.** (a) Our position; Melanie plays Bg3 (d6g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2` (ply 42) [structural distance $d=81$, QAP obj=84]: the master played `exf7+`; under μ our *Pb5* plays the role of that game’s *Pe6*; cross-type roles: our *Ph7* as their *Ra1*, our *Rh8* as their *Bc1*; transferred move `b4a3`. (c) Retrieved from `ccr1_top_p0000#2070` (ply 40) [structural distance $d=80$, QAP obj=69]: the master played `Ne3`; under μ our *Rh1* plays the role of that game’s *Nd1* (a **cross-type** correspondence); cross-type roles: our *rf2* as their *Qb2*, our *Nd7* as their *Pe2*; transferred move `h8g8`. (d) Retrieved from `ccr1_top_p0000#2613` (ply 61) [structural distance $d=80$, QAP obj=65]: the master played `Red8`; under μ our *Rh1* plays the role of that game’s *Re1*; cross-type roles: our *bc1* as their *Na5*, our *qd1* as their *pe4*; transferred move `h8g8`. Synthesis: no retrieved analog transferred to Bg3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



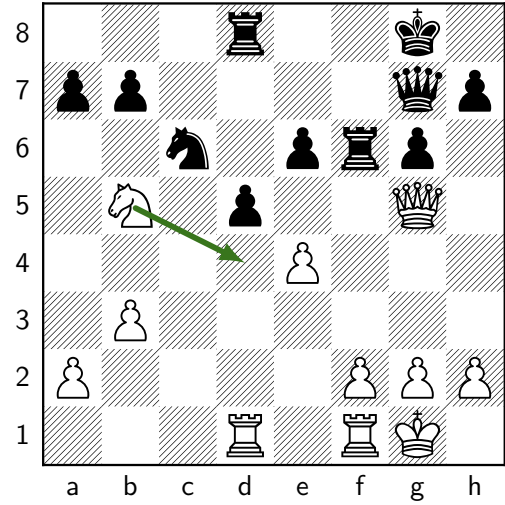
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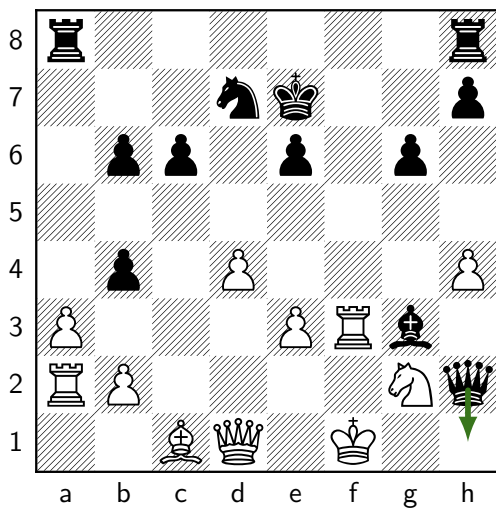


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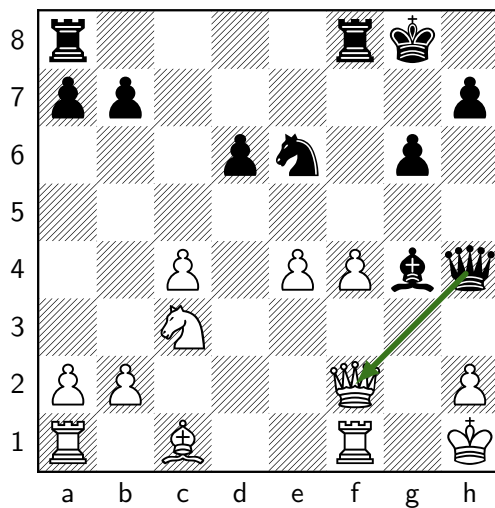


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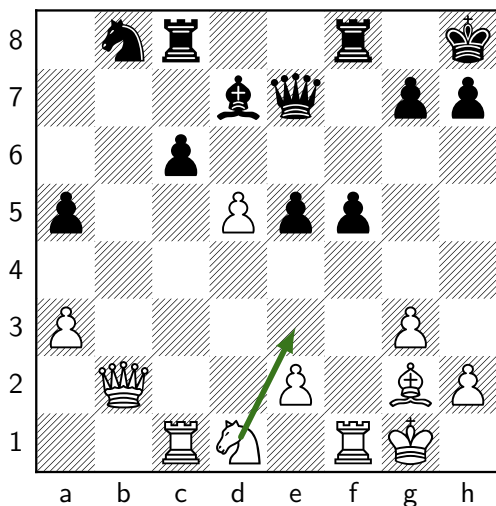
Figure 20: **Move 26, Black: Qh2+.** (a) Our position; Melanie plays Qh2+ (h3h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1692` (ply 59) [structural distance $d=87$, QAP obj=57]: the master played Bg5; under μ our Qh6 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our Pe6 as their Rf8, our Pb6 as their Rc7; transferred move h3h4. (c) Retrieved from `ccr1_top_p0001#3015` (ply 65) [structural distance $d=87$, QAP obj=55]: the master played Bc7; under μ our Bg6 plays the role of that game’s Be4; cross-type roles: our pa3 as their Rc8, our Nd7 as their Pa6; transferred move g3c7. (d) Retrieved from `ccr1_top_p0000#736` (ply 40) [structural distance $d=86$, QAP obj=40]: the master played Nd4; under μ our Nd2 plays the role of that game’s Nb5; cross-type roles: our Ra8 as their Ph2, our rf3 as their Qg5; transferred move d7e5. Synthesis: no retrieved analog transferred to Qh2+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



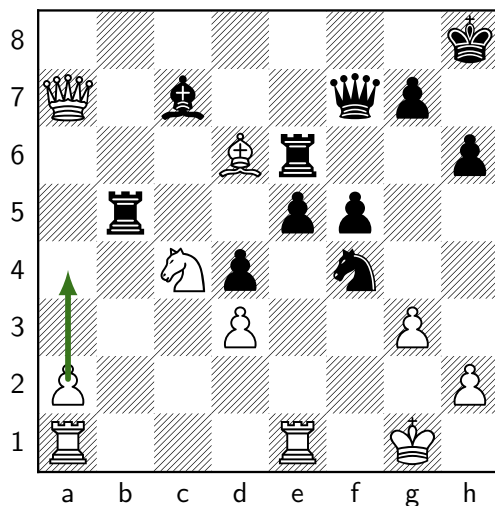
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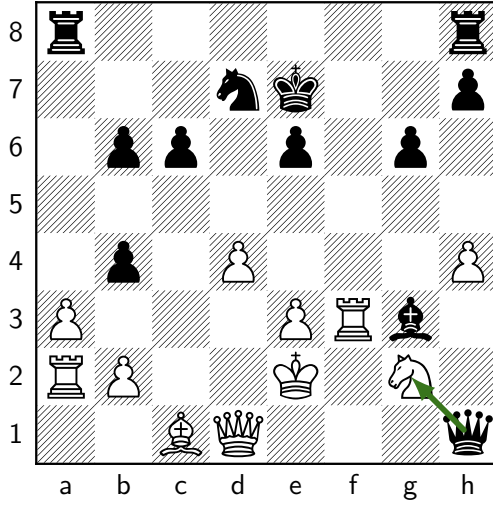


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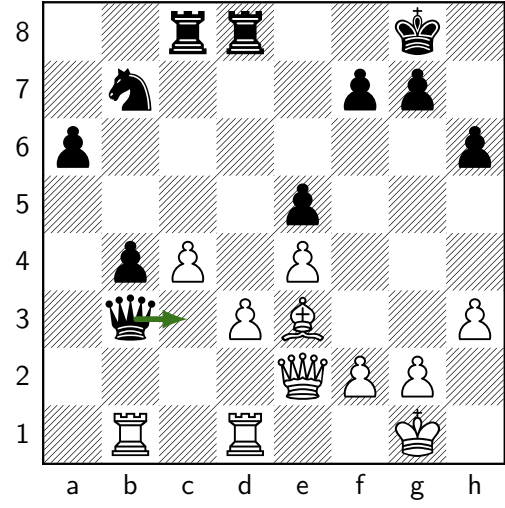


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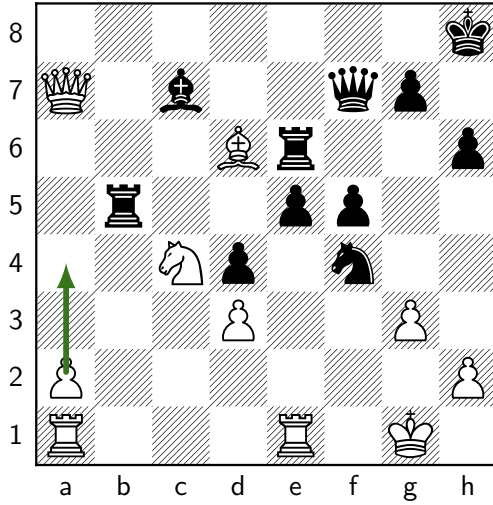
Figure 21: **Move 27, Black: Qh1+.** (a) Our position; Melanie plays Qh1+ (h2h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#400` (ply 43) [structural distance $d=60$, QAP obj=25]: the master played Qxf2; under μ our Qh7 plays the role of that game’s Qh5; cross-type roles: our pe3 as their Ne6, our rf3 as their nc3; transferred move h2h1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2070` (ply 40) [structural distance $d=87$, QAP obj=65]: the master played Ne3; under μ our Rh1 plays the role of that game’s Nd1 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Qb2, our Nd7 as their Pe2; transferred move h8c8. (d) Retrieved from `ccrl_top_p0001#1759` (ply 58) [structural distance $d=89$, QAP obj=58]: the master played a4; under μ our Rh1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Nd7 as their Re1, our Qh2 as their Pg3; transferred move h8g8. Synthesis: 1 retrieved analog transferred to Qh1+ (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



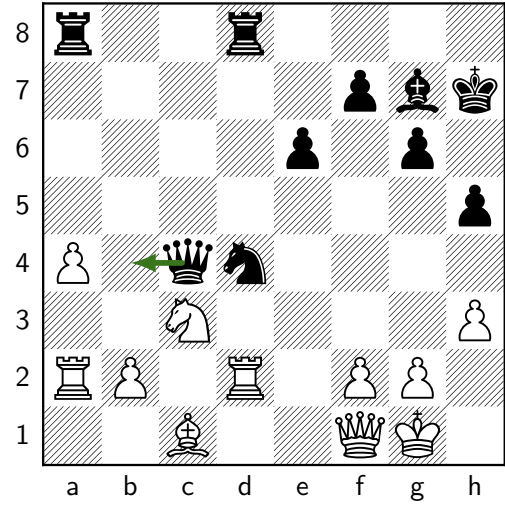
(a)



(b)



(c)



(d)

Figure 22: **Move 28, Black: Qxg2+.** (a) Our position; Melanie plays Qxg2+ (h1g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1828` (ply 61) [structural distance $d=99$, QAP obj=62]: the master played Qc3; under μ our Qh8 plays the role of that game’s Qb6; cross-type roles: our ph4 as their Rc8, our pa3 as their Nb7; transferred move h1g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1759` (ply 58) [structural distance $d=96$, QAP obj=106]: the master played a4; under μ our Rh1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our pd4 as their Re1, our Nd7 as their Ph2; transferred move h8g8. (d) Retrieved from `ccr1_top_p0002#913` (ply 57) [structural distance $d=91$, QAP obj=83]: the master played Qb4; under μ our Qh8 plays the role of that game’s Qc5; cross-type roles: our Pg6 as their Ra8, our Pe6 as their Rd8; transferred move h1h3. Synthesis: 1 retrieved analog transferred to Qxg2+ (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

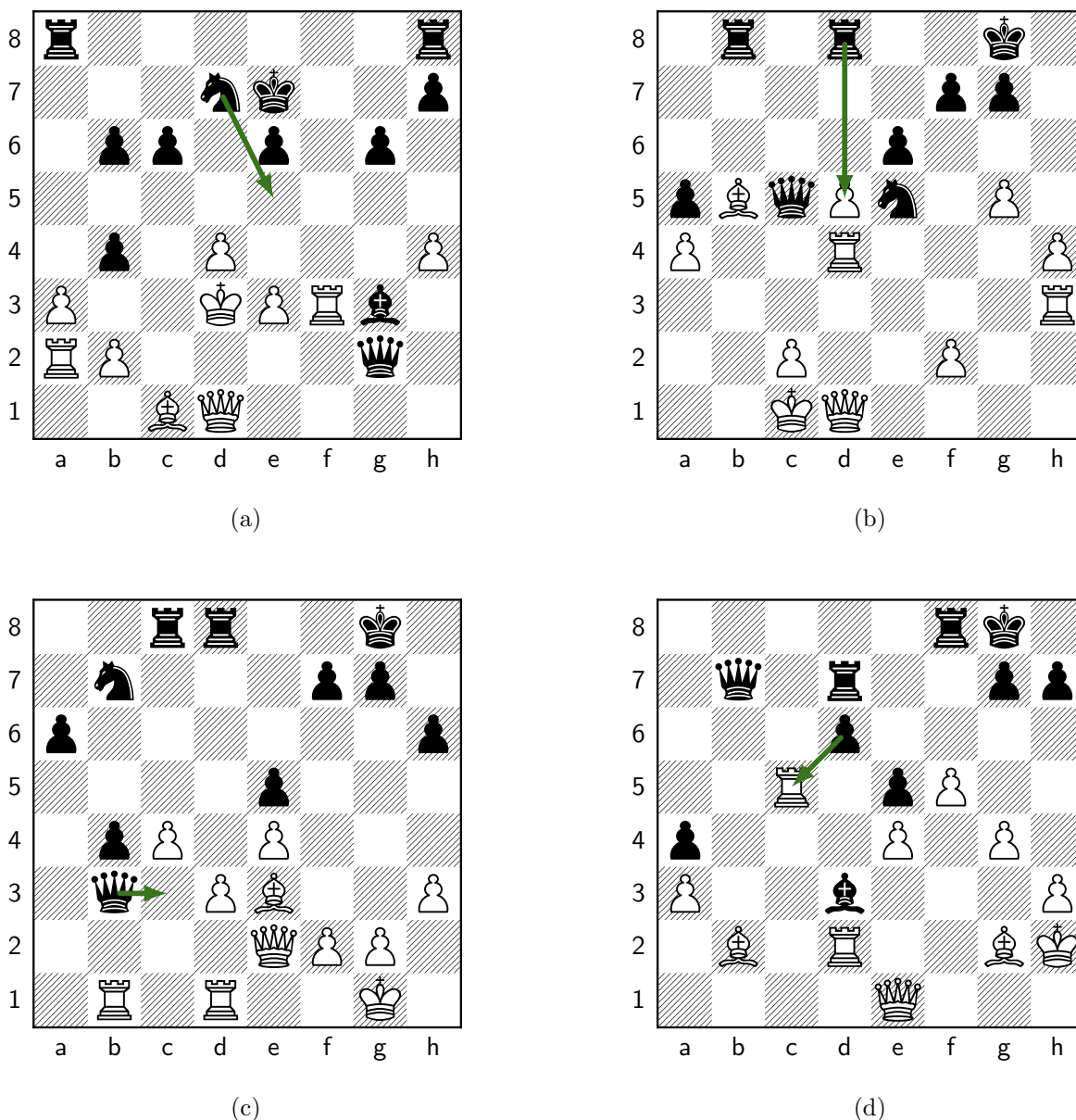
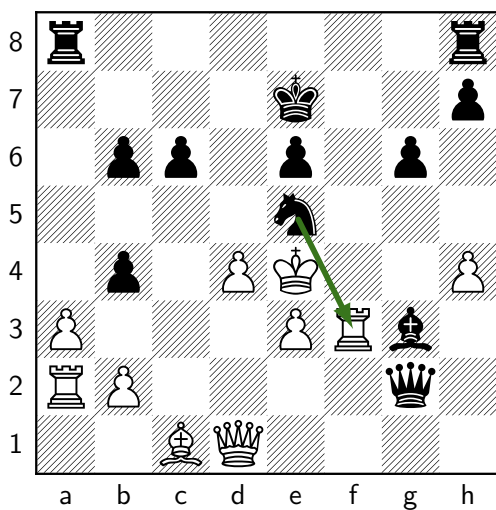
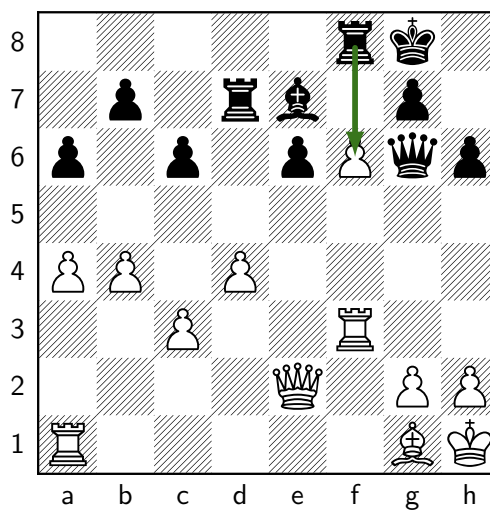


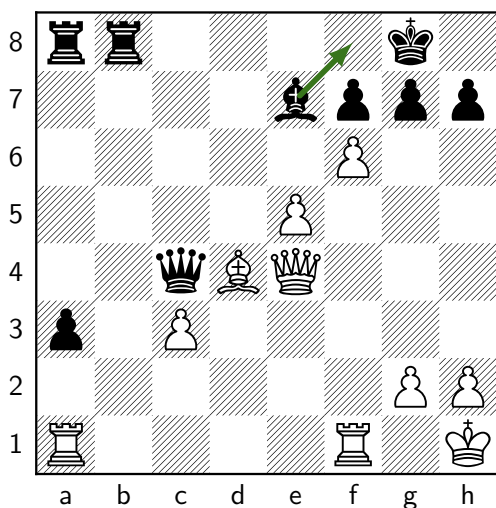
Figure 23: **Move 29, Black: Ne5+.** (a) Our position; Melanie plays Ne5+ (d7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1289` (ply 63) [structural distance $d=70$, QAP obj=78]: the master played Rxd5; under μ our Ra1 plays the role of that game’s Rd1; cross-type roles: our Nd7 as their Pg7, our rf3 as their bb5; transferred move a8a3. (c) Retrieved from `ccr1_top_p0002#1828` (ply 61) [structural distance $d=64$, QAP obj=47]: the master played Qc3; under μ our Qg7 plays the role of that game’s Qb6; cross-type roles: our Qg2 as their Rd8, our Rh8 as their Pf7; transferred move g2h2. (d) Retrieved from `ccr1_top_p0000#1477` (ply 69) [structural distance $d=64$, QAP obj=45]: the master played dxc5; under μ our Qg7 plays the role of that game’s Pd3 (a **cross-type** correspondence); cross-type roles: our Pb4 as their Rf8, our qd1 as their Rd7; transferred move g2f3. Synthesis: no retrieved analog transferred to Ne5+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



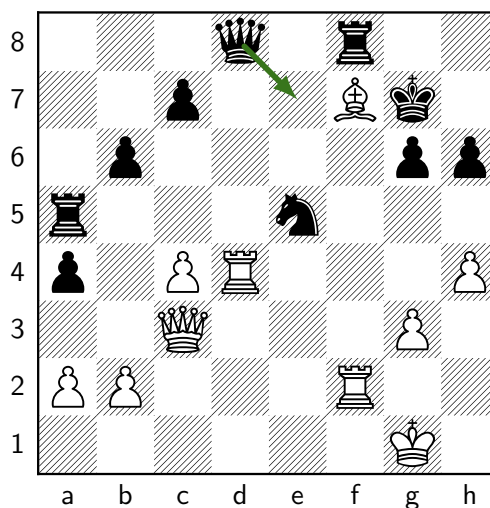
(a)



(b)

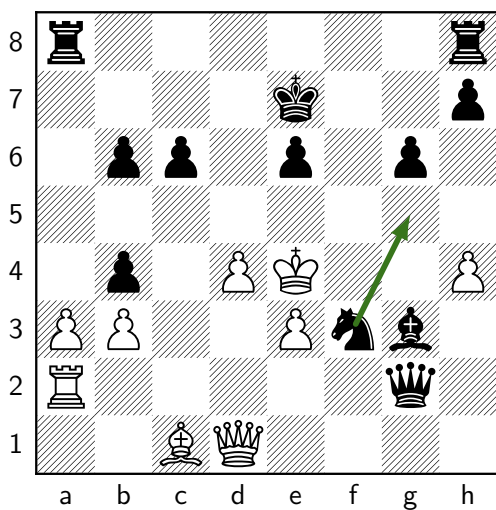


(c)

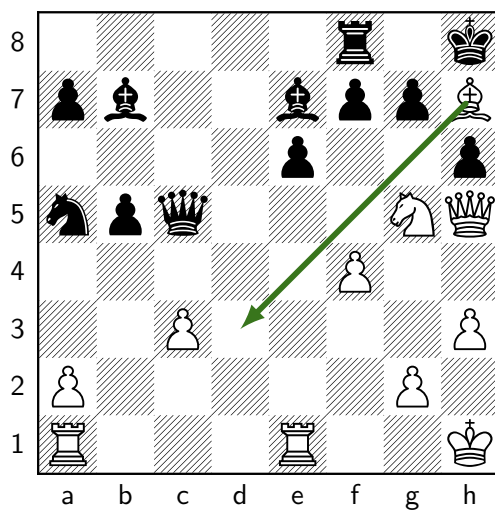


(d)

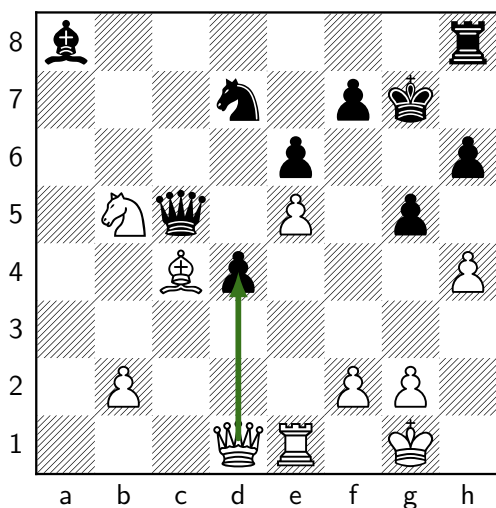
Figure 24: **Move 30, Black: Nxf3.** (a) Our position; Melanie plays Nxf3 (e5f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3135` (ply 45) [structural distance $d=106$, QAP obj=52]: the master played Rxf6; under μ our Qg7 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pb4 as their Rd7, our Ne5 as their Pg7; transferred move g2h2. (c) Retrieved from `ccr1_top_p0001#2029` (ply 55) [structural distance $d=109$, QAP obj=47]: the master played Bf8; under μ our Ne4 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our pa3 as their Rb8, our rf3 as their bd4; transferred move e5d7. (d) Retrieved from `ccr1_top_p0001#850` (ply 75) [structural distance $d=111$, QAP obj=45]: the master played Qe7; under μ our Bg6 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Qg2 as their Rf8, our Rh8 as their Pc7; transferred move g3h4. Synthesis: no retrieved analog transferred to Nxf3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



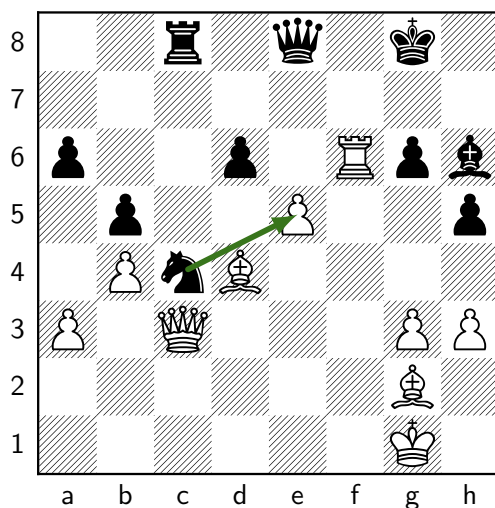
(a)



(b)

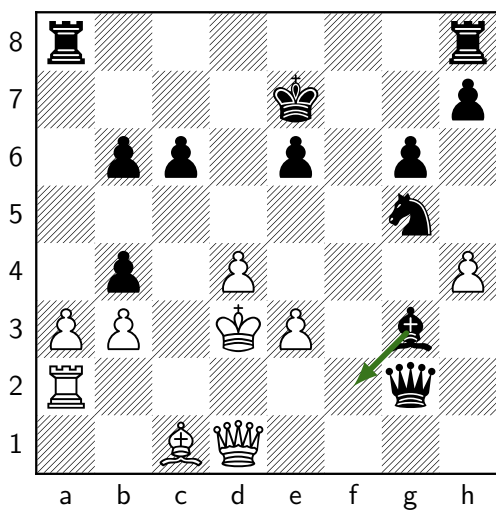


(c)

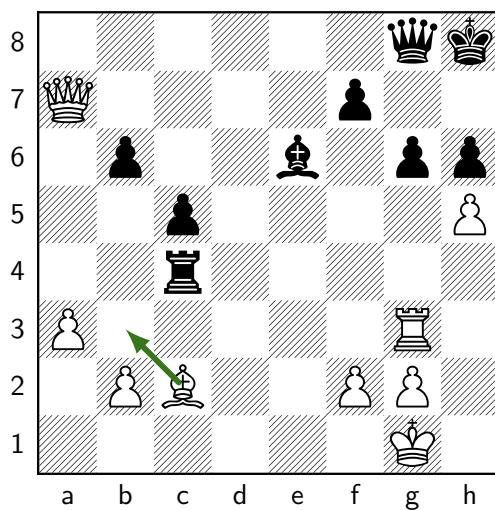


(d)

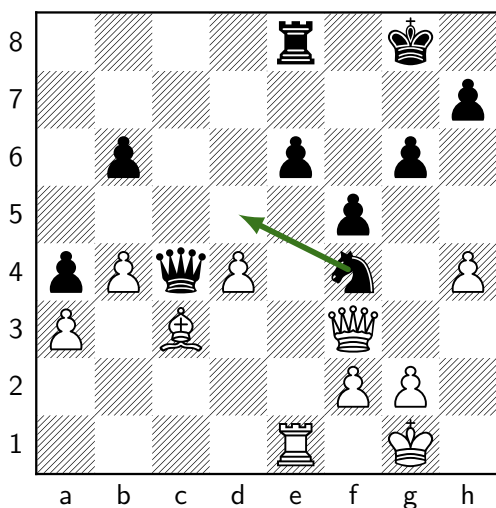
Figure 25: **Move 31, Black: Ng5+.** (a) Our position; Melanie plays Ng5+ (f3g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1109` (ply 60) [structural distance $d=91$, QAP obj=54]: the master played Bd3; under μ our Nf6 plays the role of that game’s Bh7 (a **cross-type** correspondence); cross-type roles: our pa3 as their Ra1, our pb3 as their Re1; transferred move f3g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3011` (ply 44) [structural distance $d=98$, QAP obj=55]: the master played Qxd4; under μ our Ra1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Bc4, our Qg2 as their Ph4; transferred move a8a4. (d) Retrieved from `ccr1_top_p0001#860` (ply 71) [structural distance $d=98$, QAP obj=36]: the master played Nx e5; under μ our Nf6 plays the role of that game’s Nc5; cross-type roles: our Pc6 as their Rc8, our Ph7 as their Qe8; transferred move f3h4. Synthesis: 2 retrieved analogs transferred to Ng5+ (summed QAP objective 108, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



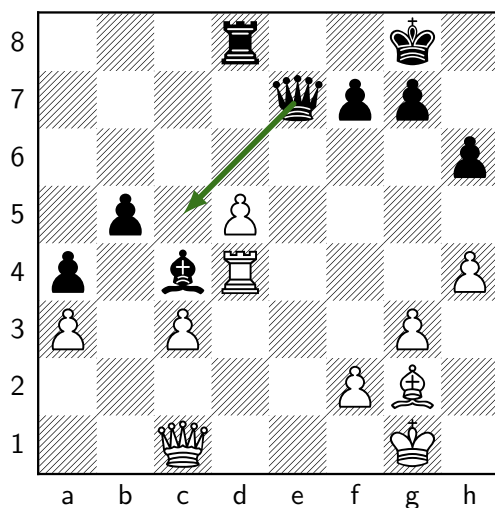
(a)



(b)

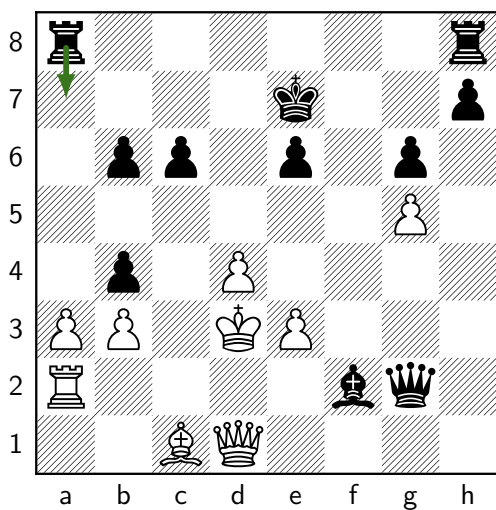


(c)

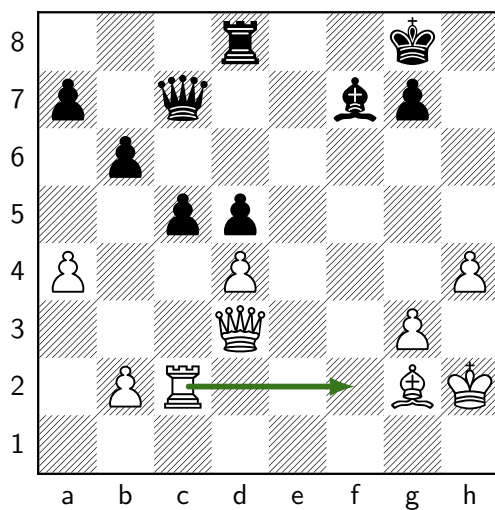


(d)

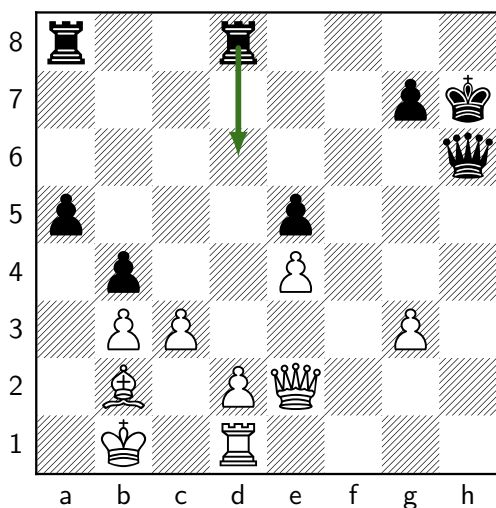
Figure 26: **Move 32, Black: Bf2**. (a) Our position; Melanie plays Bf2 (g3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3371` (ply 78) [structural distance $d=73$, QAP obj=56]: the master played Bb3; under μ our Ng4 plays the role of that game’s Bc2 (a **cross-type** correspondence); cross-type roles: our Rh8 as their Pf2, our ph4 as their rc4; transferred move g5h3. (c) Retrieved from `ccr1_top_p0000#2186` (ply 69) [structural distance $d=74$, QAP obj=54]: the master played Nd5; under μ our Ng4 plays the role of that game’s Nf5; cross-type roles: our bc1 as their Re8, our Rh8 as their Ph7; transferred move g5f3. (d) Retrieved from `ccr1_top_p0002#2962` (ply 71) [structural distance $d=70$, QAP obj=36]: the master played Qc5; under μ our Rh1 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our Qg2 as their Pa4, our Ng5 as their Bc4; transferred move h8g8. Synthesis: no retrieved analog transferred to Bf2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



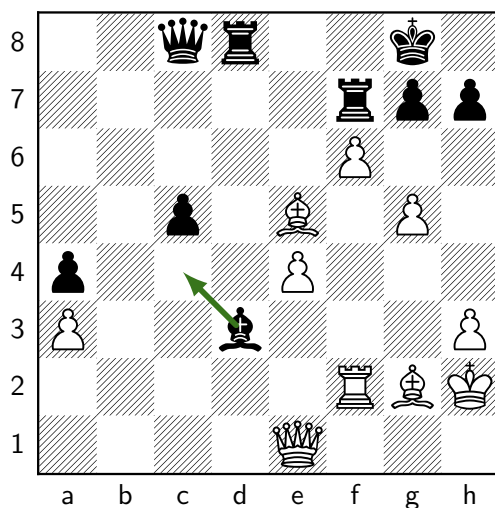
(a)



(b)

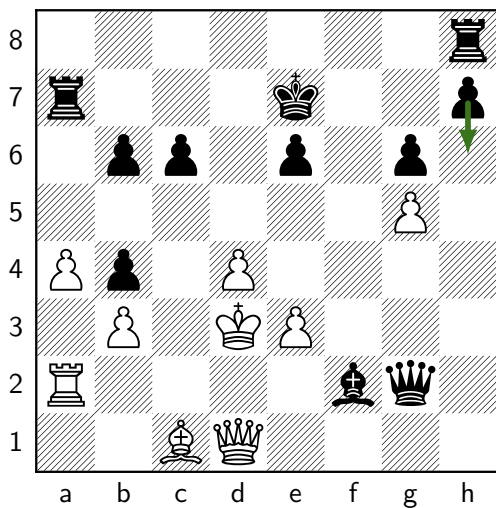


(c)

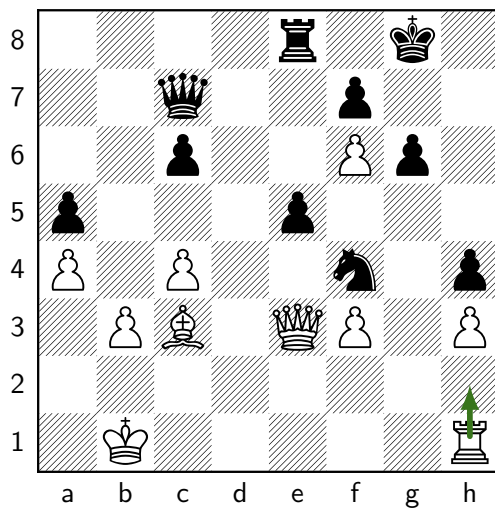


(d)

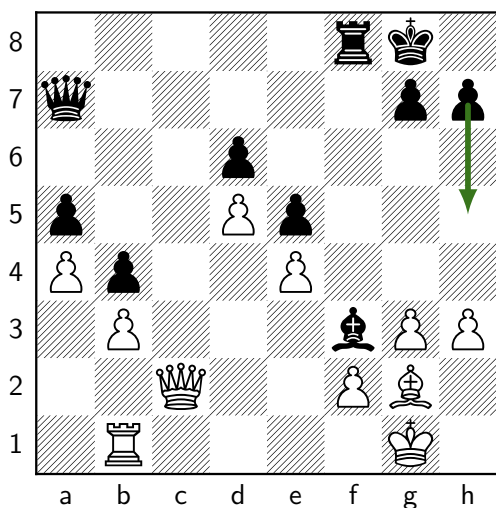
Figure 27: **Move 33, Black: Ra7 (a8a7)**. (a) Our position; Melanie plays Ra7 (a8a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1442` (ply 64) [structural distance $d=62$, QAP obj=82]: the master played Rf2; under μ our Ra1 plays the role of that game’s Rc2; cross-type roles: our Rh8 as their Pb2, our Qg2 as their Bg2; transferred move a8a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#260` (ply 71) [structural distance $d=52$, QAP obj=52]: the master played Rd6; under μ our Ra1 plays the role of that game’s Rd1; cross-type roles: our Pb4 as their Qh6, our Qg2 as their Pa5; transferred move a8a7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1477` (ply 79) [structural distance $d=55$, QAP obj=73]: the master played Bc4; under μ our Bf7 plays the role of that game’s Bd6; cross-type roles: our pd4 as their Qc8, our Qg2 as their Rd8; transferred move f2g3. Synthesis: 2 retrieved analogs transferred to Ra7 (summed QAP objective 134, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



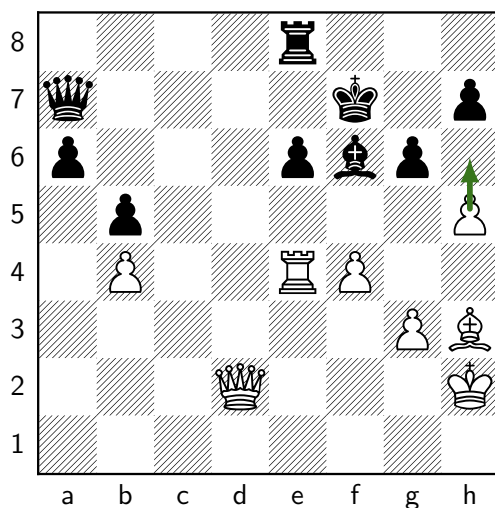
(a)



(b)

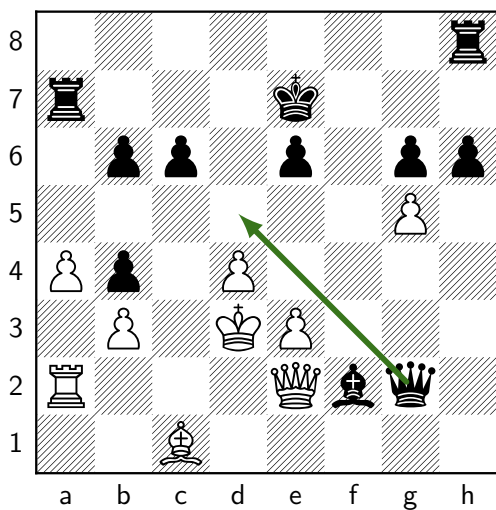


(c)

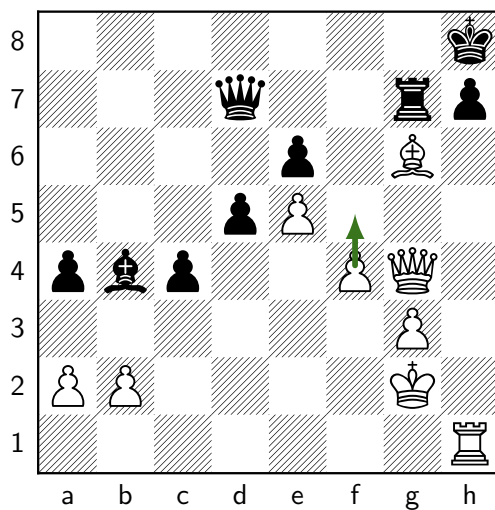


(d)

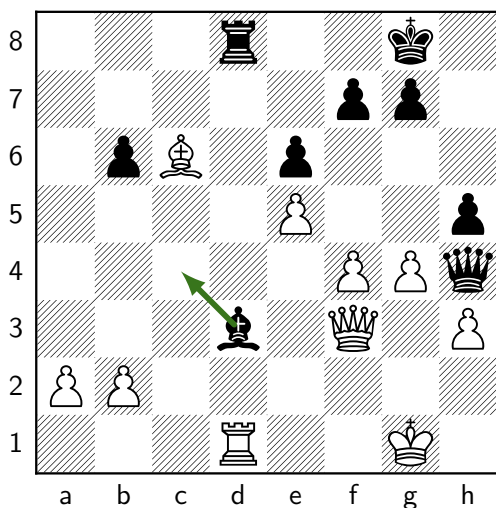
Figure 28: **Move 34, Black: h6**. (a) Our position; Melanie plays h6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1410` (ply 62) [structural distance $d=53$, QAP obj=33]: the master played Rh2; under μ our Ph2 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our pa4 as their Bc3, our ra2 as their Qe3; transferred move h7h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1359` (ply 75) [structural distance $d=50$, QAP obj=21]: the master played h5; under μ our Ph2 plays the role of that game’s Ph2; cross-type roles: our Qg2 as their Rf8, our Ra7 as their Qa7; transferred move h7h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1356` (ply 82) [structural distance $d=54$, QAP obj=50]: the master played h6; under μ our Bf7 plays the role of that game’s Ph5 (a **cross-type** correspondence); cross-type roles: our Ra7 as their Qd2, our ra2 as their Bh3; transferred move f2h4. Synthesis: 2 retrieved analogs transferred to h6 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



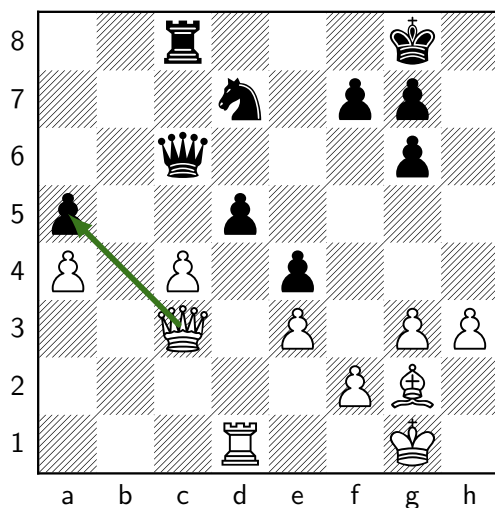
(a)



(b)

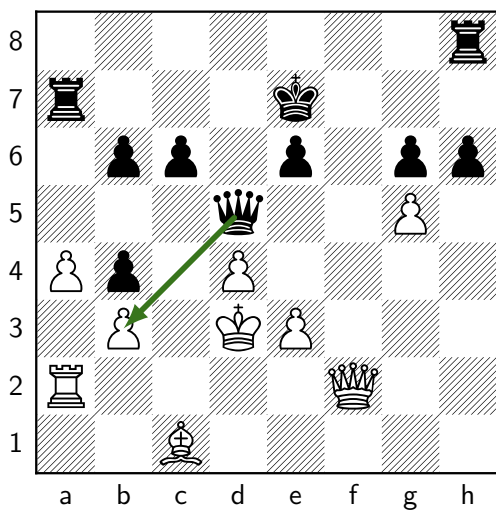


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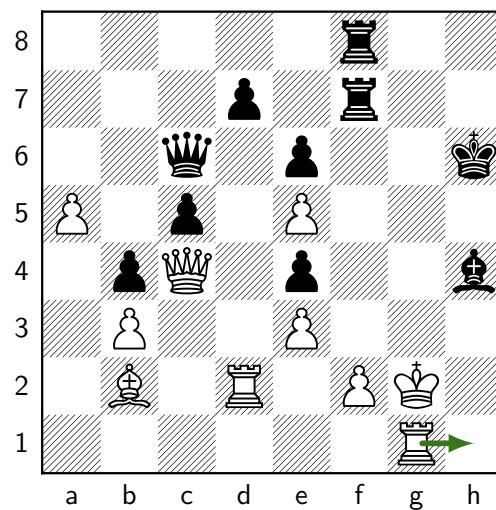


(d)

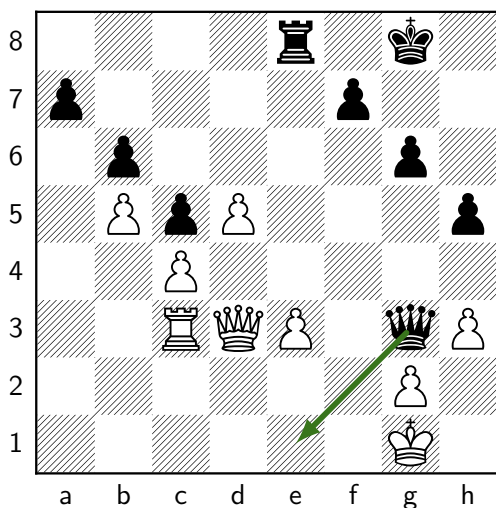
Figure 29: **Move 35, Black: Qd5.** (a) Our position; Melanie plays Qd5 (g2d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#75` (ply 72) [structural distance $d=63$, QAP obj=80]: the master played f5; under μ our *Ph3* plays the role of that game’s *Pf4*; cross-type roles: our *Pb4* as their *Rh1*, our *bc1* as their *Pg3*; transferred move h6h5. (c) Retrieved from `ccr1_top_p0001#922` (ply 57) [structural distance $d=63$, QAP obj=51]: the master played Bc4; under μ our *Bf7* plays the role of that game’s *Bd6*; cross-type roles: our *Qg2* as their *Rd8*, our *ra2* as their *bc6*; transferred move f2h4. (d) Retrieved from `ccr1_top_p0003#1324` (ply 68) [structural distance $d=62$, QAP obj=46]: the master played Qxa5; under μ our *Qg7* plays the role of that game’s *Qc3*; cross-type roles: our *Pb6* as their *Rd1*, our *Ph6* as their *Bg2*; transferred move g2h3. Synthesis: no retrieved analog transferred to Qd5 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



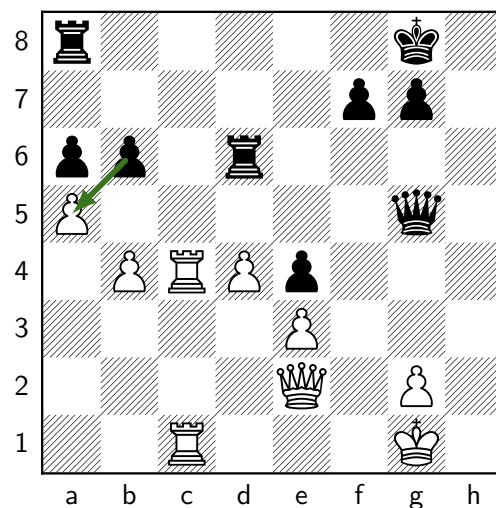
(a)



(b)

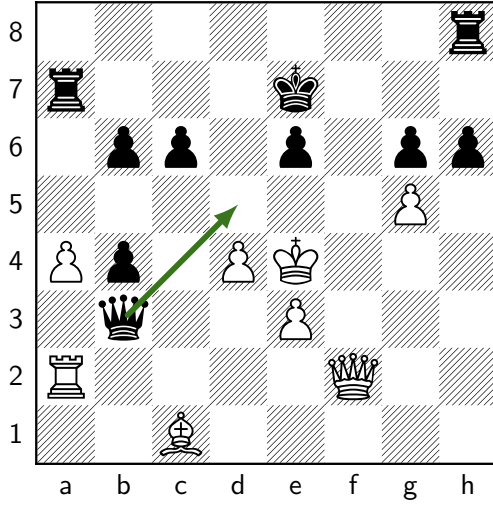


(c)

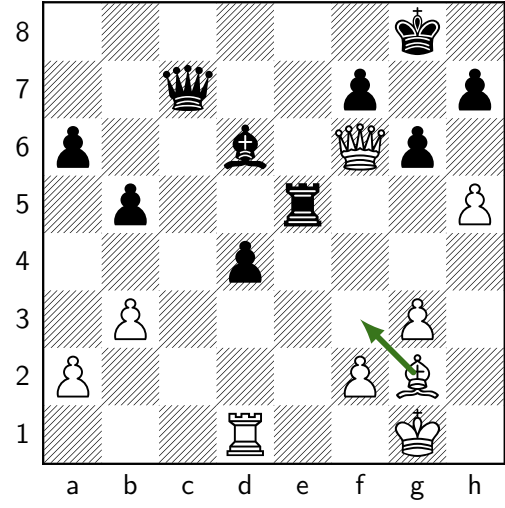


(d)

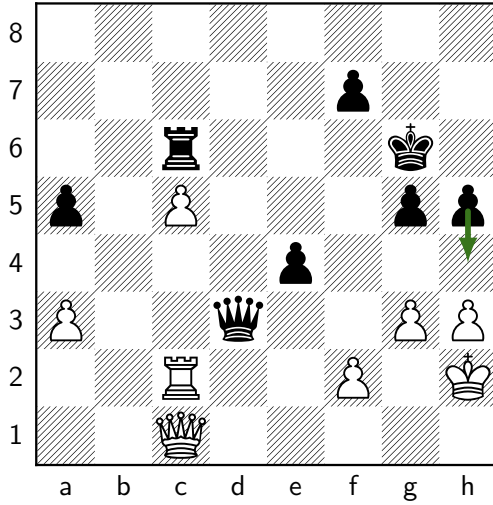
Figure 30: **Move 36, Black: Qxb3+.** (a) Our position; Melanie plays Qxb3+ (d5b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3167` (ply 70) [structural distance $d=66$, QAP obj=201]: the master played Rh1; under μ our *Ra2* plays the role of that game’s *Rg1*; cross-type roles: our *Pb6* as their *Bb2*, our *pa4* as their *Rd2*; transferred move a7d7. (c) Retrieved from `ccr1_top_p0003#2926` (ply 71) [structural distance $d=51$, QAP obj=132]: the master played Qe1+; under μ our *Qd4* plays the role of that game’s *Qg6*; cross-type roles: our *qf2* as their *pb5*, our *Ph6* as their *qd3*; transferred move d5h1. (d) Retrieved from `ccr1_top_p0001#3257` (ply 65) [structural distance $d=64$, QAP obj=127]: the master played bxa5; under μ our *Ph3* plays the role of that game’s *Pb3*; cross-type roles: our *Pc6* as their *rc1*; transferred move h6g5. Synthesis: no retrieved analog transferred to Qxb3+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



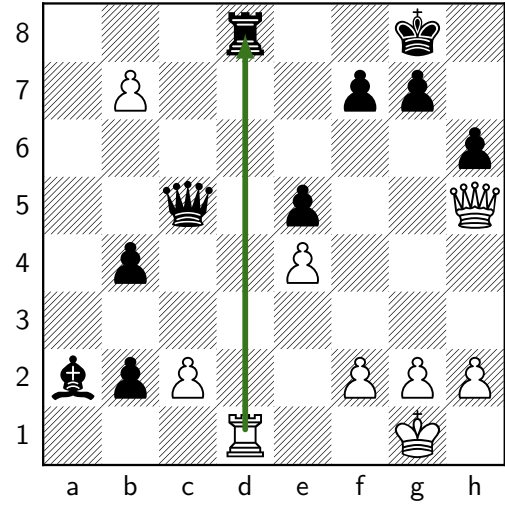
(a)



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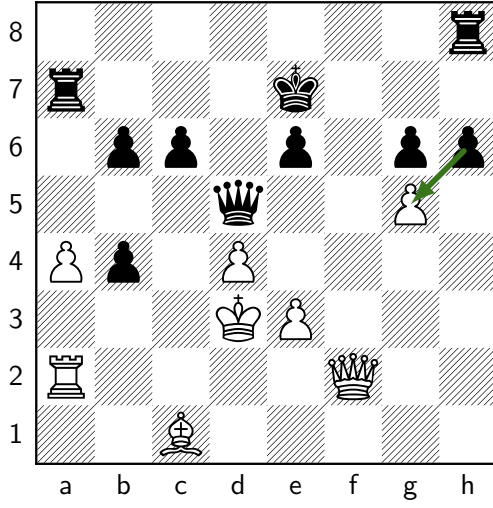


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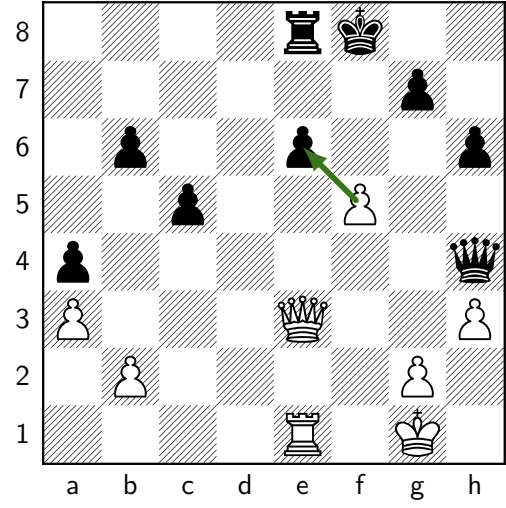


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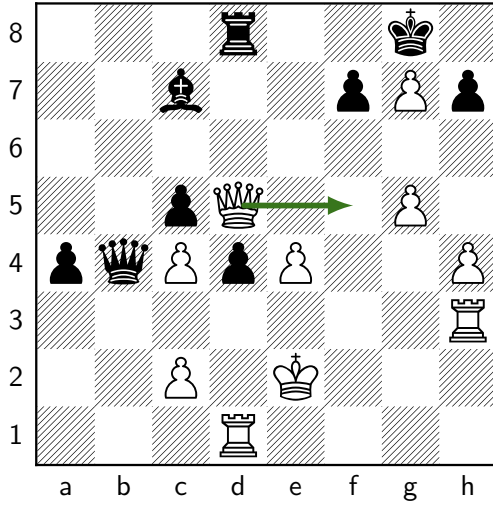
Figure 31: **Move 37, Black: Qd5+.** (a) Our position; Melanie plays Qd5+ (b3d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2541` (ply 62) [structural distance $d=61$, QAP obj=51]: the master played Bf3; under μ our *Pc3* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *Rh8* as their *pa6*, our *ra2* as their *bd6*; transferred move c6c5. (c) Retrieved from `ccr1_top_p0003#1520` (ply 89) [structural distance $d=48$, QAP obj=37]: the master played h4; under μ our *Pc3* plays the role of that game’s *Ph4*; cross-type roles: our *bc1* as their *pf2*; transferred move c6c5. (d) Retrieved from `ccr1_top_p0003#2307` (ply 66) [structural distance $d=58$, QAP obj=32]: the master played Rxd8+; under μ our *Qb6* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *Pg6* as their *ba2*, our *bc1* as their *pb2*; transferred move b3b1. Synthesis: no retrieved analog transferred to Qd5+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



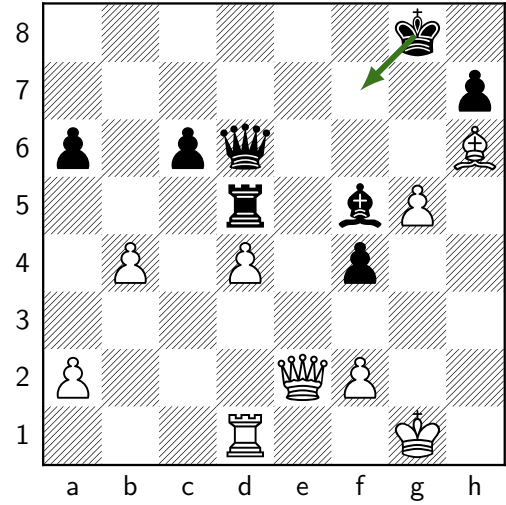
(a)



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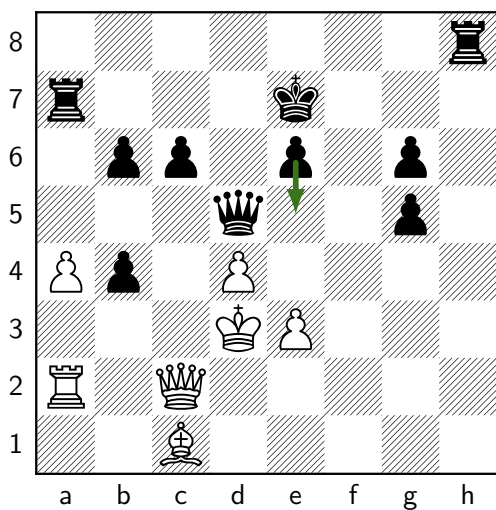


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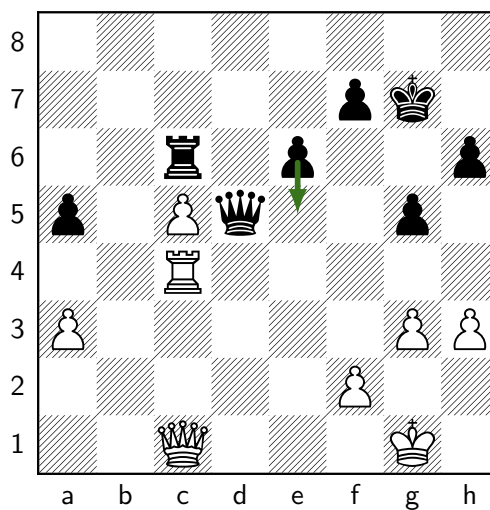


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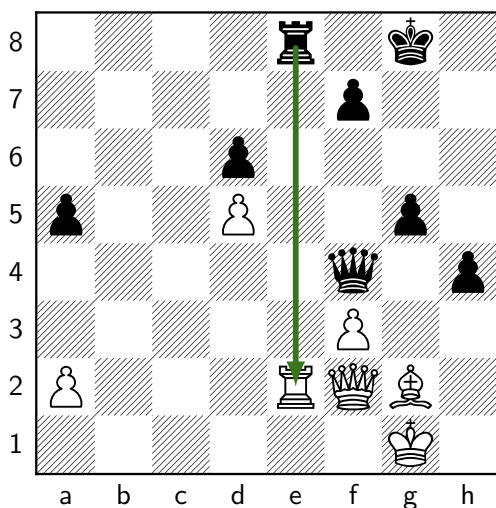
Figure 32: **Move 38, Black: hxg5.** (a) Our position; Melanie plays hxg5 (h6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1989` (ply 58) [structural distance $d=54$, QAP obj=61]: the master played `fxe6`; under μ our *Ph3* plays the role of that game’s *Pf5*; cross-type roles: our *bc1* as their *Pb2*, our *qf2* as their *pb6*; transferred move `h6g5` (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3527` (ply 72) [structural distance $d=52$, QAP obj=51]: the master played `Qf5`; under μ our *Qd4* plays the role of that game’s *Qd5*; cross-type roles: our *bc1* as their *Pc2*, our *Pg6* as their *Rh3*; transferred move `d5a5`. (d) Retrieved from `ccrl_top_p0003#2606` (ply 67) [structural distance $d=57$, QAP obj=46]: the master played `Kf7`; under μ our *Ke2* plays the role of that game’s *Kg1*; cross-type roles: our *Ph6* as their *Qd6*, our *pg5* as their *bh6*; transferred move `e7e8`. Synthesis: 1 retrieved analog transferred to `hxg5` (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



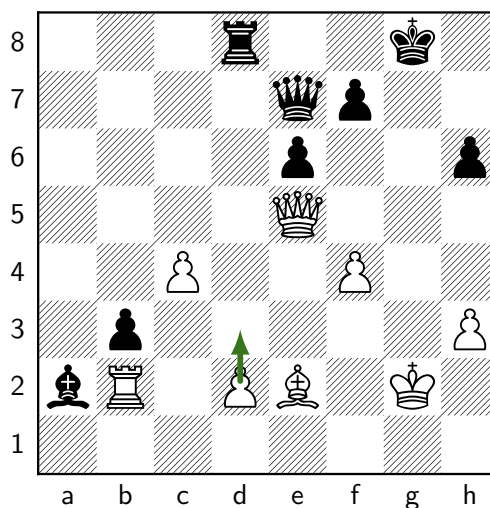
(a)



(b)

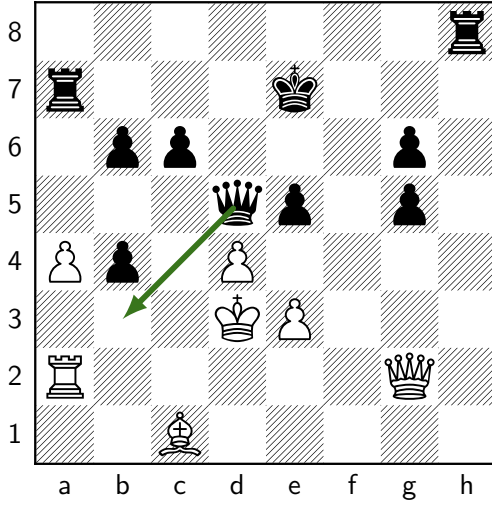


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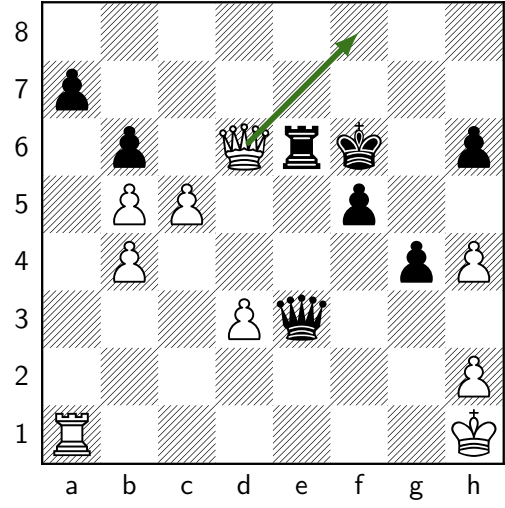


(d)

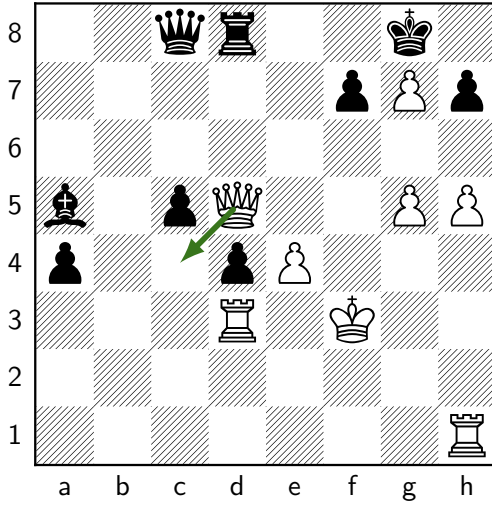
Figure 33: **Move 39, Black: e5**. (a) Our position; Melanie plays **e5** (**e6e5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1520` (ply 65) [structural distance $d=59$, QAP obj=32]: the master played **e5**; under μ our *Pe3* plays the role of that game’s *Pe3*; cross-type roles: our *bc1* as their *Pf7*, our *Qd5* as their *Rc6*; transferred move **e6e5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3095` (ply 71) [structural distance $d=52$, QAP obj=50]: the master played **Rxe2**; under μ our *Qd4* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Rh8* as their *Pa5*, our *Ra7* as their *pa2*; transferred move **d5a2**. (d) Retrieved from `ccrl_top_p0001#857` (ply 76) [structural distance $d=59$, QAP obj=45]: the master played **d3**; under μ our *Pc3* plays the role of that game’s *Pd2*; cross-type roles: our *ra2* as their *ba2*, our *Qd5* as their *Rb2*; transferred move **c6c5**. Synthesis: 1 retrieved analog transferred to **e5** (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



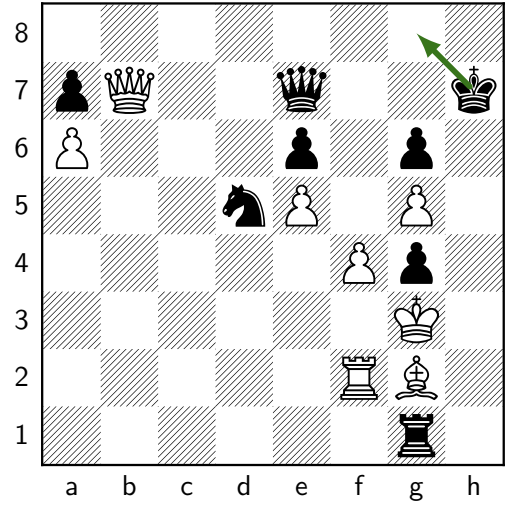
(a)



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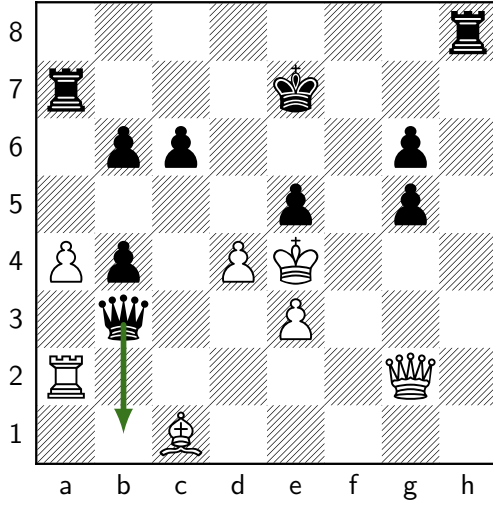


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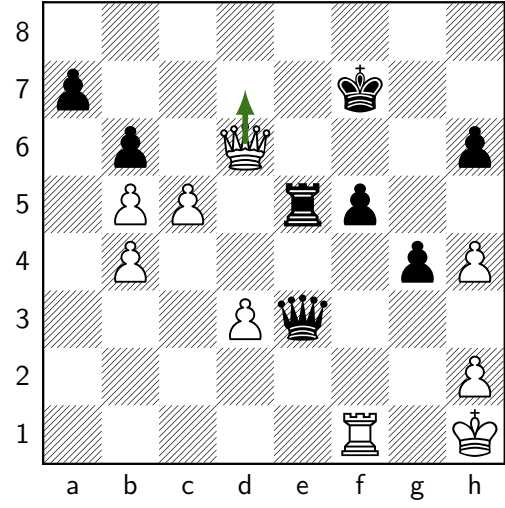


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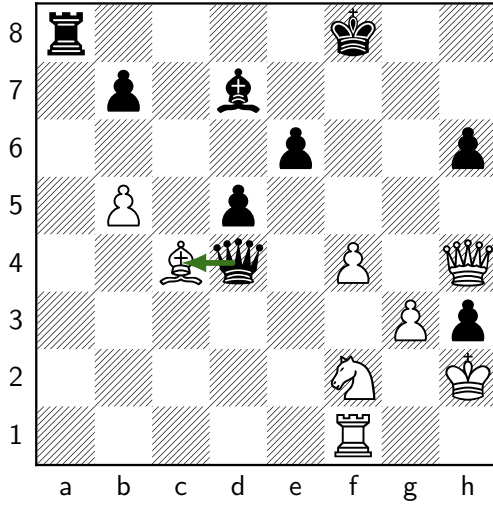
Figure 34: **Move 40, Black: Qb3+.** (a) Our position; Melanie plays Qb3+ (d5b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1509` (ply 76) [structural distance $d=60$, QAP obj=30]: the master played Qf8+; under μ our Qd4 plays the role of that game’s Qd6; cross-type roles: our ra2 as their qe3, our Rh8 as their Pb4; transferred move d5b3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3527` (ply 94) [structural distance $d=55$, QAP obj=129]: the master played Qc4; under μ our Qd4 plays the role of that game’s Qd5; cross-type roles: our Qd5 as their Rd3, our ra2 as their pd4; transferred move d5f3. (d) Retrieved from `ccrl_top_p0000#2683` (ply 81) [structural distance $d=65$, QAP obj=78]: the master played Kg8; under μ our Ke2 plays the role of that game’s Kh2; cross-type roles: our ra2 as their Nd5, our Pg6 as their rf2; transferred move e7f8. Synthesis: 1 retrieved analog transferred to Qb3+ (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



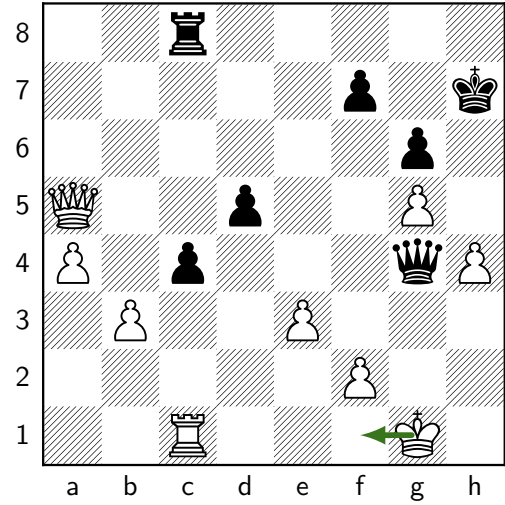
(a)



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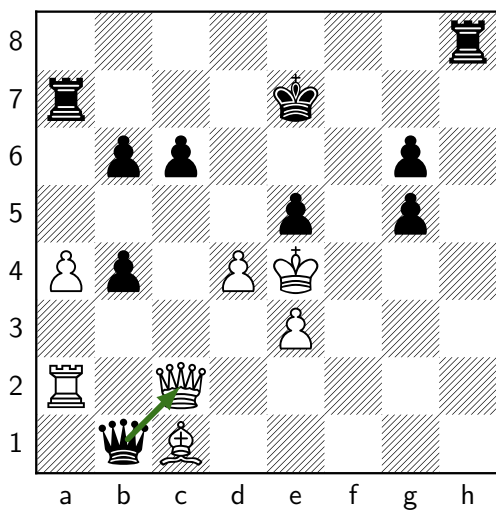


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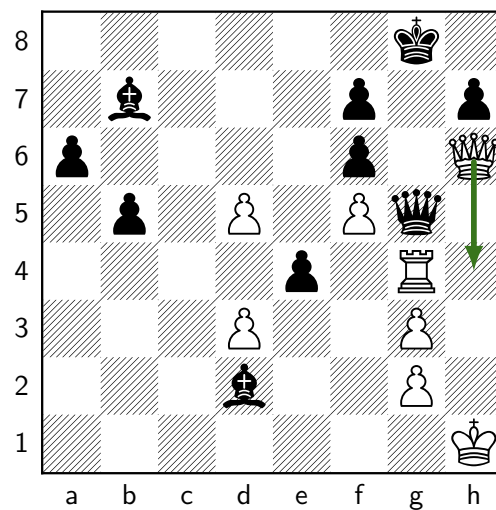


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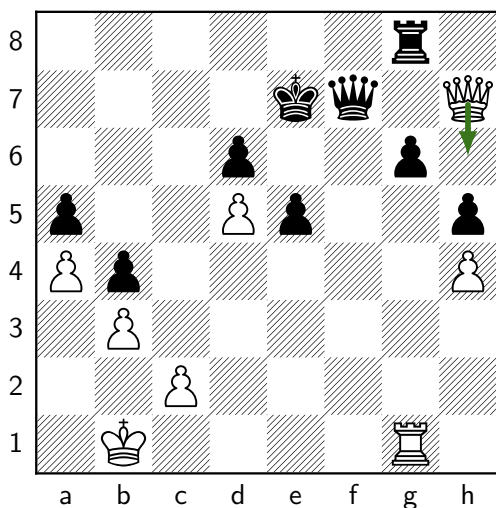
Figure 35: **Move 41, Black: Qb1+.** (a) Our position; Melanie plays Qb1+ (b3b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1509` (ply 92) [structural distance $d=68$, QAP obj=44]: the master played Qd7+; under μ our Qb6 plays the role of that game’s Qd6; cross-type roles: our pd4 as their qe3, our Rh8 as their pg4; transferred move b3b1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2383` (ply 71) [structural distance $d=73$, QAP obj=31]: the master played Qxc4; under μ our Ra2 plays the role of that game’s Qd5 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Bd7, our bc1 as their Pe6; transferred move a7a4. (d) Retrieved from `ccrl_top_p0002#3183` (ply 78) [structural distance $d=75$, QAP obj=27]: the master played Kf1; under μ our Ke2 plays the role of that game’s Kg1; cross-type roles: our Qb3 as their Pb3, our bc1 as their Pe3; transferred move e7f8. Synthesis: 1 retrieved analog transferred to Qb1+ (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



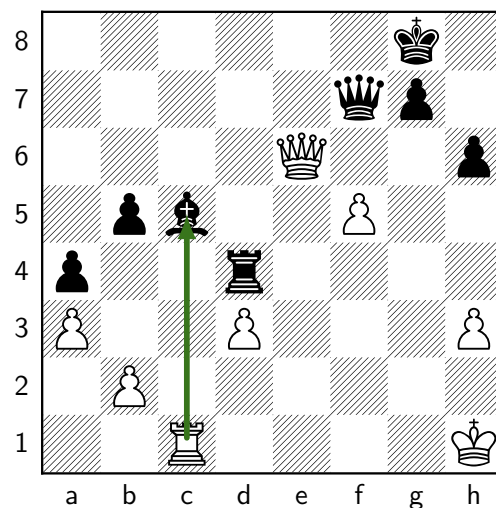
(a)



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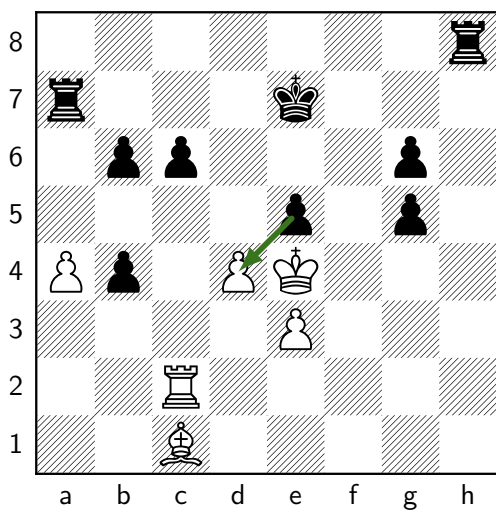


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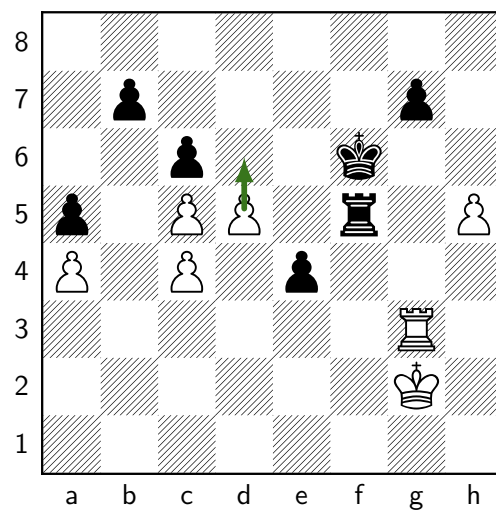


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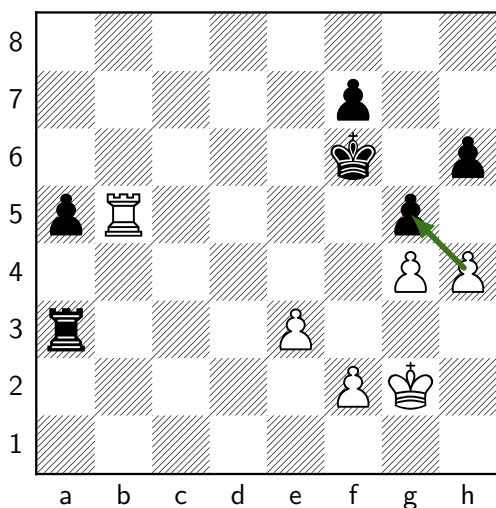
Figure 36: **Move 42, Black: Qxc2+.** (a) Our position; Melanie plays Qxc2+ (b1c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3626` (ply 72) [structural distance $d=81$, QAP obj=108]: the master played Qh4; under μ our Qb8 plays the role of that game’s Qh6; cross-type roles: our Pb6 as their bd2, our Ra7 as their Pg2; transferred move b1b3. (c) Retrieved from `ccr1_top_p0000#3518` (ply 94) [structural distance $d=73$, QAP obj=94]: the master played Qh6; under μ our Qb8 plays the role of that game’s Qh7; cross-type roles: our Pb6 as their Rg1, our Ra7 as their pe5; transferred move b1c1. (d) Retrieved from `ccr1_top_p0000#227` (ply 72) [structural distance $d=72$, QAP obj=73]: the master played Rxc5; under μ our Rh1 plays the role of that game’s Rc1; cross-type roles: our pe3 as their rd4, our Ra7 as their pb5; transferred move h8h6. Synthesis: no retrieved analog transferred to Qxc2+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



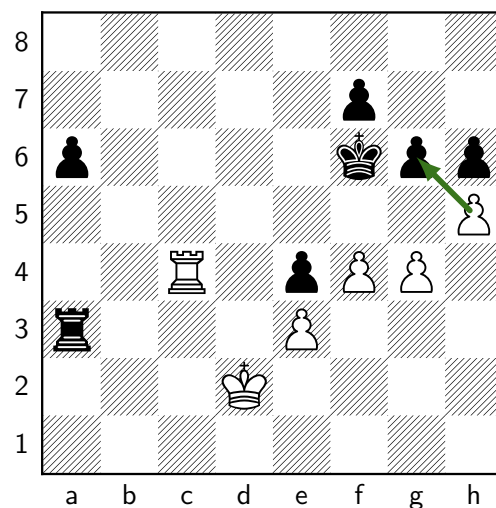
(a)



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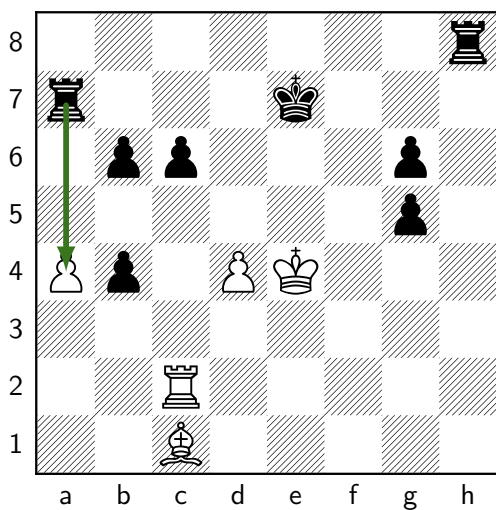


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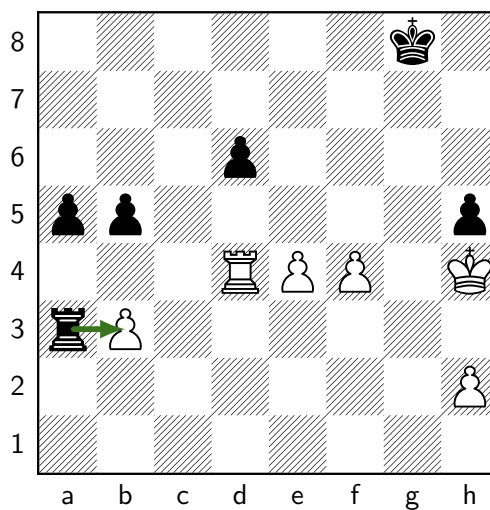


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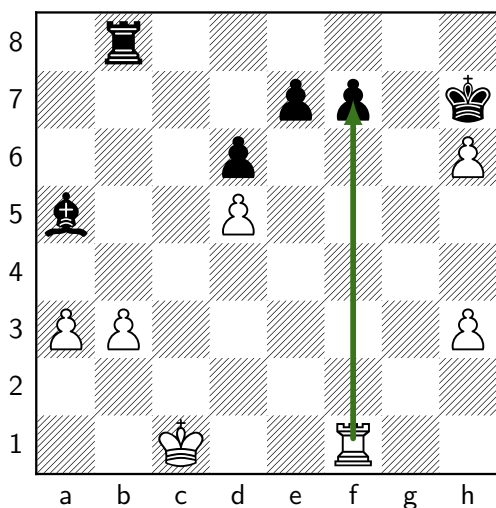
Figure 37: **Move 43, Black: exd4**. (a) Our position; Melanie plays **exd4** (**e5d4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#954` (ply 90) [structural distance $d=44$, QAP obj=59]: the master played **d6**; under μ our Pe_4 plays the role of that game’s Pd_5 ; cross-type roles: our Rh_8 as their pe_4 , our rc_2 as their pa_5 ; transferred move **e5d4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1146` (ply 90) [structural distance $d=49$, QAP obj=54]: the master played **hxg5+**; under μ our Pe_4 plays the role of that game’s Ph_4 ; cross-type roles: our Pe_5 as their Rb_5 ; transferred move **e5d4** (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2941` (ply 96) [structural distance $d=45$, QAP obj=52]: the master played **hxg6**; under μ our Pe_4 plays the role of that game’s Ph_5 ; cross-type roles: our bc_1 as their pa_6 ; transferred move **e5d4** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **exd4** (summed QAP objective 184, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



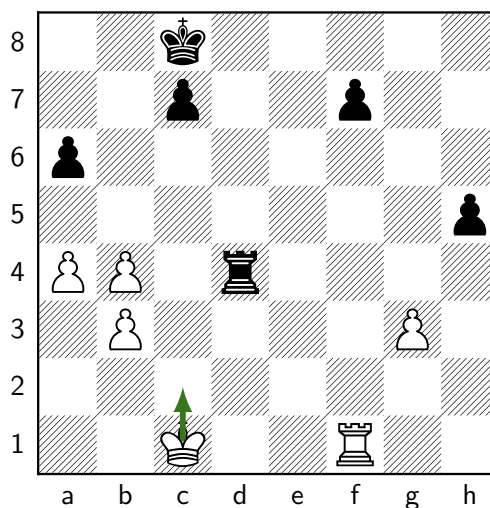
(a)



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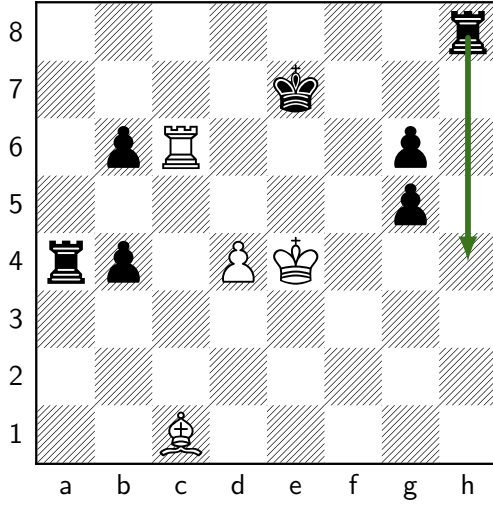


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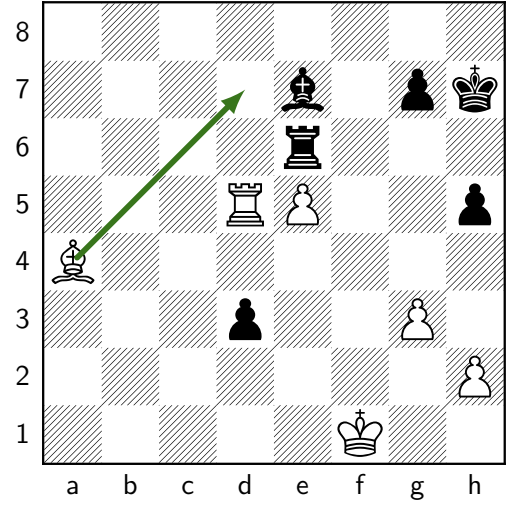


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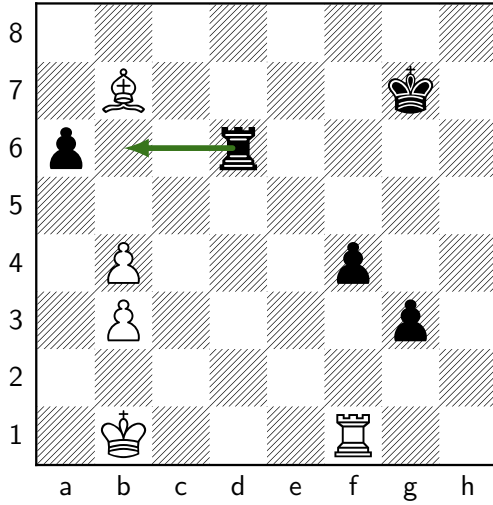
Figure 38: **Move 44, Black: Rxa4**. (a) Our position; Melanie plays **Rxa4** (a7a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#254` (ply 71) [structural distance $d=25$, QAP obj=12]: the master played **Rxb3**; under μ our *Ra2* plays the role of that game’s *Ra6*; cross-type roles: our *bc1* as their *pe4*, our *Rh8* as their *ph2*; transferred move **a7a4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2538` (ply 82) [structural distance $d=17$, QAP obj=4]: the master played **Rxf7+**; under μ our *Ra2* plays the role of that game’s *Rf1*; cross-type roles: our *Rh8* as their *Ph3*, our *pd4* as their *ba5*; transferred move **a7a4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2778` (ply 72) [structural distance $d=23$, QAP obj=19]: the master played **Kc2**; under μ our *Ke2* plays the role of that game’s *Kc1*; cross-type roles: our *bc1* as their *Rf1*, our *Rh8* as their *ph5*; transferred move **e7f7**. Synthesis: 2 retrieved analogs transferred to **Rxa4** (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



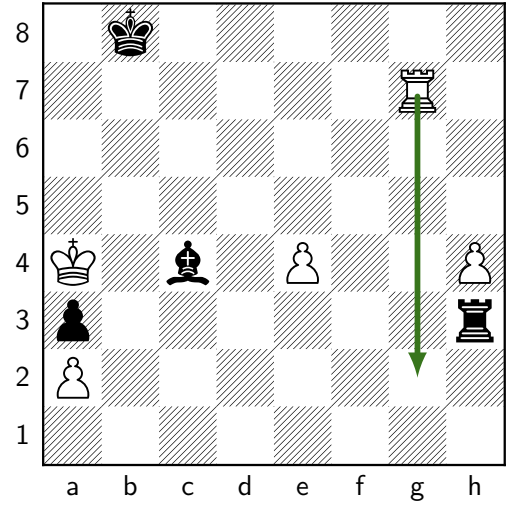
(a)



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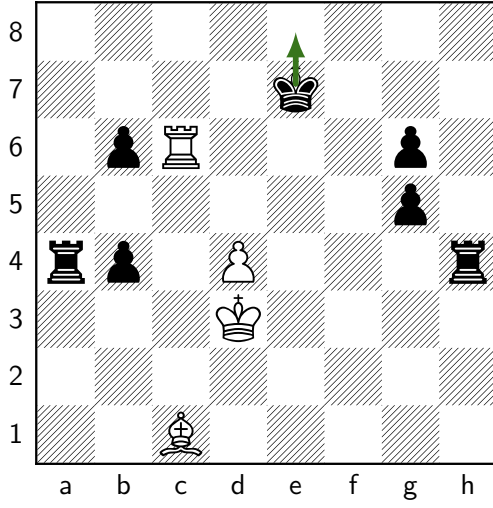


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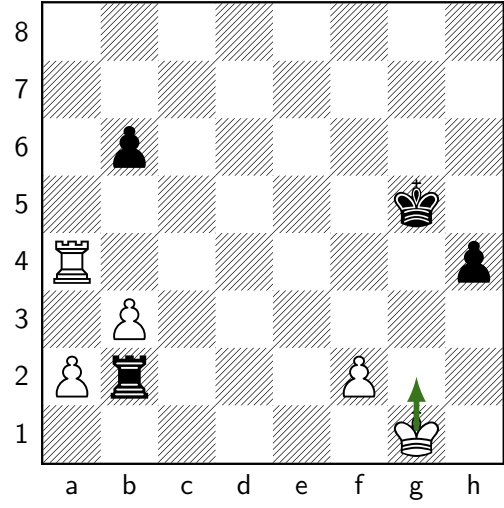


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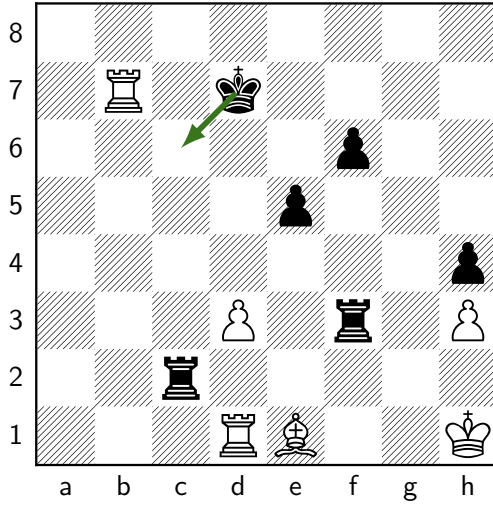
Figure 39: **Move 45, Black: Rh4+.** (a) Our position; Melanie plays Rh4+ (h8h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#845 (ply 80) [structural distance $d=24$, QAP obj=24]: the master played Bd7; under μ our Rh1 plays the role of that game’s Ba4 (a **cross-type** correspondence); cross-type roles: our bc1 as their Rd5, our Ra4 as their be7; transferred move h8h4 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#2413 (ply 93) [structural distance $d=22$, QAP obj=25]: the master played Rb6; under μ our Pb5 plays the role of that game’s Rd3 (a **cross-type** correspondence); cross-type roles: our rc6 as their bb7, our Rh8 as their pb3; transferred move b4b3. (d) Retrieved from ccr1_top_p0003#885 (ply 90) [structural distance $d=24$, QAP obj=23]: the master played Rg2; under μ our Ra5 plays the role of that game’s Rg7; transferred move a4a7. Synthesis: 1 retrieved analog transferred to Rh4+ (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



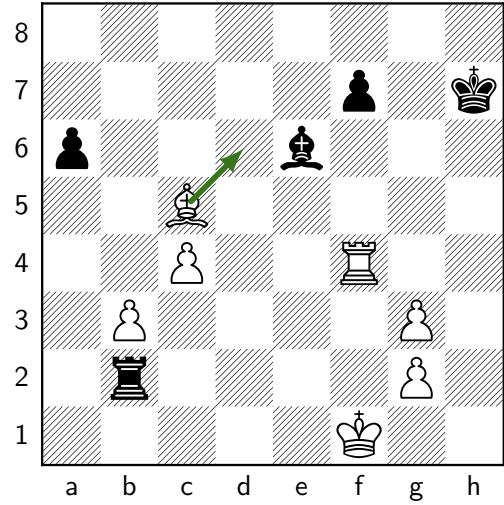
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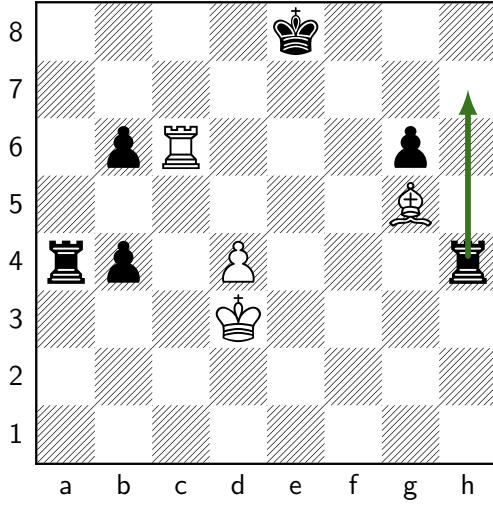


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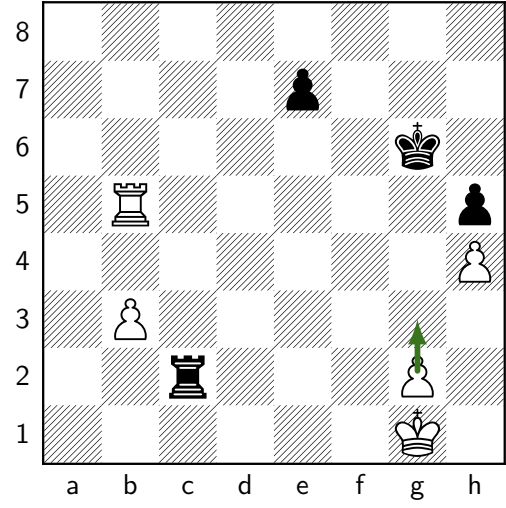


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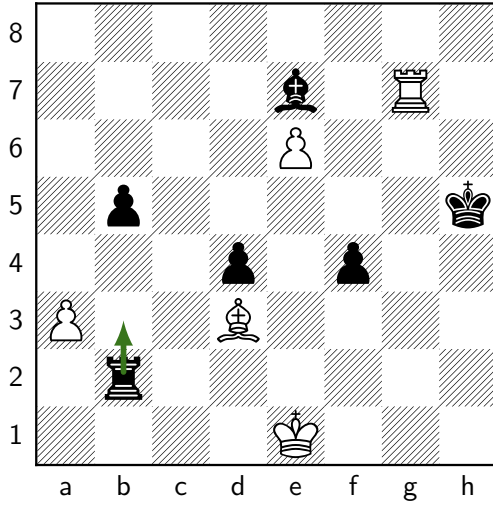
Figure 40: **Move 46, Black: Ke8 (e7e8).** (a) Our position; Melanie plays Ke8 (e7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#83` (ply 82) [structural distance $d=27$, QAP obj=39]: the master played Kg2; under μ our Ke2 plays the role of that game’s Kg1; transferred move e7e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2174` (ply 81) [structural distance $d=25$, QAP obj=3]: the master played Kc6; under μ our Ke2 plays the role of that game’s Kd2; cross-type roles: our bc1 as their Pf6, our Rh4 as their Pe5; transferred move e7e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1798` (ply 90) [structural distance $d=27$, QAP obj=30]: the master played Bd6; under μ our Ra5 plays the role of that game’s Bc5 (a **cross-type** correspondence); cross-type roles: our bc1 as their Rf4, our Rh4 as their be6; transferred move a4a5. Synthesis: 2 retrieved analogs transferred to Ke8 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



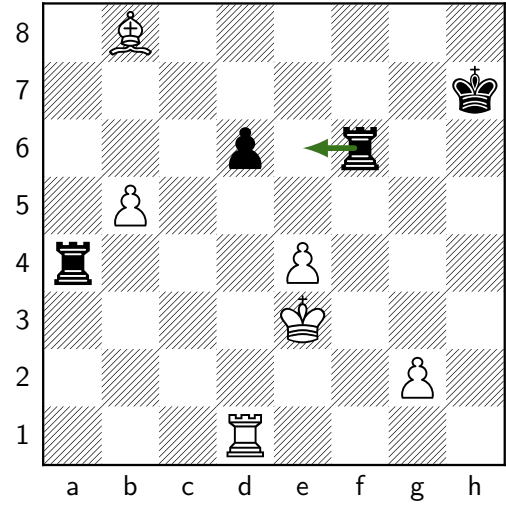
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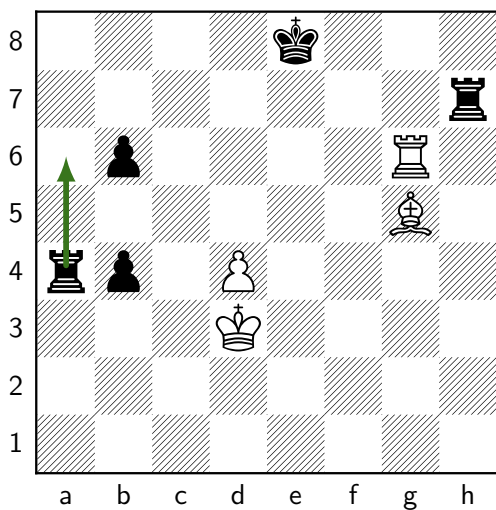


(c)

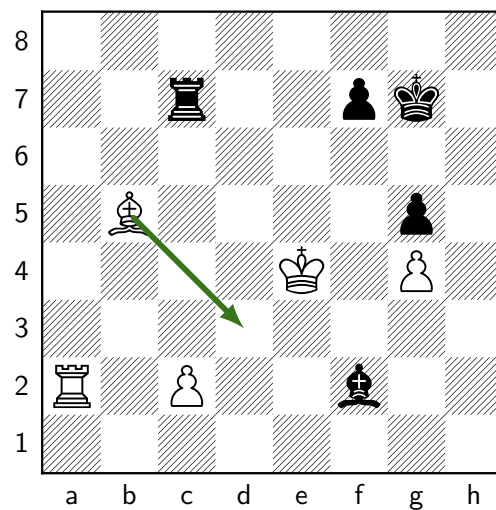


(d)

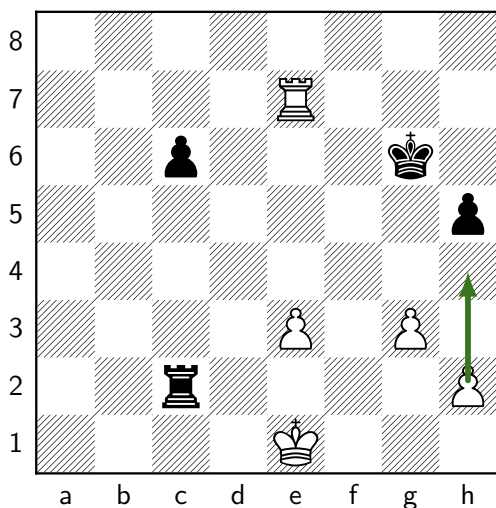
Figure 41: **Move 47, Black: Rh7 (h4h7)**. (a) Our position; Melanie plays Rh7 (h4h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1633` (ply 100) [structural distance $d=16$, QAP obj=3]: the master played g3; under μ our *Rh5* plays the role of that game’s *Pg2* (a **cross-type** correspondence); cross-type roles: our *bg5* as their *rc2*, our *rc6* as their *pe7*; transferred move h4h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2041` (ply 105) [structural distance $d=18$, QAP obj=16]: the master played Rb3; under μ our *Ra5* plays the role of that game’s *Rb7*; cross-type roles: our *Rh4* as their *Be7*, our *bg5* as their *rg7*; transferred move a4a2. (d) Retrieved from `ccr1_top_p0002#2288` (ply 95) [structural distance $d=16$, QAP obj=5]: the master played Re6; under μ our *Ra5* plays the role of that game’s *Rf3*; cross-type roles: our *rc6* as their *bb8*, our *bg5* as their *rd1*; transferred move a4a6. Synthesis: 1 retrieved analog transferred to Rh7 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



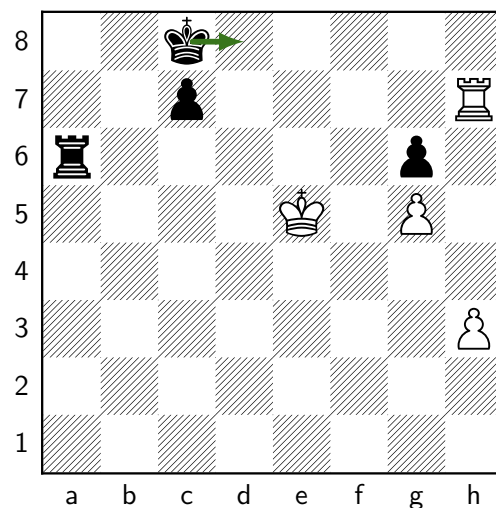
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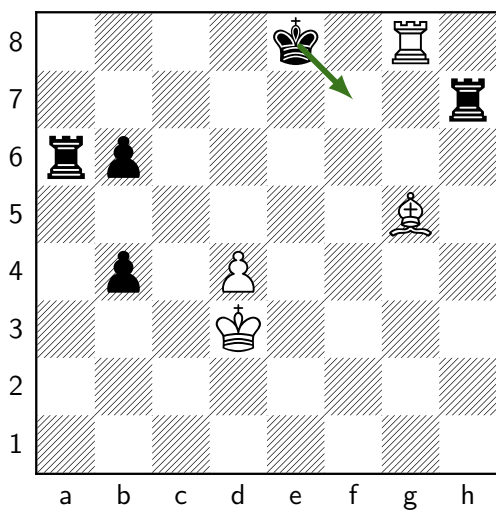


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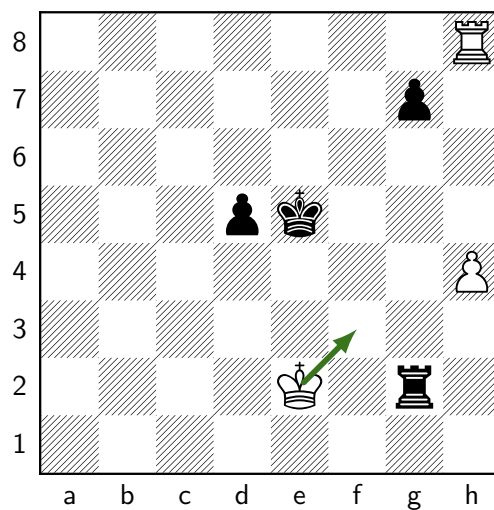


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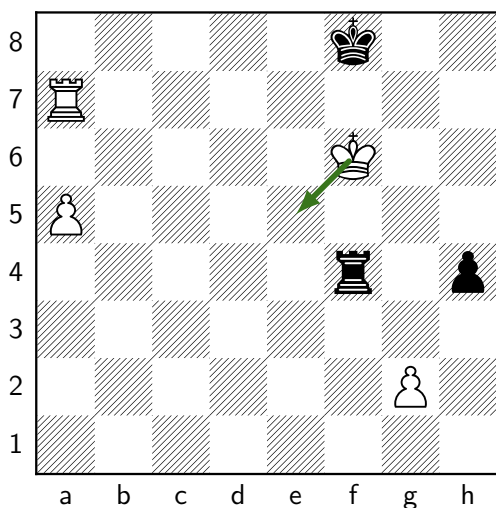
Figure 42: **Move 48, Black: Ra6.** (a) Our position; Melanie plays Ra6 (a4a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1699` (ply 108) [structural distance $d=10$, QAP obj=20]: the master played Bd3; under μ our Ra5 plays the role of that game’s Bb5 (a **cross-type** correspondence); transferred move a4a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#671` (ply 86) [structural distance $d=12$, QAP obj=20]: the master played h4; under μ our Pb3 plays the role of that game’s Ph2; cross-type roles: our Rh7 as their ph5, our bg5 as their pc6; transferred move b6b5. (d) Retrieved from `ccr1_top_p0002#3600` (ply 91) [structural distance $d=13$, QAP obj=19]: the master played Kd8; under μ our Kc1 plays the role of that game’s Kc1; cross-type roles: our bg5 as their ph3; transferred move e8f8. Synthesis: 1 retrieved analog transferred to Ra6 (summed QAP objective 20, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



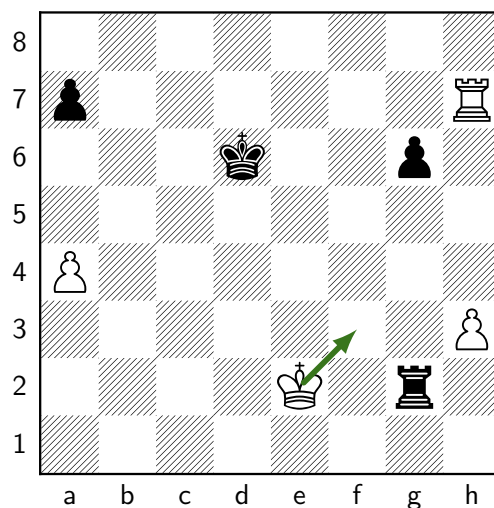
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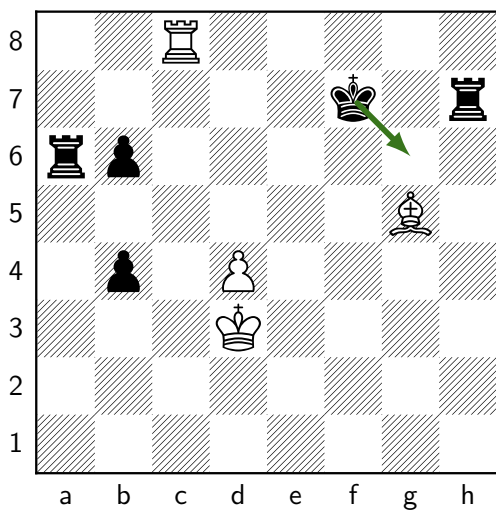


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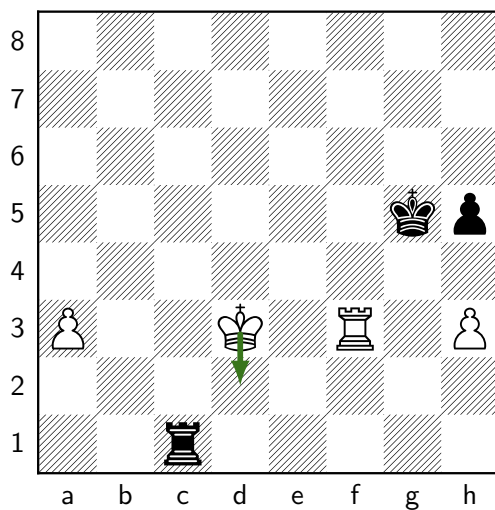


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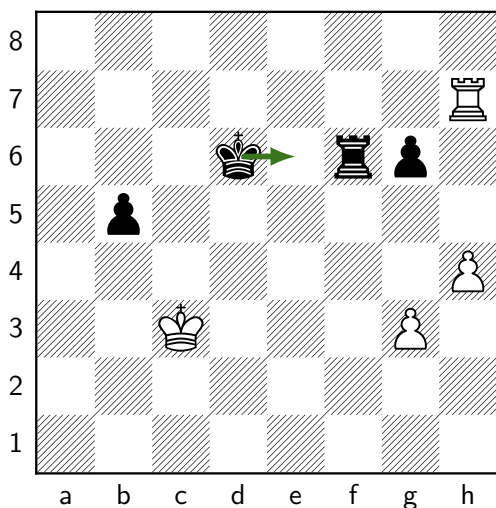
Figure 43: **Move 49, Black: Kf7.** (a) Our position; Melanie plays Kf7 (e8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1880` (ply 104) [structural distance $d=14$, QAP obj=68]: the master played Kf3; under μ our Ke1 plays the role of that game’s Ke2; cross-type roles: our Rh7 as their pg7; transferred move e8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#585` (ply 110) [structural distance $d=12$, QAP obj=68]: the master played Ke5; under μ our Ke1 plays the role of that game’s Kf6; cross-type roles: our Rh7 as their Pg2, our bg5 as their ph4; transferred move e8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3528` (ply 92) [structural distance $d=13$, QAP obj=67]: the master played Kf3; under μ our Ke1 plays the role of that game’s Ke2; cross-type roles: our Rh7 as their Ph3, our bg5 as their pg6; transferred move e8f7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf7 (summed QAP objective 251, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



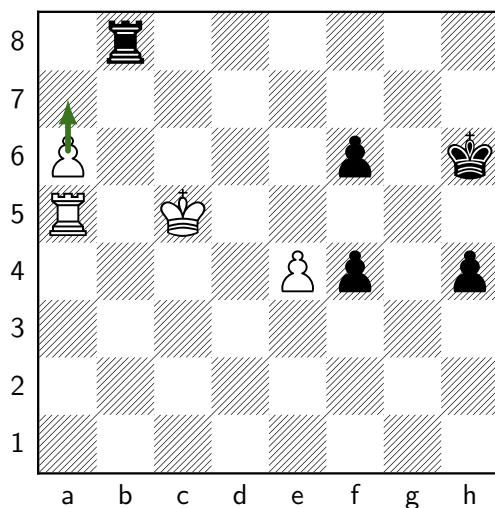
(a)



(b)



(c)



(d)

Figure 44: **Move 50, Black: Kg6.** (a) Our position; Melanie plays Kg6 (f7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#925` (ply 94) [structural distance $d=8$, QAP obj=2]: the master played Kd2; under μ our Kf2 plays the role of that game’s Kd3; cross-type roles: our bg5 as their Pa3; transferred move f7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#789` (ply 95) [structural distance $d=12$, QAP obj=4]: the master played Ke6; under μ our Kf2 plays the role of that game’s Kd3; cross-type roles: our bg5 as their pg3; transferred move f7g7. (d) Retrieved from `ccr1_top_p0002#1072` (ply 100) [structural distance $d=11$, QAP obj=3]: the master played a7; under μ our Pb5 plays the role of that game’s Pa6; cross-type roles: our Ra6 as their Pe4, our bg5 as their ph4; transferred move b4b3. Synthesis: 2 retrieved analogs transferred to Kg6 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

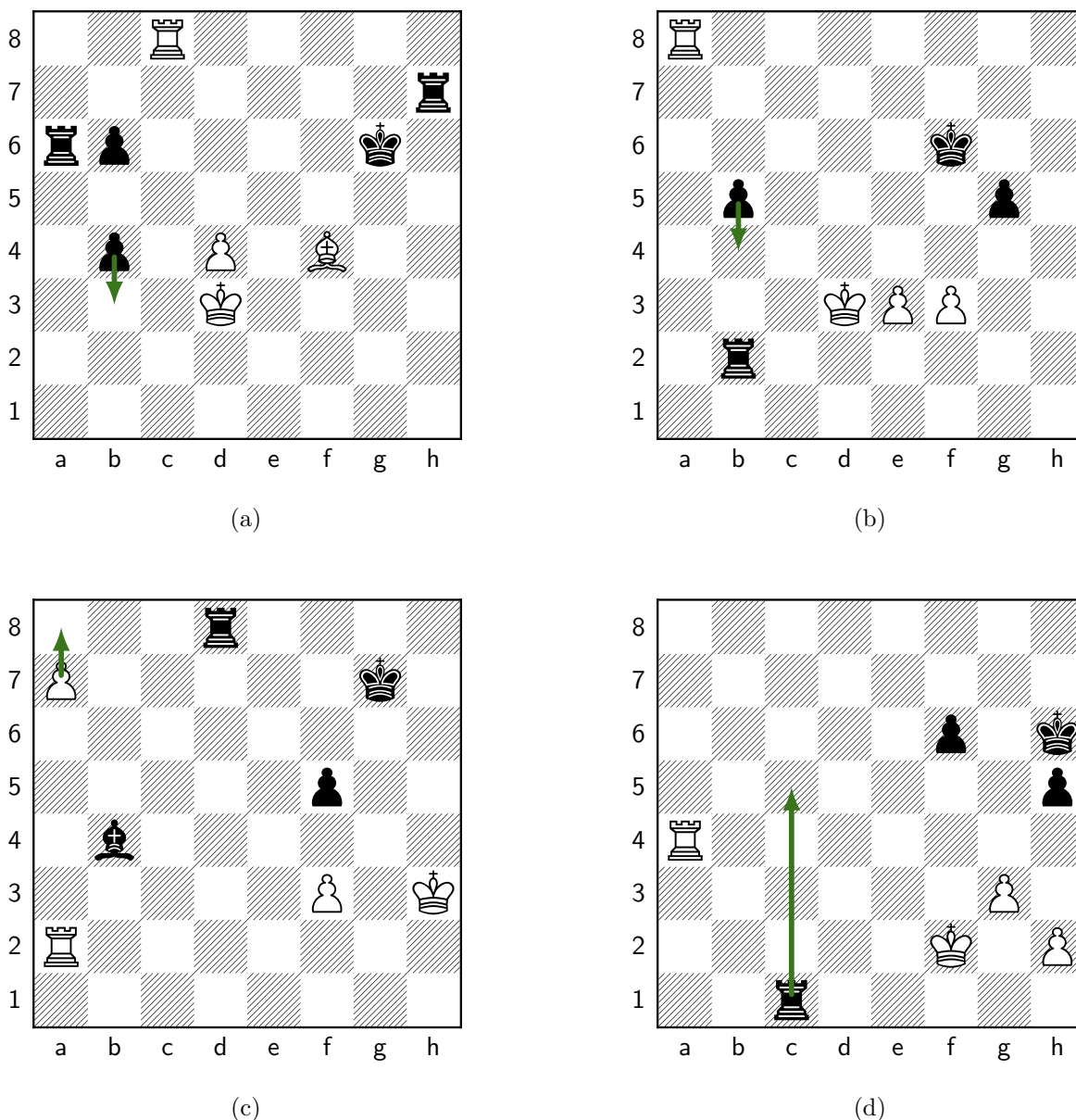
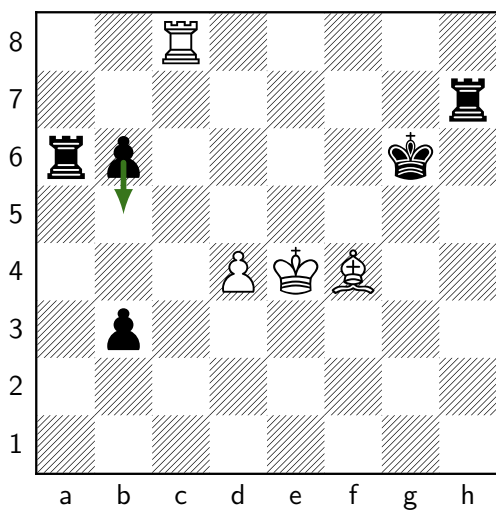
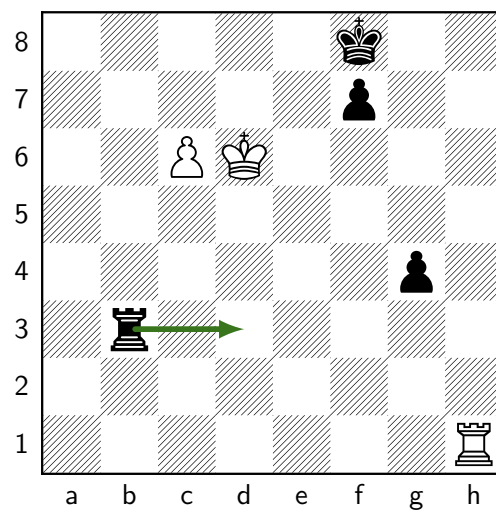


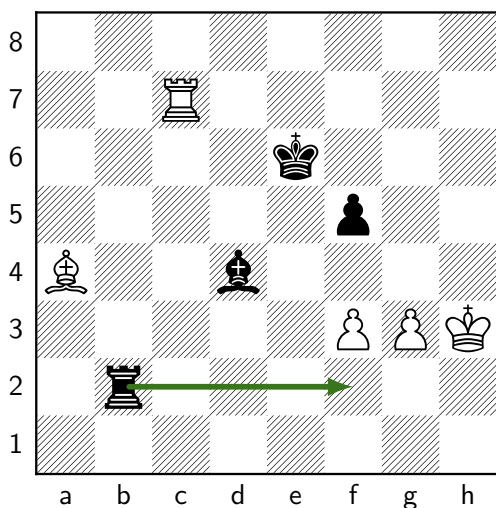
Figure 45: **Move 51, Black: b3**. (a) Our position; Melanie plays **b3** (**b4b3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2896` (ply 99) [structural distance $d=10$, QAP obj=2]: the master played **b4**; under μ our *Pb5* plays the role of that game’s *Pb4*; cross-type roles: our *bf4* as their *pf3*; transferred move **b4b3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3067` (ply 108) [structural distance $d=11$, QAP obj=1]: the master played **a8=Q**; under μ our *Pb5* plays the role of that game’s *Pa7*; transferred move **b4b3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2733` (ply 113) [structural distance $d=12$, QAP obj=1]: the master played **Rc5**; under μ our *Pb5* plays the role of that game’s *Rc8* (a **cross-type** correspondence); cross-type roles: our *Rh7* as their *Pf6*, our *Ra6* as their *ph2*; transferred move **b4b3** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **b3** (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



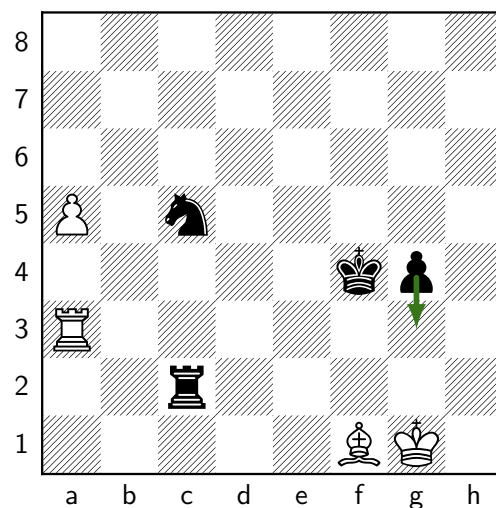
(a)



(b)

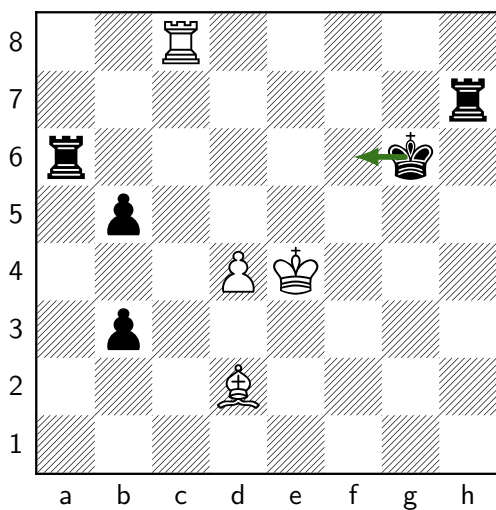


(c)

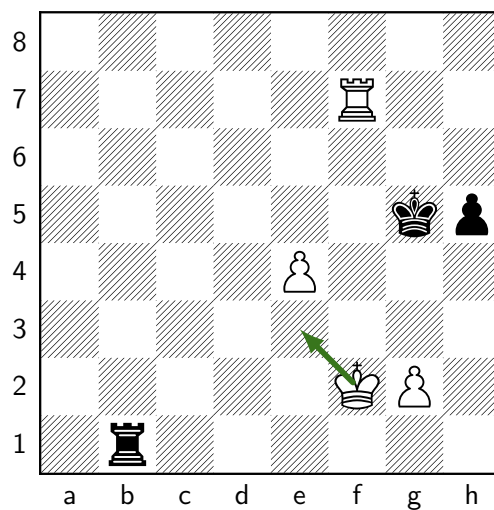


(d)

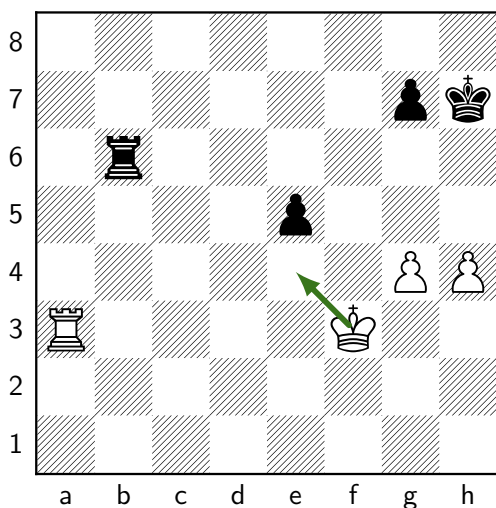
Figure 46: **Move 52, Black: b5.** (a) Our position; Melanie plays b5 (b6b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#42` (ply 99) [structural distance $d=16$, QAP obj=2]: the master played `Rd3+`; under μ our *Pb3* plays the role of that game’s *Rb6* (a **cross-type** correspondence); cross-type roles: our *bf4* as their *Pf7*; transferred move `b6b5` (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3332` (ply 111) [structural distance $d=15$, QAP obj=1]: the master played `Rf2`; under μ our *Pb3* plays the role of that game’s *Rb7* (a **cross-type** correspondence); cross-type roles: our *bf4* as their *rc7*, our *rc8* as their *Pf5*; transferred move `b6b5` (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#633` (ply 101) [structural distance $d=16$, QAP obj=2]: the master played `g3`; under μ our *Pb6* plays the role of that game’s *Pg5*; cross-type roles: our *Pb6* as their *Nc5*; transferred move `b3b2`. Synthesis: 2 retrieved analogs transferred to b5 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



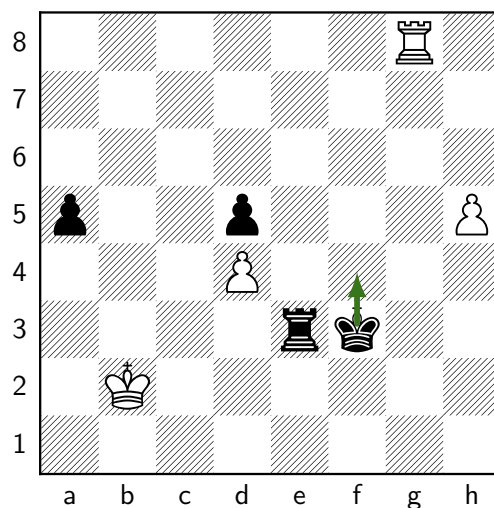
(a)



(b)

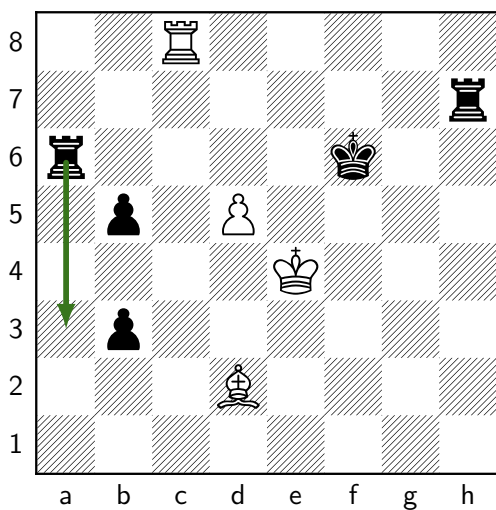


(c)

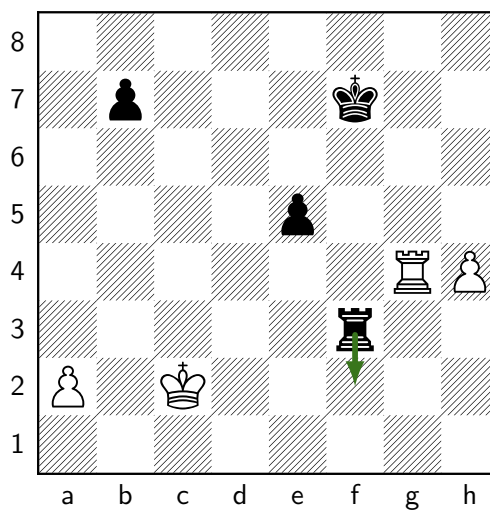


(d)

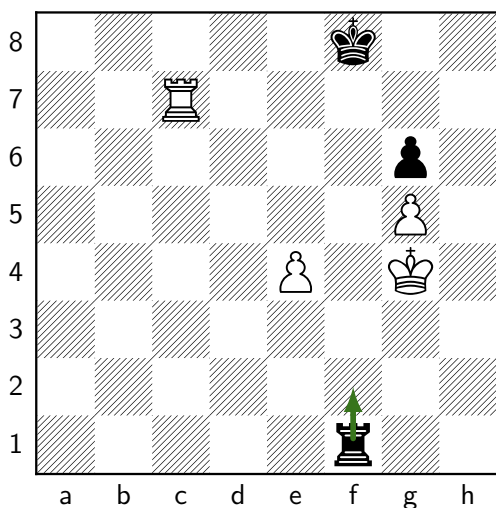
Figure 47: **Move 53, Black: Kf6**. (a) Our position; Melanie plays Kf6 (g6f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#63` (ply 118) [structural distance $d=12$, QAP obj=3]: the master played Kc3; under μ our Kg3 plays the role of that game’s Kf2; cross-type roles: our bd2 as their rb1, our Rh7 as their Pg2; transferred move g6f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3464` (ply 98) [structural distance $d=12$, QAP obj=2]: the master played Ke4; under μ our Kg3 plays the role of that game’s Kf3; cross-type roles: our Rh7 as their Pg4, our bd2 as their pe5; transferred move g6f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1638` (ply 115) [structural distance $d=10$, QAP obj=1]: the master played Kf4; under μ our Kg3 plays the role of that game’s Kf6; cross-type roles: our bd2 as their rg8, our Rh7 as their Pa5; transferred move g6f6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf6 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



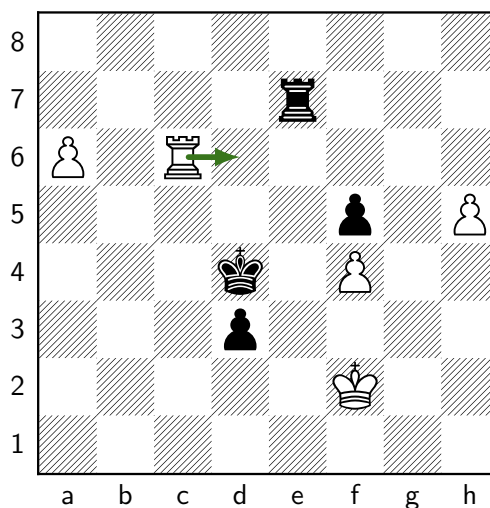
(a)



(b)

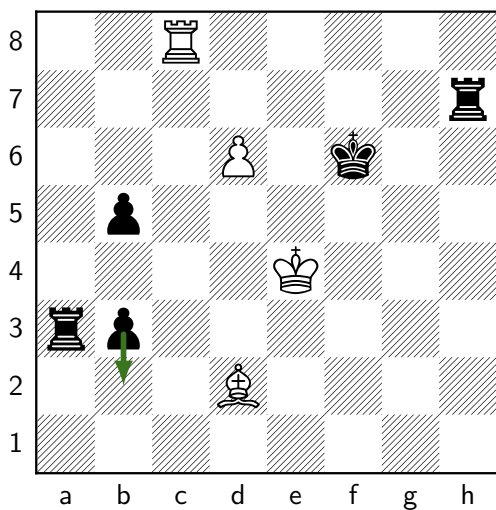


(c)

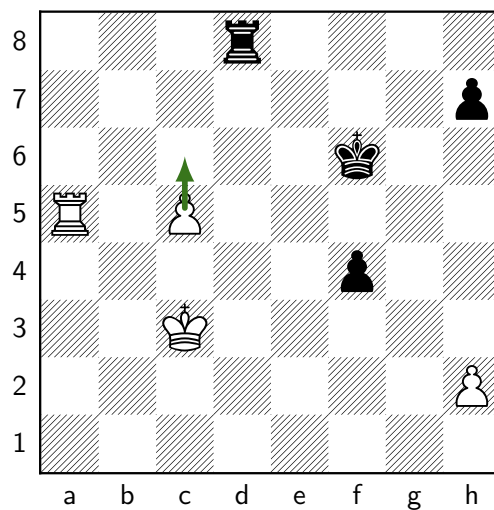


(d)

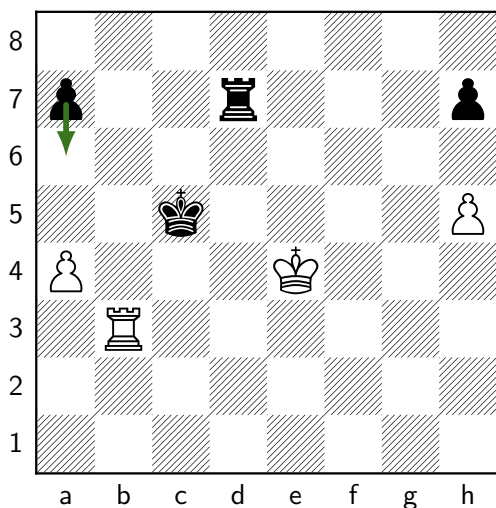
Figure 48: **Move 54, Black: Ra3**. (a) Our position; Melanie plays Ra3 (a6a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2310` (ply 99) [structural distance $d=9$, QAP obj=2]: the master played Rf2+; under μ our Ra3 plays the role of that game’s Rf6; cross-type roles: our rc8 as their Pb7, our bd2 as their rg4; transferred move a6a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3817` (ply 99) [structural distance $d=11$, QAP obj=2]: the master played Rf2; under μ our Ra3 plays the role of that game’s Rf8; cross-type roles: our pd5 as their rc7, our rc8 as their pg5; transferred move a6a3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3427` (ply 112) [structural distance $d=12$, QAP obj=1]: the master played Rd6+; under μ our Ra3 plays the role of that game’s Rc6; cross-type roles: our bd2 as their pf5, our Rh7 as their Ph5; transferred move a6a3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ra3 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



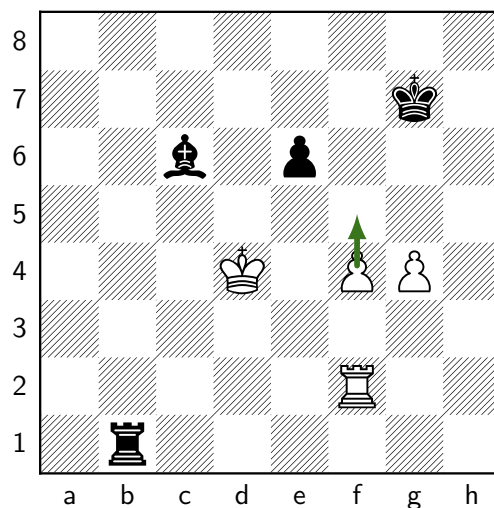
(a)



(b)

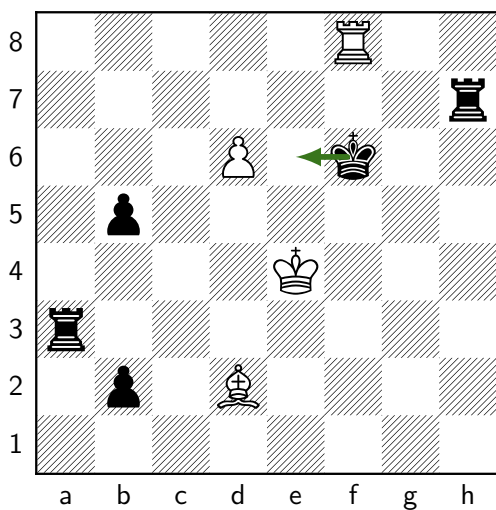


(c)

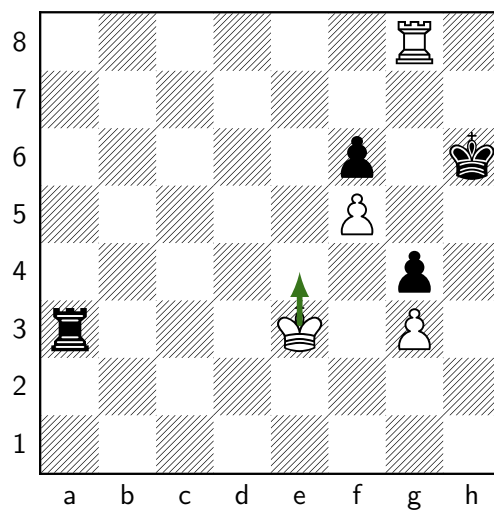


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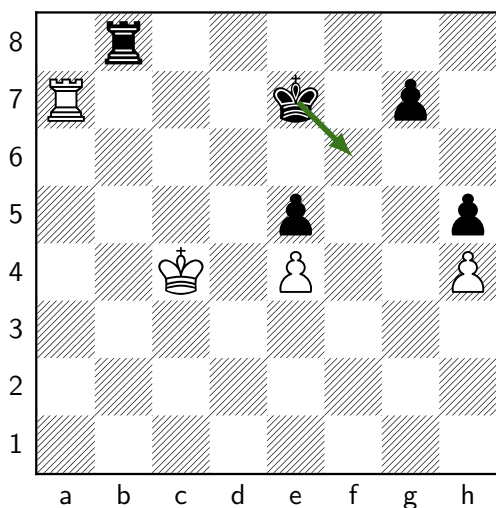
Figure 49: **Move 55, Black: b2.** (a) Our position; Melanie plays b2 (b3b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2213` (ply 122) [structural distance $d=8$, QAP obj=1]: the master played c6; under μ our $Pb6$ plays the role of that game’s $Pc5$; cross-type roles: our $Rh7$ as their $pf4$; transferred move b3b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1166` (ply 95) [structural distance $d=11$, QAP obj=1]: the master played a6; under μ our $Pb6$ plays the role of that game’s $Pa2$; cross-type roles: our $bd2$ as their $Ph7$; transferred move b3b2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1202` (ply 98) [structural distance $d=5$, QAP obj=1]: the master played f5; under μ our $Pb6$ plays the role of that game’s $Pf4$; cross-type roles: our $Rh7$ as their $Pg4$; transferred move b3b2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to b2 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



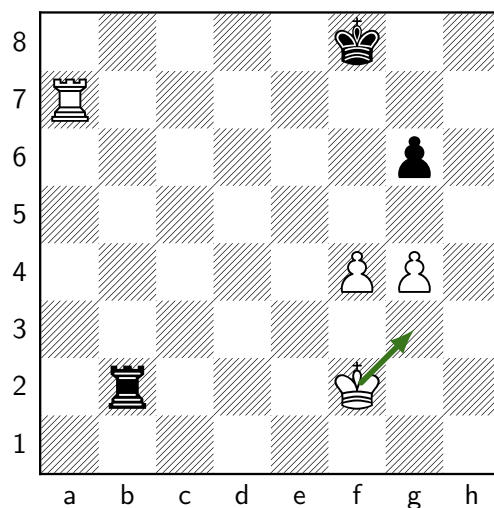
(a)



(b)

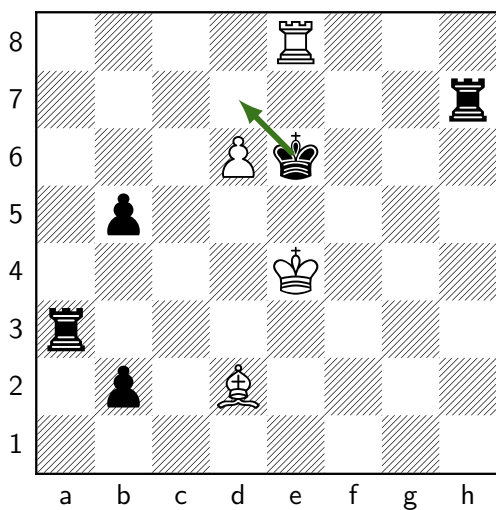


(c)

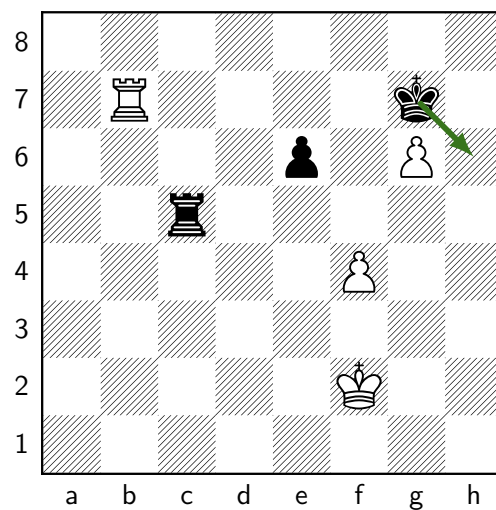


(d)

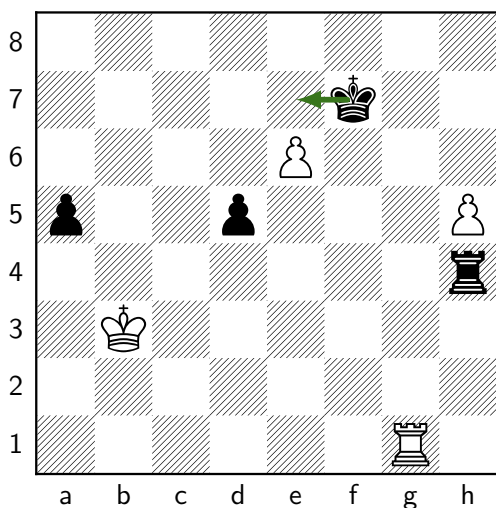
Figure 50: **Move 56, Black: Ke6**. (a) Our position; Melanie plays Ke6 (f6e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#1137 (ply 112) [structural distance $d=12$, QAP obj=34]: the master played Ke4; under μ our Kf3 plays the role of that game’s Ke3; cross-type roles: our Rh7 as their Pg3, our bd2 as their pf6; transferred move f6e6 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#507 (ply 127) [structural distance $d=11$, QAP obj=34]: the master played Kf6; under μ our Kf3 plays the role of that game’s Ke2; cross-type roles: our bd2 as their pe4, our Rh7 as their ph4; transferred move f6e6 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#2860 (ply 106) [structural distance $d=12$, QAP obj=33]: the master played Kg3; under μ our Kf3 plays the role of that game’s Kf2; cross-type roles: our Rh7 as their Pg4; transferred move f6g6. Synthesis: 2 retrieved analogs transferred to Ke6 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



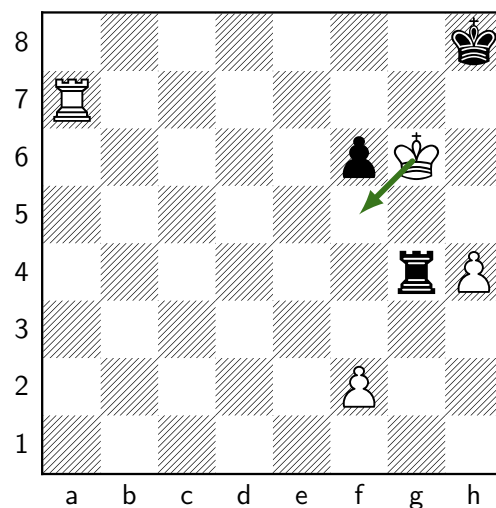
(a)



(b)

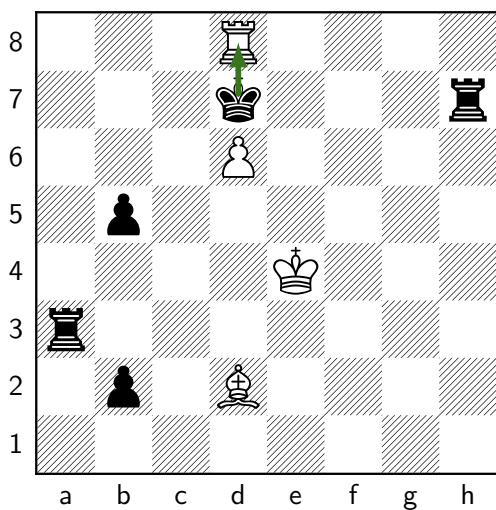


(c)

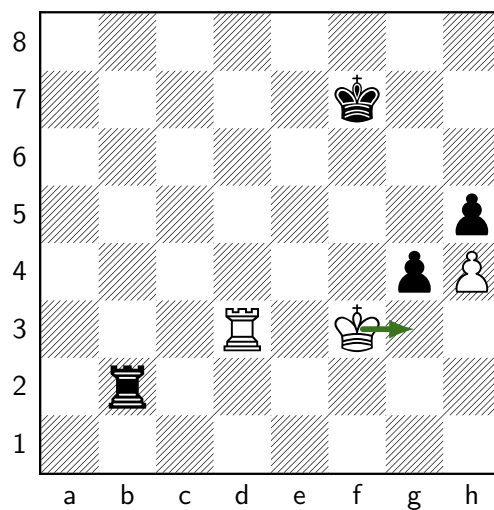


(d)

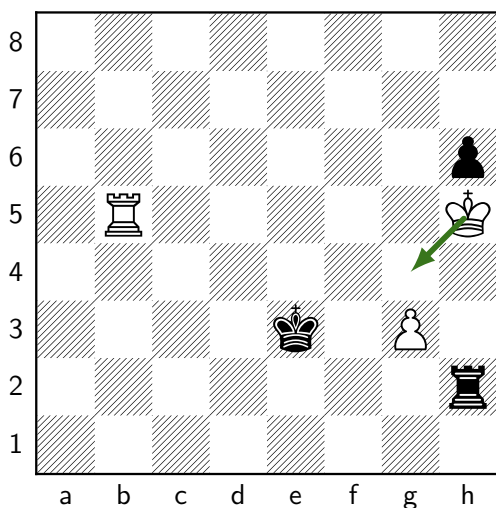
Figure 51: **Move 57, Black: Kd7.** (a) Our position; Melanie plays Kd7 (e6d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3970` (ply 101) [structural distance $d=12$, QAP obj=35]: the master played Kh6; under μ our Ke3 plays the role of that game’s Kg2; cross-type roles: our bd2 as their pf4; transferred move e6d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2105` (ply 121) [structural distance $d=15$, QAP obj=34]: the master played Ke7; under μ our Ke3 plays the role of that game’s Kf2; cross-type roles: our re8 as their pe6, our Rh7 as their Pa5; transferred move e6d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2015` (ply 106) [structural distance $d=15$, QAP obj=84]: the master played Kf5; under μ our Ke3 plays the role of that game’s Kg6; cross-type roles: our Rh7 as their Ph4; transferred move e6d6. Synthesis: 2 retrieved analogs transferred to Kd7 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



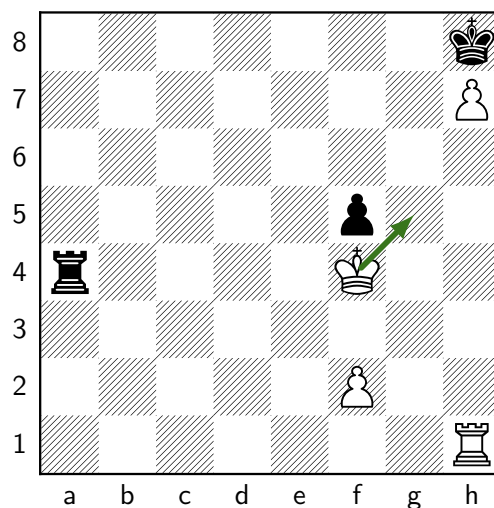
(a)



(b)

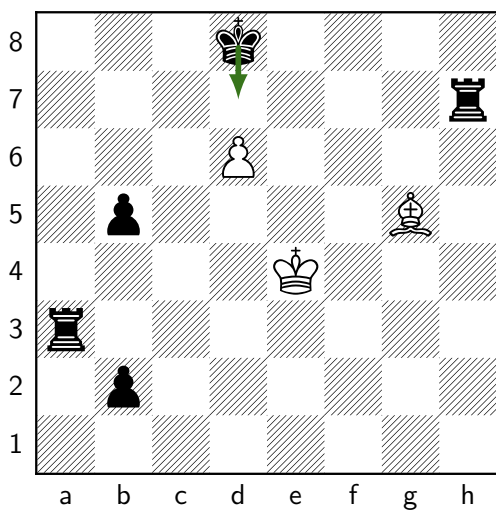


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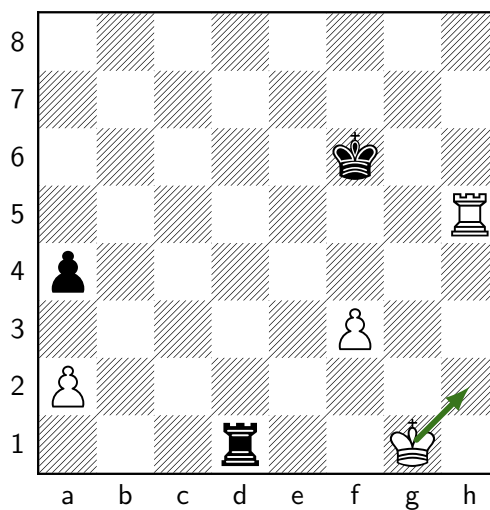


(d)

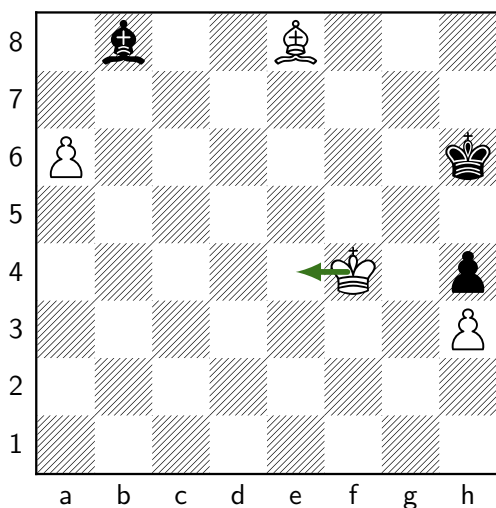
Figure 52: **Move 58, Black: Kxd8.** (a) Our position; Melanie plays Kxd8 (d7d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1333` (ply 114) [structural distance $d=20$, QAP obj=69]: the master played Kg3; under μ our Kd2 plays the role of that game’s Kf3; cross-type roles: our pd6 as their rb2, our rd8 as their pg4; transferred move d7d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2013` (ply 110) [structural distance $d=17$, QAP obj=29]: the master played Kg4; under μ our Kd2 plays the role of that game’s Kh5; cross-type roles: our Ra3 as their Pg3; transferred move d7d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2938` (ply 124) [structural distance $d=20$, QAP obj=18]: the master played Kg5; under μ our Kd2 plays the role of that game’s Kf4; cross-type roles: our Pb5 as their Rh1, our Rh7 as their Pf2; transferred move d7d8 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kxd8 (summed QAP objective 125, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



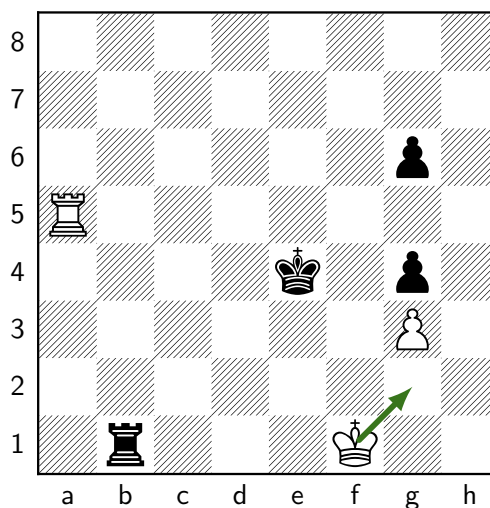
(a)



(b)

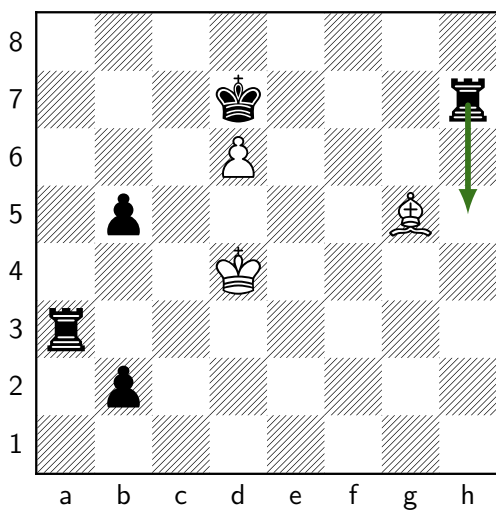


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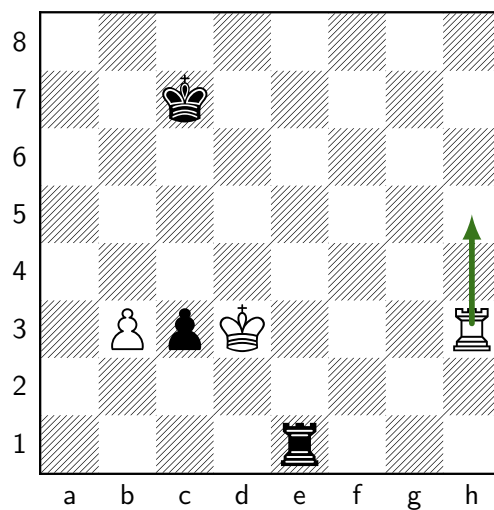


(d)

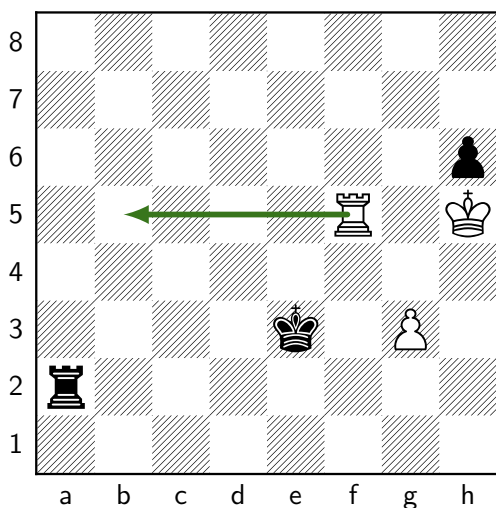
Figure 53: **Move 59, Black: Kd7.** (a) Our position; Melanie plays Kd7 (d8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2490` (ply 128) [structural distance $d=13$, QAP obj=66]: the master played Kh2; under μ our *Kd1* plays the role of that game’s *Kg1*; cross-type roles: our *bg5* as their *rd1*, our *Rh7* as their *pa4*; transferred move d8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1943` (ply 114) [structural distance $d=12$, QAP obj=51]: the master played Ke4; under μ our *Kd1* plays the role of that game’s *Kf4*; cross-type roles: our *Rh7* as their *Ph3*, our *Ra3* as their *Be8*; transferred move d8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1746` (ply 132) [structural distance $d=12$, QAP obj=50]: the master played Kg2; under μ our *Kd1* plays the role of that game’s *Kf1*; cross-type roles: our *bg5* as their *rb1*, our *Ra3* as their *Pg3*; transferred move d8d7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kd7 (summed QAP objective 266, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



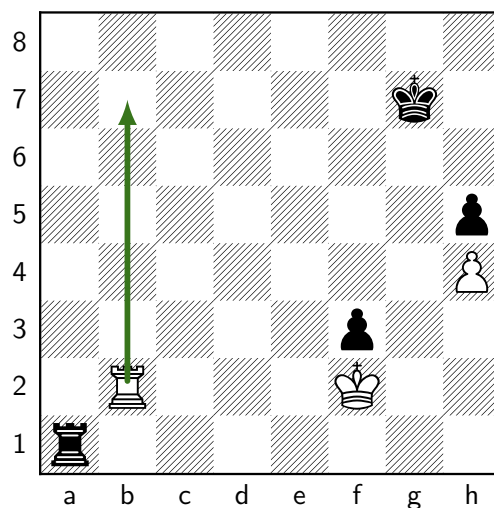
(a)



(b)

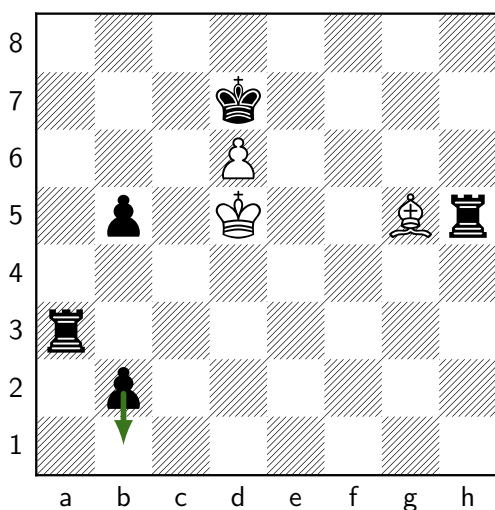


(c)

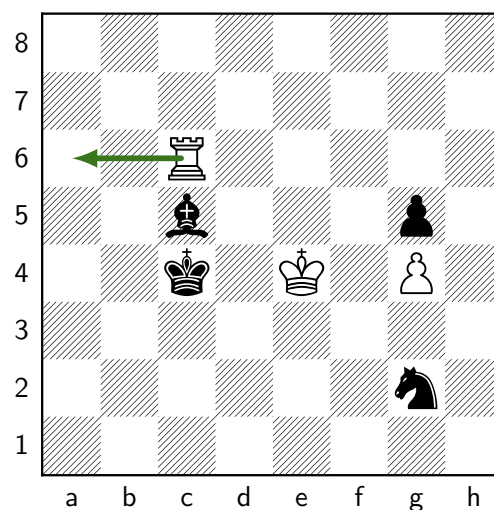


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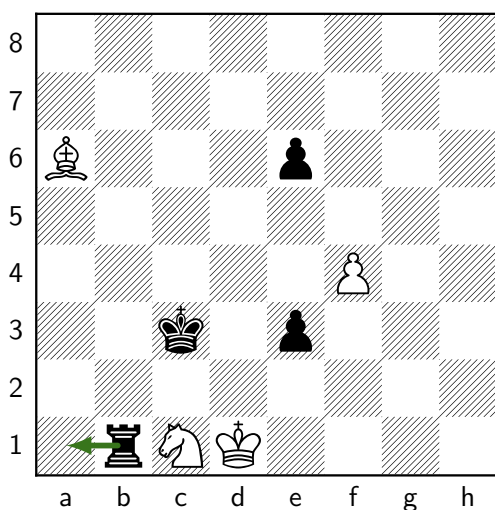
Figure 54: **Move 60, Black: Rh5**. (a) Our position; Melanie plays Rh5 (**h7h5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3359` (ply 110) [structural distance $d=14$, QAP obj=22]: the master played Rh5; under μ our Rh2 plays the role of that game’s Rh3; cross-type roles: our bg5 as their rel1; transferred move **h7h5 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#2013` (ply 108) [structural distance $d=10$, QAP obj=21]: the master played Rb5; under μ our Rh2 plays the role of that game’s Rf5; cross-type roles: our bg5 as their ra2; transferred move **h7h5 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#2445` (ply 124) [structural distance $d=10$, QAP obj=23]: the master played Rb7+; under μ our Rh2 plays the role of that game’s Rb2; cross-type roles: our bg5 as their ra1, our Ra3 as their ph5; transferred move **h7h2**. Synthesis: 2 retrieved analogs transferred to Rh5 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



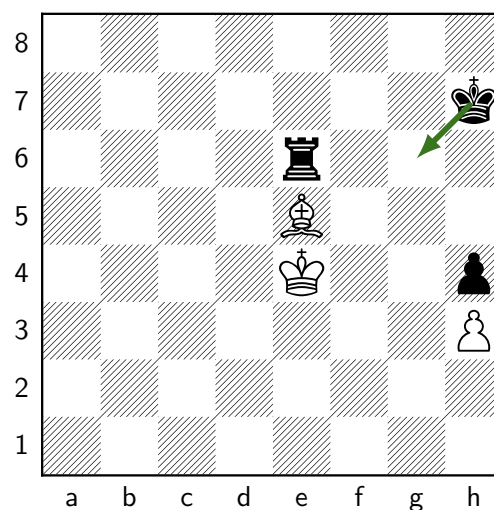
(a)



(b)



(c)



(d)

Figure 55: **Move 61, Black: b1=Q.** (a) Our position; Melanie plays **b1=Q** (b2b1q). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0001#2045` (ply 122) [structural distance $d=10$, QAP obj=39]: the master played **Ra6**; under μ our *Rh4* plays the role of that game’s *Rc6*; cross-type roles: our *pd6* as their *ng2*, our *Ra3* as their *pg5*; transferred move **h5h2**. (c) Retrieved from `ccr1_top.p0000#2908` (ply 129) [structural distance $d=15$, QAP obj=39]: the master played **Ra1**; under μ our *Rh4* plays the role of that game’s *Rb8*; cross-type roles: our *pd6* as their *ba6*, our *Ra3* as their *Pe6*; transferred move **h5h1**. (d) Retrieved from `ccr1_top.p0003#2261` (ply 115) [structural distance $d=15$, QAP obj=39]: the master played **Kg6**; under μ our *Kd2* plays the role of that game’s *Kh2*; transferred move **d7c8**. Synthesis: no retrieved analog transferred to **b1=Q** – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

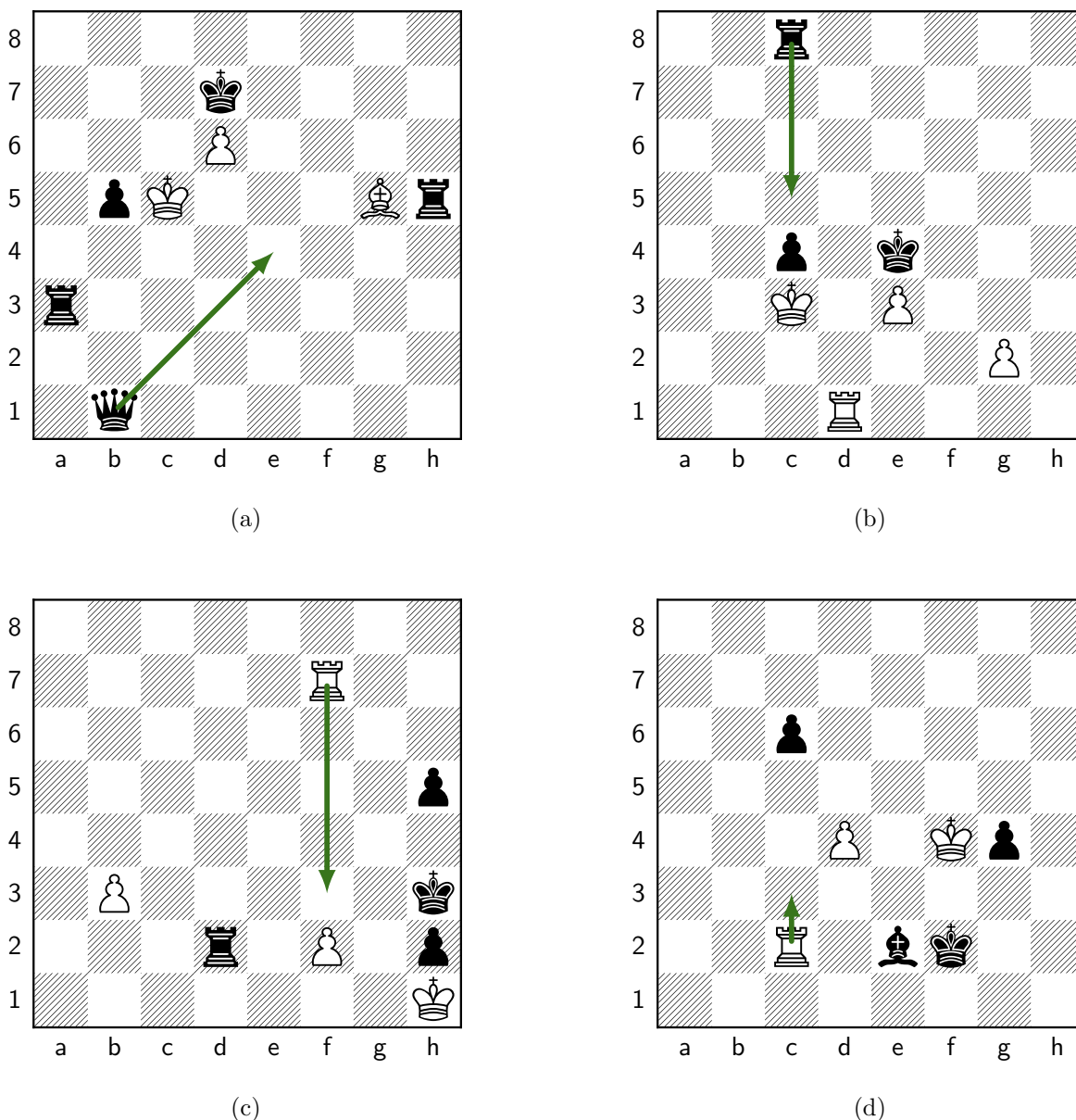
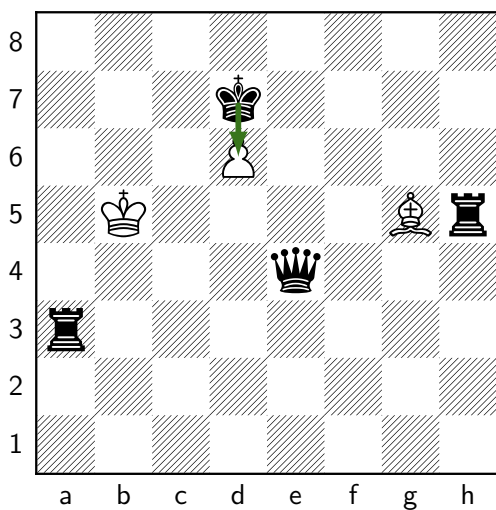
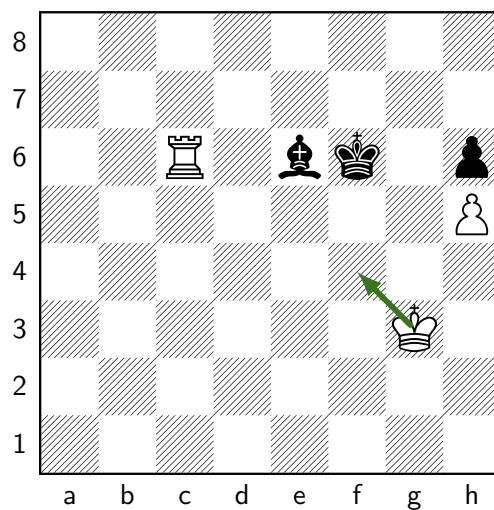


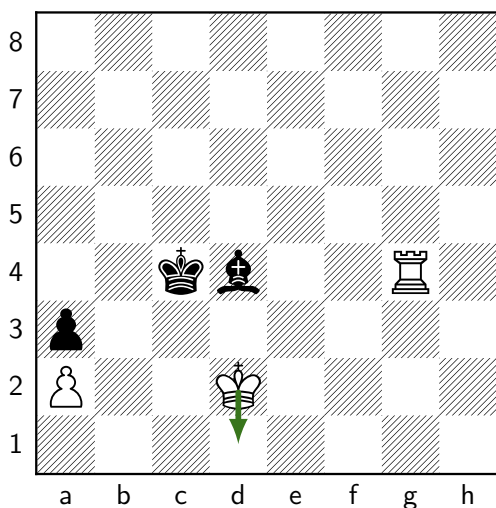
Figure 56: **Move 62, Black: Qe4**. (a) Our position; Melanie plays **Qe4** (b1e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#316` (ply 123) [structural distance $d=20$, QAP obj=16]: the master played **Rc5**; under μ our **Qb8** plays the role of that game's **Rc1** (a **cross-type** correspondence); cross-type roles: our **Rh5** as their **pg2**, our **bg5** as their **rd1**; transferred move **b1e4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#519` (ply 106) [structural distance $d=19$, QAP obj=10]: the master played **Rf3+**; under μ our **Qb8** plays the role of that game's **Rf7** (a **cross-type** correspondence); cross-type roles: our **bg5** as their **Pb3**, our **Ra3** as their **ph5**; transferred move **b1e4** (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2476` (ply 130) [structural distance $d=17$, QAP obj=27]: the master played **Rc3**; under μ our **Rh4** plays the role of that game's **Rc2**; cross-type roles: our **Ra3** as their **Pd4**; transferred move **h5h8**. Synthesis: 2 retrieved analogs transferred to **Qe4** (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



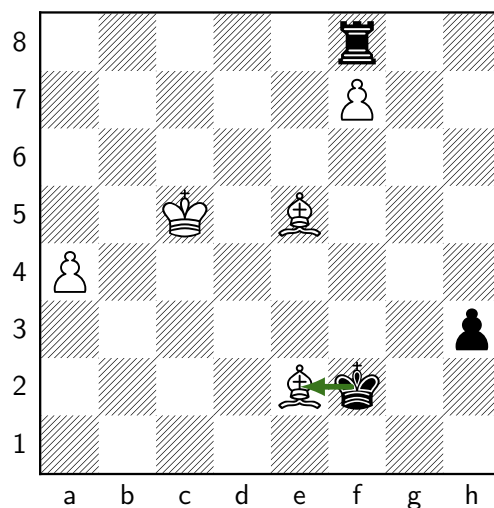
(a)



(b)

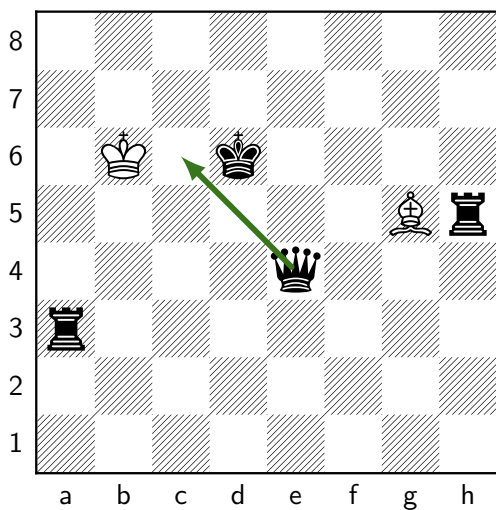


(c)

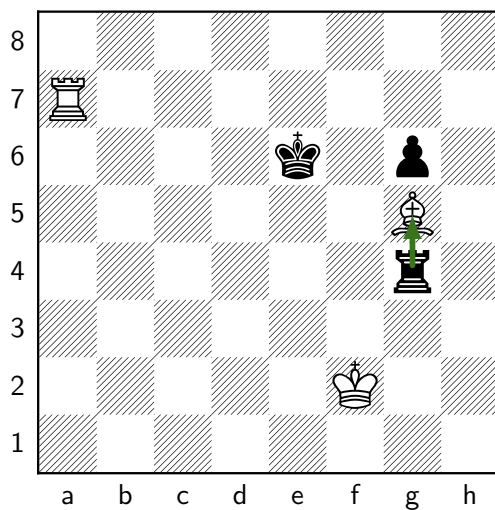


(d)

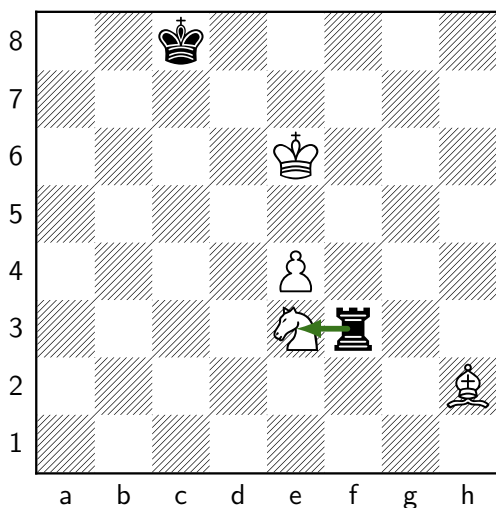
Figure 57: **Move 63, Black: Kxd6**. (a) Our position; Melanie plays Kxd6 (d7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1676` (ply 130) [structural distance $d=15$, QAP obj=20]: the master played Kf4; under μ our Kd2 plays the role of that game’s Kg3; cross-type roles: our Qe4 as their Ph5; transferred move d7d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1213` (ply 128) [structural distance $d=15$, QAP obj=14]: the master played Kd1; under μ our Kd2 plays the role of that game’s Kd2; cross-type roles: our Qe4 as their Pa2; transferred move d7d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3337` (ply 121) [structural distance $d=14$, QAP obj=12]: the master played Kxe2; under μ our Kd2 plays the role of that game’s Kf7; cross-type roles: our bg5 as their pf7, our Qe4 as their be5; transferred move d7d6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kxd6 (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



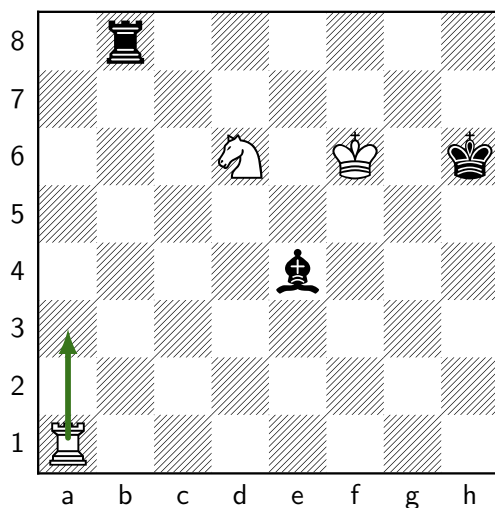
(a)



(b)



(c)



(d)

Figure 58: **Move 64, Black: Qc6#.** (a) Our position; Melanie plays Qc6# (e4c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1242` (ply 127) [structural distance $d=6$, QAP obj=19]: the master played Rxc5; under μ our Rh4 plays the role of that game’s Rg5; cross-type roles: our Qe4 as their Pg6; transferred move h5g5. (c) Retrieved from `ccr1_top_p0000#3049` (ply 135) [structural distance $d=8$, QAP obj=19]: the master played Rxe3; under μ our Rh4 plays the role of that game’s Rf6; cross-type roles: our Ra3 as their pe4, our bg5 as their ne3; transferred move h5h2. (d) Retrieved from `ccr1_top_p0001#3765` (ply 136) [structural distance $d=9$, QAP obj=10]: the master played Ra3; under μ our Qe5 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Rh5 as their Nd6; transferred move e4g6. Synthesis: no retrieved analog transferred to Qc6# – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.